Integration of III-V Optical Devices and Interconnects on Si Using SiGe Virtual Substrates

by

Vicky Kung-Fan Yang

S.B. Materials Science and Engineering
Massachusetts Institute of Technology, 1997

Submitted to the Department of Materials Science and Engineering
in Partial Fulfillment of the Requirements for the Degree of

Doctor of Philosophy in Electronic Materials

at the
Massachusetts Institute of Technology

June 2002

Signature of Author: ______________________
Department of Materials Science and Engineering
May 10, 2002

Certified by: ______________________
Eugene A. Fitzgerald
Professor of Materials Science and Engineering
Thesis Supervisor

Accepted by: ______________________
Harry L. Tuller
Professor of Ceramics and Electronic Materials
Chair, Departmental Committee on Graduate Students

Integration of III-V Optical Devices and Interconnects on Si Using SiGe Virtual Substrates

by

Vicky Kung-Fan Yang

Submitted to the Department of Materials Science and Engineering
on May 10, 2002 in Partial Fulfillment of the Requirements for the
Degree of Doctor of Philosophy in Electronic Materials

ABSTRACT

Because of the limitations to the functionality that Si can provide, integration of light emitting materials such as GaAs and other III-V materials provides the promise for the combination of electrical and optical devices onto a Si platform. Monolithic integration, compare to hybrid integration, is the more cost effective and reliable way to combine these dissimilar materials. However, materials compatibility issues such as lattice mismatch, polar-on-non-polar epitaxy, and thermal mismatch make the direct growth of high quality GaAs-based material on Si difficult. However, with the use of SiGe virtual substrates, Si substrates that have the Ge lattice constant, this goal can be achieved since Ge and GaAs have a small lattice mismatch of only 0.07%. High quality GaAs on Si having a threading dislocation density lower than 3×10^{6} cm^{-2} on offcut SiGe virtual substrates has been demonstrated.

While utilizing SiGe virtual substrate technology to enable high quality GaAs on Si, we have studied the crack formation in GaAs layers on SiGe resulting from the thermal mismatch between the film and the substrate. GaAs has a thermal expansion coefficient of 5.8×10^{-6} /°C while the value for Si and the SiGe substrate is 2.6×10^{-6} /°C. The thermal expansion coefficient of the SiGe virtual substrate is mainly dictated by the underlying Si substrate. As a result of the thermal mismatch, crack arrays can form in GaAs layers grown over a critical thickness after a temperature change. These cracks can act as electrical shorting paths and can also limit the usable area for device fabrication, and therefore need to be eliminated. We have reported the experimental and calculated critical cracking thickness of GaAs on Si and SiGe in this work and have discussed possible strategies in the control of crack formation.

We have also studied the luminescent characteristics of $In_{0.2}Ga_{0.8}As$ quantum wells grown on the SiGe virtual substrates and other substrates such as GaAs, Si, and Ge. The luminescent intensities of InGaAs quantum wells on SiGe approach that of ones grown on GaAs substrates and out performs structures on Si and Ge substrates, an unexpected advantage for InGaAs quantum well integration on SiGe.

In this work, we have also reported the first working optical interconnect on Si using SiGe virtual substrate technology. This relatively simple device structure proves the feasibility of monolithically integrated optical circuit architectures on the Si microelectronic platform. The performance of various optical interconnect structures is analyzed, and an alternative waveguide structure is proposed.

Thesis Supervisor: Eugene A. Fitzgerald
Title: Professor of Materials Science and Engineering

Table of Contents

LIST OF FIGURES **8**

LIST OF TABLES **15**

ACKNOWLEDGEMENTS **16**

1 OVERVIEW OF MATERIALS ISSUES IN III-V INTEGRATION ON SI .. **19**

1.1 MOTIVATION FOR III-V INTEGRATION ON SI 20

1.2 CHALLENGES OF MONOLITHIC INTEGRATION OF III-V MATERIALS ON SI 23

1.2.1 Lattice Mismatch *23*

1.2.2 Polar on Non-Polar Epitaxy *30*

1.2.3 Thermal Mismatch *32*

1.3 III-V ON SI INTEGRATION STRATEGY 34

1.3.1 SiGe Virtual Substrates *34*

1.3.2 Suppression of Antiphase Boundaries *40*

1.3.3 GaAs on Si Integration Results *41*

1.4 EXPERIMENTAL SYNOPSIS 45

2 GROWTH AND CHARACTERIZATION OF III-V MATERIALS ON SIGE VIRTUAL SUBSTRATES **47**

2.1 METALORGANIC CHEMICAL VAPOR DEPOSITION OVERVIEW 48

2.2 MOCVD REACTOR 51

2.3 GROWTH PROCEDURE 53

2.4 CHARACTERIZATION TECHNIQUES 55

2.4.1 Surface Morphology Characterization *56*

2.4.2 Dislocation Characterization *56*

2.2.3 Other Characterization Techniques *58*

3 GROWTH AND FABRICATION OF INAS/GAAS/SI AVALANCHE PHOTODIODES **61**

3.1 OVERVIEW 62

3.2 DEVICE STRUCTURE 66

3.3 EXPERIMENTAL PROCEDURE 69

3.4 MATERIAL GROWTH QUALITY 70

3.4.1 Study of InAs/Si, GaAs/Si, and InAs/GaAs 70

3.4.2 InAs/GaAs/Si APD Structure 75

3.5 DEVICE RESULTS 76

3.6 CONCLUSIONS 78

4 CRITICAL THICKNESS FOR CRACK FORMATION IN GAAS HETEROEXPITAXIAL FILMS ON SI AND SIGE VIRTUAL SUBSTRATES 81

4.1 OVERVIEW 82

4.2 MATHEMATICAL MODELS 83

4.2.1 Crack Channels with no Substrate Fracture 85

4.2.2 Crack Channels with Substrate Fracture 89

4.2.3 Crack Channel Arrays 89

4.3 ANALYTICAL CALCULATIONS 90

4.3.1 Crack Channels with no Substrate Fracture 90

4.3.2 Crack Channels with Substrate Fracture 93

4.3.3 Crack Channel Arrays 94

4.4 EXPERIMENTAL PROCEDURE 95

4.5 EXPERIMENTAL RESULTS 96

4.6 FINITE ELEMENT MODELING 102

4.7 CONCLUSION 104

5 LUMINESCENT EFFICIENCY OF INGAAS QUANTUM WELL STRUCTURES GROWN ON SI, GAAS, GE, AND SIGE VIRTUAL SUBSTRATES 107

5.1 OVERVIEW 108

5.2 EXPERIMENTAL PROCEDURE 109

5.3 LUMINESCENCE RESULTS 112

5.3.1 Emission Wavelength Characteristics 112

5.3.2 Emission Intensity Characteristics 121

5.3.3 Misfit Dislocation Densities in InGaAs on Ge and SiGe Substrates 126

5.4 CONCLUSION 129

6 GAAS-BASED OPTICAL INTERCONNECTS ON SIGE VIRTUAL SUBSTRATES **131**

6.1 OVERVIEW 132

6.2 DESIGN OF GAAS-BASED OPTICAL INTERCONNECTS 133

6.3 DEVICE GROWTH AND FABRICATION 138

6.4 DEVICE RESULTS 138

6.5 CONCLUSION 143

7 CONCLUSIONS AND SUGGESTIONS **145**

7.1 SUMMARY OF EXPERIMENTAL WORK 146

7.2 SUGGESTIONS FOR FUTURE WORK 147

BIBLIOGRAPHY **150**

List of Figures

FIGURE 1.1 EXAMPLE OF ON-CHIP AND CHIP-TO-CHIP COMMUNICATION USING OPTICAL INTERCONNECTS. THE ELECTRICAL SIGNALS ARE FIRST CONVERTED TO OPTICAL SIGNALS THAT ARE TRANSMITTED VIA THE OPTICAL INTERCONNECTS AND THEN CONVERTED BACK TO ELECTRICAL SIGNALS... 22

FIGURE 1.2 THE LATTICE CONSTANT VS. ENERGY GAP DIAGRAM FOR GROUP IV AND III-V SEMICONDUCTORS. SIGE ALLOY LATTICE CONSTANTS SPAN THE GAP BETWEEN SI AND GAAS. 24

FIGURE 1.3 RELAXATION OF A MISMATCHED HETEROEPITAXIAL FILM. THE FILM IS INITIALLY PSEUDOMORPHICALLY STRAINED TO THE SUBSTRATE LATTICE PARAMETER *(LEFT)*. WHEN THE LAYER THICKNESS EXCEEDS THE CRITICAL THICKNESS, MISFIT DISLOCATIONS PROVIDE STRAIN RELIEF *(RIGHT)*. 26

FIGURE 1.4 SCHEMATIC OF MISFIT AND THREADING DISLOCATIONS IN MISMATCHED HETEROEPITAXY. BECAUSE THE MISFIT SEGMENTS CANNOT TERMINATE WITHIN THE CRYSTAL, THEY LEAD TO THE FORMATION OF THREADING SEGMENTS THAT EXTEND TO THE FILM SURFACES. 28

FIGURE 1.5 SCHEMATIC OF HALF LOOP FORMATION FROM MISFIT SEGMENTS AT THE HETEROEPITAXY INTERFACE. 28

FIGURE 1.6 FORMATION OF ANTIPHASE BOUNDARIES AND DOMAINS RESULTING FROM GROUP III AND GROUP V SUBLATTICE MISALIGNMENT AT THE III-V/IV INTERFACE. THE SUBLATTICE MISALIGNMENT IS PRODUCED BY THE ODD-NUMBER STEP STRUCTURE OF THE SUBSTRATE SURFACE.[1] 31

FIGURE 1.7 X-TEM MICROGRAPH OF APB AT A GAAS/GE INTERFACE....................... 32

FIGURE 1.8 THERMAL EXPANSION COEFFICIENTS OF GE, GAAS, AND SI AS A FUNCTION OF TEMPERATURE. NOTE THAT THE THERMAL EXPANSION COEFFICIENT OF GE INCREASES MUCH FASTER THAN THE OTHERS WITH INCREASING TEMPERATURE.[25,26] 33

FIGURE 1.9 BINARY PHASE DIAGRAM OF SI AND GE SHOWING COMPLETE MISCIBILITY THROUGHOUT THE COMPOSITION RANGE. 36

FIGURE 1.10 SCHEMATIC OF A RELAXED GRADED BUFFER. THREADING DISLOCATIONS FROM INITIAL LAYERS GLIDE TO RELIEVE STRAIN AT SUBSEQUENT MISMATCHED INTERFACES. BY REUSING THE EXISTING DEFECTS, NUCLEATION OF ADDITIONAL DEFECTS MAY BE AVOIDED, THUS MAINTAINING A LOW OVERALL THREADING DISLOCATION DENSITY........... 37

FIGURE 1.11 PILE-UP OF THREADING DISLOCATION SEGMENTS ON TRENCHES AT THE SURFACE OF SIGE GRADED-BUFFER. THE PILE-UP INHIBITS DISLOCATION GLIDE AND FURTHER STRAIN-RELAXATION BY EXISTING DISLOCATIONS.[12,38] 38

FIGURE 1.12 X-TEM MICROGRAPH OF THE SIGE VIRTUAL SUBSTRATE. THE MICROGRAPH SHOWS THE GRADED REGION OF THE SIGE SUBSTRATE AND THE PURE GE CAP LAYER........ 39

FIGURE 1.13 SUPPRESSION OF ANTIPHASE DISORDER IN III-V MATERIALS USING A SI OR GE SUBSTRATE WITH A DOUBLE ATOMIC-LAYER STEP SURFACE. THE DOUBLE-STEP STRUCTURE DECREASES THE CHANCES OF III-V LAYER SUBLATTICE MISALIGNMENT.[1] 40

FIGURE 1.14 SELF ANNIHILATION OF AN APB ON A SI OR GE SURFACE ENCOURAGED BY THE HIGH DENSITY OF SINGLE-STEPS. THE PROBABILITY FOR THE COALESCENCE OF NEIGHBORING APBS INCREASES WITH THE HIGH DENSITY OF SINGLE-STEPS.[1] 41

FIGURE 1.15 X-TEM MICROGRAPH OF GAAS GROWN ON A SIGE VIRTUAL SUBSTRATE. THE MICROGRAPH SHOWS MISFIT DISLOCATION SEGMENTS AT THE HETEROEXPITAXIAL INTERFACE. HOWEVER, THIS PICTURE DOES NOT SHOW ANY THREADING DISLOCATION SEGMENTS EXTENDING INTO THE GAAS LAYER. 42

FIGURE 1.16 MINORITY CARRIER LIFETIME OF GAAS AS A FUNCTION OF THREADING DISLOCATION DENSITY. THE PLOT SHOWS THE LIFETIME MEASURED FOR GAAS GROWN ON SIGE SUBSTRATES APPROACHING THE PLATEAU REGION OF THE CURVE, SUGGESTING A DECREASE IN LIFETIME IMPROVEMENT WITH FURTHER REDUCTION OF THE DISLOCATION DENSITY.[29] 43

FIGURE 1.17 SIMS ANALYSIS OF GE CONCENTRATION IN A GAAS FILM GROWN ON A SIGE VIRTUAL SUBSTRATE AT 650 °C. THE SCAN SHOWS THE DIFFUSION OF GE INTO THE GAAS LAYER, THUS UNINTENTIONALLY CHANGING THE DOPING PROFILE. 44

FIGURE 2.1 SCHEMATIC DIAGRAM OF THE MOCVD GROWTH CHAMBER USED IN THIS STUDY. THE CHAMBER IS A HORIZONTAL COLD-WALL REACTOR THAT CAN ACCOMMODATE A 1.5×2 CM SUBSTRATE ON THE GRAPHITE SUSCEPTOR. ... 51

FIGURE 2.2 PHOTOGRAPH OF THE ATMOSPHERIC-PRESSURE MOCVD REACTOR USED IN THIS STUDY. 52

FIGURE 2.3 AT GROWTH TEMPERATURES OF 400 TO 600°C, ARSENIC REPLACES THE FIRST MONOLAYER OF THE SINGLE-DOMAIN GE SURFACE (LEFT). OUTSIDE

OF THIS TEMPERATURE RANGE, GAAS ATOMS ARE ONLY ABSORBED ONTO THE GE SURFACE RATHER THAN DISPLACING THE FIRST MONOLAYER (RIGHT). THE TWO DIAGRAMS SHOW A ROTATION IN THE GAAS ALIGMENT.[1] 55

FIGURE 2.4 NOMARSKI OPTICAL MICROGRAPH OF A SELECTIVELY ETCHED GAAS SAMPLE SURFACE FOR ANALYSIS OF THREADING DISLOCATION DENSITY. HILLOCKS REPRESENT SITES WHERE THREADING DISLOCATIONS INTERSECT IN THE FILM SURFACE. 58

FIGURE 3.1 SCHEMATIC DIAGRAM OF THE INAS/GAAS/SI SAM-APD DEVICE STRUCTURE. NOTE A THIN GAAS LAYER IS INSERTED BETWEEN THE INAS ABSORPTION LAYER AND SI MULTIPLICATION LAYER TO REDUCE THE LATTICE MISMATCH BETWEEN THE INAS AND SI LAYERS. 67

FIGURE 3.2 SPREADING RESISTANCE PROFILE OF THE DOPING LEVELS FOR THE SI PIN DIODE. 67

FIGURE 3.3 BAND DIAGRAM OF THE INAS/GAAS/SI SAM-APD. CARRIERS GENERATED AT THE ABSORPTION LAYER DIFFUSE TOWARD THE SI PIN LAYERS WHERE MULTIPLICATION TAKES PLACE. 69

FIGURE 3.4 A 5×5 µM AFM MICROGRAPH OF INAS ON A SI SUBSTRATE. THE SCAN SHOWS 3D GROWTH MODE AND INAS ISLAND FORMATION ON THE SI SURFACE. 71

FIGURE 3.5 X-TEM MICROGRAPH OF AN INAS LAYER ON A SI SUBSTRATE. HIGH DENSITIES OF DEFECTS AND GRAIN BOUNDARIES ARE OBSERVED. 72

FIGURE 3.6 A 5×5 µM AFM MICROGRAPH OF GAAS SURFACE GROWN ON A SI SUBTRATE. LARGE SURFACE ROUGHNESS IS A RESULT OF THE HIGH LATTICE MISMATCH BETWEEN THE EPILAYER AND THE SUBSTRATE. 74

FIGURE 3.7 X-TEM MICROGRAPH OF A GAAS LAYER GROWN ON A SI SUBTRATE. DUE TO THE HIGH LATTICE MISMATCH, A HIGH DENSITY OF THREADING AND MISFIT DISLOCATIONS ARE PRESENT. 74

FIGURE 3.8 X-TEM MICROGRAPH OF AN INAS LAYER GROWN ON A GAAS SUBSTRATE. THE INAS LAYER IS SINGLE CRYSTAL BUT WITH A HIGH DENSITY OF DEFECTS. 75

FIGURE 3.9 X-TEM MICROGRAPH OF THE INAS/GAAS/SI SAM-APD STRUCTURE. THE SURFACE ROUGHNESS HAS BEEN SLIGHTLY IMPROVED COMPARED TO DIRECT INAS GROWTH ON SI. 76

FIGURE 3.10 I-V CHARACTERISTIC OF THE SI PIN MULTIPLICATION REGION AND THE INAS/GAAS/SI SAM-APD. LEAKAGE CURRENT INCREASES FROM THE SI I-V CURVE TO THE SAM-APD I-V CURVE AS A RESULT OF INCREASED DEFECT DENSITY. 77

FIGURE 4.1 EXAMPLES OF DIFFERENT CRACKING PATTERNS THAT ARE POSSIBLE AS A RESULT OF THERMAL MISMATCH BETWEEN THE FILM AND THE SUBSTRATE. THIS INCLUDES SURFACE CRACKS (PLANE-VIEW), CHANNELING (PLANE-VIEW), SUBSTRATE-DAMAGING (CROSS-SECTION), SPALLING (CROSS-SECTION), AND DEBONDING (CROSS-SECTION). 85

FIGURE 4.2 CALCULATED CRITICAL CRACKING THICKNESS FOR GAAS ON SI AND SIGE SUBSTRATES. NOTE THAT THE CRITICAL CRACKING THICKNESS FOR GAAS ON SIGE IS LOWER THAN GAAS ON SI BECAUSE OF THE ADDITIONAL THERMAL STRESS IN THE GE LAYER. SINCE WE HAVE ASSUMED A α_{GE} VALUE EQUAL TO THE ONE FOR α_{GAAS} IN OUR CALCULATIONS, THE THEORETICAL T_C VALUES FOR GAAS ON SIGE SHOULD BE SLIGHTLY LOWER THAN THE ONES SHOWN. THE ACTUAL α_{GE} VALUE IS SLIGHTLY HIGHER THAN α_{GAAS} AT THE GROWTH TEMPERATURE. 92

FIGURE 4.3 COMPARISON OF CALCULATED AND EXPERIMENTAL CRITICAL CRACKING THICKNESS FOR GAAS LAYERS GROWN ON SI AND SIGE SUBSTRATES. THE EXPERIMENTAL VALUES FOR GAAS ON SI ARE SLIGHTLY HIGHER THAN CALCULATED DUE TO KINETIC BARRIERS TO CRACK FORMATION. 97

FIGURE 4.4 CRACK SPACING AS A FUNCTION OF GAAS FILM THICKNESS GROWN ON SI AT 750°C. THE SPACING SHOWS AN EXPONENTIAL INCREASE WITH THE FILM THICKNESS UP TO 3 µM FOR THE ARRAY PARALLEL TO THE SUBSTRATE OFFCUT <110> DIRECTION. PAST 3 µM, THE SPACING PLATEAUS TO A CONSTANT VALUE. 99

FIGURE 4.5 IMAGES OF CRACKS IN GAAS HETEROEXPITAXIAL FILMS. THE 4 µM GAAS LAYER ON SIGE HAS CRACKS ONLY IN THE DIRECTION PARALLEL TO THE SUBSTRATE OFFCUT <110> DIRECTION *(LEFT)*. THE 8 µM GAAS FILM ON SI HAS CRACKS EXTENDING IN BOTH IN-PLANE <110> DIRECTIONS *(RIGHT)*... 99

FIGURE 4.6 THE X-TEM MICROGRAPH ON THE LEFT SHOWS A CRACK IN THE GAAS LAYER GROWN ON A SI SUBSTRATE WHERE THE CRACK TIP TERMINATES AT THE GAAS/SI INTERFACE. THE CRACK TIP EXTENDS INTO THE GE LAYER FOR GAAS GROWN ON SIGE SUBSTRATES *(RIGHT)*. DISLOCATION EMISSION FROM THE CRACK TIP IS ALSO OBSERVED IN THE X-TEM MICROGRAPH... 101

FIGURE 4.7 ANSYS SIMULATION OF A 5 μM GAAS LAYER ON SI *(LEFT)* AND A 3 μM GAAS LAYER ON A 1 μM GE LAYER ON SI *(RIGHT)*. THE CONTOURS SHOW THE EVOLUTION OF THERMAL STRESS AROUND THE CRACK PLANES. 103

FIGURE 5.1 SCHEMATIC DIAGRAM OF THE INGAAS QUANTUM LUMINESCENT TEST STRUCTURE. THE INGAAS LAYER IS PLACED CLOSE TO THE SURFACE ON TOP OF A RELATIVELY THICK GAAS LAYER. THIS ENSURES A MORE ACCURATE MEASUREMENT OF THE EFFECTS OF DISLOCATIONS ON THE LUMINESCENT EFFICIENCY. 110

FIGURE 5.2 CL SPECTRUM OF THE TEST STRUCTURE GROWN ON GAAS, GE, SIGE, AND SI SUBSTRATES. THE PEAKS AT THE LONGER WAVELENGTHS REPRESENT EMISSION FROM THE INGAAS QUANTUM WELL AND THE PEAKS AT 870 NM END REPRESENT GAAS EMISSION. A) SHOWS THE SPECTRUM FOR THE 60 Å QUANTUM WELL SET AND B) SHOWS THE SPECTRUM FOR THE 120 Å QUANTUM WELL SET. 113

FIGURE 5.3 PLOT OF THE CL INTEGRATED INTENSITY AS AS FUNCTION OF WAVELENGTH. THE INTENSITY INCREASES AS THE QUANTUM WELL THICKNESS CHANGES FROM 60 Å TO 120 Å.. 114

FIGURE 5.4 PV-TEM OF INGAAS QUANTUM WELL INTERFACE ON A GE SUBSTRATE *(LEFT)* AND ON A SIGE SUBSTRATE *(RIGHT)*. A HIGHER MISFIT DENSITY IS SEEN IN THE SAMPLE ON THE GE SUBSTRATE. 118

FIGURE 5.5 CHANGE IN THE STRESS LEVELS IN THE GAAS CLADDING LAYERS GROWN ON GE, SI, AND SIGE. THE GAAS LAYER ON THE SIGE SUBSTRATE AT THE GROWTH TEMPERATURE IS NEARLY STRESS-FREE BECAUSE OF THE INTENTIONAL COMPRESSIVE STRAIN ADDED TO THE SIGE SUBSTRATE. ALL GAAS LAYERS ARE IN COMPRESSION AT THE GROWTH TEMPERATURE..... 120

FIGURE 5.6 X-TEM MICROGRAPH OF THE LUMINESCENCE TEST STRUCTURE. THE INGAAS QUANTUM WELL IS CLOSE TO THE STRUCTURE SURFACE. BELOW THE QUANTUM WELL IS APPROXIMATELY 5000 Å OF GAAS. THIS TEST STRUCTURE IS DESIGNED FOR INCREASED SENSITIVITY TO DISLOCATIONS. 121

FIGURE 5.7 THE CHANGE IN THE LUMINESCENT INTENSITY OF THE INGAAS QUANTUM WELL TEST STRUCTURES AS A FUNCTION OF THE MISFIT DISLOCATION DENSITY. THE INTENSITY DECREASES WITH INCREASES IN THE MISFIT DENSITY AND DECREASES IN THE QUANTUM WELL THICKNESS. NOTE THAT DATA POINTS WITH MISFIT DENSITIES RANGING FROM 100 TO 1000 CM^{-1} ARE FROM SAMPLES GROWN ON SIGE SUBSTRATES AND THE POINTS NEAR 10000 CM^{-1} ARE FROM SAMPLES GROWN ON GE SUBSTRATES. 124

FIGURE 5.8 THE CHANGE IN THE LUMINESCENT INTENSITY OF THE InGaAs QUANTUM WELL TEST STRUCTURE AS A FUNCTION OF THE THREADING DISLOCATION DENSITY. THE INTENSITY DECREASES WITH INCREASES IN THE MISFIT DENSITY AND DECREASES IN THE QUANTUM WELL THICKNESS. NOTE THAT DATA POINTS WITH THREADING DENSITIES NEAR 1×10^4 CM^{-2} ARE FROM SAMPLES GROWN ON GaAs SUBSTRATES AND THE POINTS NEAR 1×10^9 CM^{-2} ARE FROM SAMPLES GROWN ON Si SUBSTRATES. 124

FIGURE 5.9 CHANGE IN THE RELATIVE STRESS LEVELS IN THE InGaAs QUANTUM WELL GROWN ON Ge AND SiGe. THE InGaAs QUANTUM WELL ON A Ge SUBSTRATE EXPERIENCES AN INCREASE IN THE COMPRESSIVE STRAIN FROM THE TEMPERATURE DROP WHILE THE QUANTUM WELL ON THE SiGe SUBSTRATE EXPERIENCES A RELATIVE TENSILE STRAIN AS THE TEMPERATURE FALLS. 127

FIGURE 6.1 SCHEMATIC DIAGRAM OF THE OPTICAL INTERCONNECT UNIT WITH AN GaAs LED AS THE LIGHT EMITTER CONNECTED BY THE WAVEGUIDE TO A GaAs DETECTOR. THE TWO DEVICES ARE BUTT-COUPLED TO THE $Al_{0.15}Ga_{0.85}As$ WAVEGUIDE. OXIDE AND METAL LAYERS ACT AS THE TOP CLADDING LAYERS FOR THE WAVEGUIDE AND AN $Al_{0.9}Ga_{0.1}As$ LAYER ACTS AS THE BOTTOM CLADDING LAYER. 134

FIGURE 6.2 TWO TYPES OF WAVEGUIDE COUPLING SCHEMES. THE ONE ON TOP SHOWS A BUTT, OR VERTICAL, COUPLING SCHEME. IN THIS COUPLING METHOD, THE DEVICE SITS DIRECTLY ON TOP OF THE WAVEGUIDE. ON THE BOTTOM, AN EDGE COUPLING SCHEME IS SHOWN WHERE THE WAVEGUIDE CONNECTS DIRECTLY TO THE ACTIVE REGION OF THE DEVICE.................................. 134

FIGURE 6.3 EQUIVALENT CIRCUIT DIAGRAM OF THE OPTICAL INTERCONNECT UNIT. THE ADDITIONAL PN JUNCTION UNDER THE DEVICES MINIMIZES ELECTRICAL LEAKAGE THROUGH THE SUBSTRATE. 136

FIGURE 6.4 TOP VIEW OF THE MASK DESIGN USED FOR THE FABRICATION OF THE OPTICAL INTERCONNECTS. DIFFERENT INTERCONNECT GEOMETRIES ARE SHOWN, INCLUDING Y-JUNCTIONS WITH DIFFERENT JUNCTION ANGLES, BENDS WITH VARIOUS BEND ANGLES, AND STRAIGHT WAVEGUIDES RANGING FROM 10 μM TO 1 CM IN LENGTH.. 137

FIGURE 6.5 OPTICAL MICROGRAPHS OF FABRICATED OPTICAL INTERCONNECT UNITS. INTERCONNECTS WITH A 10 μM STRAIGHT WAVEGUIDE *(LEFT)*, Y-JUNCTIONS *(CENTER)*, AND BENDS *(RIGHT)*. 139

FIGURE 6.6 I-V CHARACTERISTIC OF THE LED AND THE DETECTOR RESPONSE TO THE LED EMISSION. THE DETECTOR DIODE OUTPUT CURRENT STARTS

INCREASING AS THE LED IS TURNED ON. THE I-V CURVES SHOW HIGH SERIES RESISTANCE AT HIGH VOLTAGES. 140

FIGURE 6.7 THE DETECTOR OUTPUT CURRENT TO LED INPUT CURRENT RATIO OF THE OPTICAL INTERCONNECT UNIT AS A FUNCTION OF LENGTH A). NOTE THAT ONLY A NOISE LEVEL SIGNAL IS DETECTED FOR A WAVEGUIDE LENGTH OF 1 CM. THIS RATIO ALSO DECREASES WITH INCREASING Y-JUNCTION ANGLE B). 141

FIGURE 7.1 SCHEMATIC DIAGRAM OF AN ALTERNATIVE OPTICAL INTERCONNECT STRUCTURE. THE LIGHT EMITTER IS A LASER WHICH IS EDGE-COUPLED TO THE INTERCONNECT. THE INTERCONNECT IS SURROUNDED BY SiO_2 CLADDING LAYERS TO INCREASE THE INDEX CONTRAST. 149

List of Tables

TABLE 1.1 MATERIALS PARAMETERS OF GAAS, GE, AND SI. 23

TABLE 4.1 MECHANICAL PROPERTIES OF GE, SI, AND GAAS[25] 91

TABLE 5.1 COMPARISON OF INTEGRATED LUMINESCENT INTENSITY OF INGAAS QUANTUM WELL AND GAAS CLADDING LAYERS FOR TEST STRUCTURES GROWN ON SI, GE, SIGE, AND GAAS. 114

TABLE 5.2 THREADING AND MISFIT DISLOCATION DENSITIES IN INGAAS QUANTUM WELLS GROWN ON THE SUBSTRATES STUDIED IN THIS WORK. 122

Acknowledgements

First and foremost, I would like to thank my advisor Prof. Eugene Fitzgerald for all his help and his encouragement. It has been a great pleasure having him as a mentor for the past seven and a half years. He has helped me become a better researcher and a better person. I would also like to thank my undergraduate advisor Prof. Kirk Kolenbrander who had first interested me in electronic materials and for being a great teacher and advisor.

My graduate career would have been unbearable if not for the members and ex-members of the Fitzgerald Group. I would not have known half as much as I do now if not for Steve Ting. He has taught me everything that I needed to know to survive these few years, from growth (including the correct choice of music if one wants to grow good GaAs), CL, TEM, to processing. Lab just wasn't the same without him practicing kung-fu in lab.

Lab also wasn't the same without Mayank Bulsara talking to himself. Mayank made working in lab more interesting, not only because of his fun personality, but also because of the random placement of tools all over the lab.

Mike Groenert has been a pleasure$_{\text{vab}}$ to work with. I'm glad that I was able to share a lot of the reactor pains with him. I have never in the past few years heard a crossed-word from him. He is a patient and knowledgeable person. But most of all, he possesses $_{\text{vab}}$ a great sense of humor.

Another ex-member of the group that I must thank is Tom Langdo. He made the group a lot more fun to work in. He has also been a great inspiration in terms of work ethics and eating ethics! One day I will be able to beat him in an eating contest.

Arthur Pitera is a smart guy who also possesses a great sense of humor. He made working in the lab fun and enjoyable, especially with the Family Guy episodes that somehow got into our computer. Too bad he decided to desert the III-V$_{\text{ab}}$ group and join the SiGe group.

I would also like to thank Chris Leitz for sharing the orthogonal humor and for keeping the computer room$_{\text{vab}}$ neat. We had to go through the thesis stress together and he is someone to look up to.

I would like to thank Gianni Taraschi for his humor and for his help with the waveguide modeling and many hours of brainstorming with me to help me come up with a better link structure.

Lisa McGill has been another person that shared our reactor pain. She has also been designated as the group editor, and I thank her for helping me improve the writing quality of this document.

Other members of the group have given me valuable support during the struggle. I would like to thank Nate Quitoriano, Larry Lee, Nava Ariel, and Anabela Afonso. They all made my five years in the Fitzgerald group as good as it can be.

The one member of the group that I owe the most thanks to is Matthew Currie. He not only is a great colleague but also my best friend. He was the person to bring laughters to our group with is evil sense of humor. This thesis would not have been possible without his constant support and all the long nights of editing! Matt's family has also been a great support for me and I have enjoyed the time spent with them. I give my thanks to Mr. and Mrs. Currie, Opa und Oma Schatte, Betsy, and Barkley.

I must thank Mike Frongillo who had to listen to all my complaining and who had to clean up after us when we have "accidents" on the microscopes. I'm glad that he won't be getting rid of me for awhile. Hopefully my microscopy skills won't be deteriorating too much. I would also like to thank Dr. Chris Chua for his continuous guidance in research.

Many thanks must be given to all my friends from back home, their parents, and my friends from elementary school that I grew up with. I thank them for all their support and love. They are the Lin, Mo, and Lee family, Tien Tien, Hsu Jen, Chen Yi Kuang, Amy Dai, Wayne Chang, Lee Hsu Pung, Lee Bing Shuo, and Gan Nian Tsu. Many thanks to my friends from Taipei American School who lasted with me until now: Susan Lee, Jenny Su, Ike Lin, George Sya, Alice Yeh, Alan Hsu, and Ming Loo. And I wouldn't have been able to accomplish all that I have hoped to without the guidance from my high school teachers. They were great teachers then and great friends now. I give my thanks to Mr. Orensky, Mr. Olson, Mr. Bos, Mr. and Mrs. Cox, Mr. Augustin, Mr. Fredette, and Mrs. McDowell.

I would like to thank my piano teacher Giavanna Confalone for opening up my eyes to the all the beautiful music and for helping me achieve something I've always dreamed of doing.

This past summer in Japan had opened up my mind to a different culture and had helped me establish friendships that I would always treasure. I thank them for their friendship and for their support. I give my thanks to Karen Bottom, Mike Rasiej, Snow Tseng, Morris Green, William Nichols, Lillian Chuang, the Nagashima family, and all the friends from Stanley Electric especially my mentor Keizo Kawaguchi and Mr. Maruyama.

I owe much to my relatives and family friends. I thank them for believing in me and for being proud of me, especially my grandparents Yang and grandparents Liu. My aunts and uncles in the US had to be my sere gate parents in the US. Many thanks to my uncles and aunts in California, Texas, and Taipei and my cousins from both sides of the family in Texas, California, Utah, and Taipei. Yuan Gon-Gon, Yuan Nai-Nai, Yi Poa-Poa, Man Yi Wai-Poa, Kan Wai-Gon, and Kan Wai-Poa have also always been supportive and had always believed in me. Uncle Ming, Manny, Matthew, Nat, aunt Hsu, and my god-parents have also taken care of me for the past nine years like a member of their own family. I cannot forget my big brother Daniel Yu and his family for taking care of me that summer in LA.

My dogs and cats went through so much with me. They were and are my companions who helped me through the most stressful times. I give my love to Candy, Er-Tong, Lo-Lo, Shi-Lo, Shiao-Lo, Chairman Miao, and Margaret Catcher.

And finally, I would like to give all my thanks and love to my parents, and my sister Lulu. Nothing would have been possible without them. I am so lucky to have them as my family and as my best friends. This thesis is dedicated to them.

Chapter 1

Overview of Materials Issues in III-V Integration on Si

1.1 Motivation for III-V Integration on Si

Very Large Scale Integration (VLSI) technology and processing advancements are based primarily on the silicon materials platform. The advantages that silicon provides over other semiconductor materials include its highly developed processing and design technologies, the availability of inexpensive large-area substrates, and a stable native oxide.[1] Because of these features, huge progress in electronic device design and fabrication has been achieved. In spite of these benefits, silicon's use in optoelectronic device applications has been limited by one fundamental property of silicon, specifically its indirect bandgap. Instead, direct bandgap materials such as GaAs and other III-V compounds are required for the fabrication of optoelectronic devices.

The vision of combining conventional electronic devices with optoelectronic devices on the same platform is one shared by many researchers. The distinguishing property of optoelectronic devices that separates them from conventional electronic devices is their ability to manipulate photons. In contrast, electronic devices such as field effect transistors form the backbone of modern integrated circuits, and their functions cannot yet be performed by optoelectronic devices. This type of technology integration promises to take advantage of both optical and electronic properties provided by the combination of dissimilar materials. This ambitious platform is termed the optoelectronic integrated circuit (OEIC).

With advances in VLSI technology through device processing and scaling, silicon integrated circuit performance has improved drastically in the past few decades. However, future gains in performance by simply scaling will decrease as fundamental physical and economic limits are reached. Further improvements will most likely come from new architectures, better circuit designs, and the introduction of new technologies such as optical interconnects.[2]

Today electrical interconnection delay has become a limiting factor in circuit performance. The replacement of electrical interconnects with optical interconnects will decrease power consumption while increasing data transmission rates.[3,4] Optical interconnects, unlike electrical interconnects, are immune to electromagnetic interference such as cross-talk, and can provide voltage isolation between different parts of the chip.[5] Furthermore, electrical interconnects suffer the skin effect at high data rates.[6]

In an OEIC, optical devices and optical interconnects can provide higher speed and higher bandwidth signal transmission for on-chip electronic devices, chip-to-chip, and even chip-to-board communication, by replacing conventional electrical interconnects. Optical integration will enable novel VLSI architecture and remove IC packaging constraints imposed by conventional perimeter I/O pinouts.[1] Figure 1.1 illustrates the concept of an OEIC where electrical signals are first converted to optical signals and then transmitted through optical interconnects on-chip and between chips. Once received, the optical signals are then converted back to electrical signals.

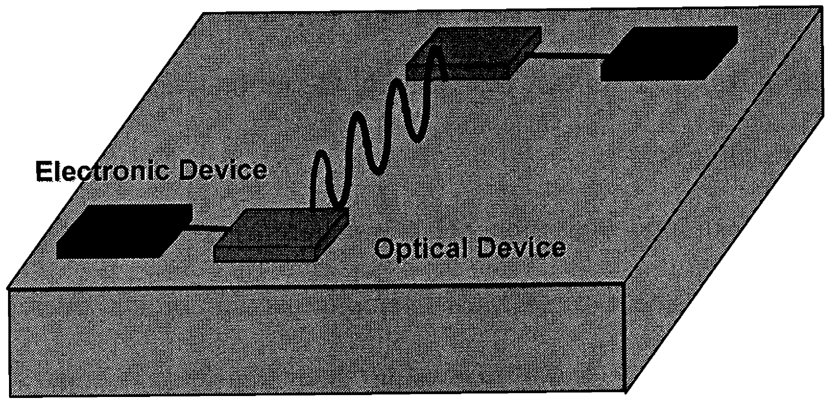


Figure 1.1 **Example of on-chip and chip-to-chip communication using optical interconnects. The electrical signals are first converted to optical signals that are transmitted via the optical interconnects and then converted back to electrical signals.**

Before this vision of optoelectronic and electronic device integration can be realized, the challenge of integrating III-V materials and other light-emitting materials onto silicon must be met. There are two ways to achieve the integration of III-V materials on silicon: hybrid integration and monolithic integration. Hybrid integration involves the fabrication of discrete optical components that are later integrated with the host IC via flip-chip bonding,[7] polyimide bonding,[8] or wafer bonding.[9] This is a volume-limited and costly procedure. On the other hand, monolithic integration involves the heteroepitaxy of device layers directly onto the host silicon substrate. Devices are self-aligned and can be co-processed with the silicon devices on large-area wafers. Hence, monolithic integration is a high throughput approach utilizing economy of scale, an advantage it enjoys over the labor-intensive hybrid integration method. Monolithic integration can also provide higher reliability, yield, and functionality. However, there are unique challenges involved in achieving monolithic integration of high quality III-V materials on silicon that must be solved before monolithic integration becomes viable.

1.2 Challenges of Monolithic Integration of III-V Materials on Si

Because of the materials issues involved in epitaxial growth of high-quality, low-defect-density III-V layers on silicon can be difficult to achieve. Some important materials issues that must be considered include: the difference in lattice constants, the epitaxy of polar material on non-polar substrates, and the difference between thermal expansion coefficients. Each of these materials issues will be discussed in the following sections. Some materials parameters for gallium arsenide (GaAs), germanium (Ge), and silicon (Si) are listed in Table 1.1.

Table 1.1 Materials parameters of GaAs, Ge, and Si.

	Crystal Structure	Lattice Constant (Å)	Coefficient of Thermal Expansion (K^{-1}) @ 300K
Si	diamond	5.430	2.6×10^{-6}
Ge	diamond	5.657	5.86×10^{-6}
GaAs	zinc-blende	5.653	5.8×10^{-6}

1.2.1 Lattice Mismatch

Lattice mismatch, or the difference in the equilibrium lattice constants between film and substrate, may result in the nucleation of defects that if present at high densities can be detrimental to the performance of minority carrier devices. There is a 4.1% lattice mismatch between GaAs and Si, a difference large enough to cause an extremely high threading dislocation density >10^9 cm^{-2} in GaAs epitaxial layers grown directly on Si. There are a limited number of substrate materials that can accommodate GaAs growth without the creation of high defect densities due to lattice mismatch. Figure 1.2 shows the bandgap and lattice constant of various semiconductor materials. The large difference between the lattice constants of Si and GaAs is evident.

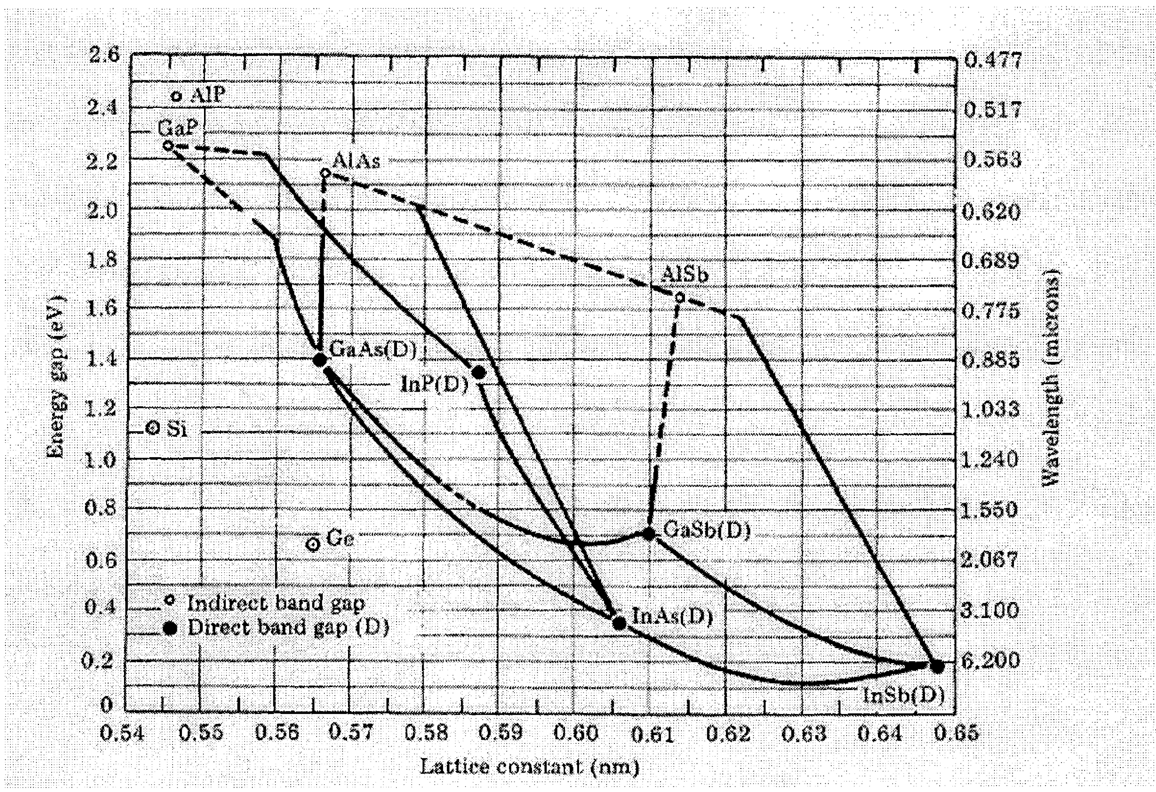


Figure 1.2 **The lattice constant vs. energy gap diagram for group IV and III-V semiconductors. SiGe alloy lattice constants span the gap between Si and GaAs.**

The misfit strain between mismatched layers may be expressed as

$$f = \frac{a_s - a_o}{a_o} = \varepsilon + \delta \qquad \textbf{Equation 1.1}$$

where a_o and a_s are the lattice constants of the overlayer and the substrate, respectively, and ε represents the elastic component of the strain whereas δ represents the plastic component of the strain. Initially, the overlayer film is elastically strained, remaining coherent with the substrate. The elastic strain value ε, is a function of the elastic properties of the film and the substrate. Beyond a certain film thickness, termed the critical thickness, plastic deformation takes place in the form of misfit dislocation introduction. Defect nucleation may be energetically favored to achieve strain relaxation across a mismatched interface as illustrated in Figure 1.3. Matthews and Blakeslee have derived an expression for this critical thickness by minimizing the sum of the areal energy of an orthogonal misfit dislocation array and the elastic strain energy of the lattice mismatched film. The energy expressions for the dislocation and the elastic strain energies are, respectively,[10]

$$E_d = D(b/b_{eff})(1-\nu\cos^2\alpha)(f-\varepsilon)\left[ln(\frac{h}{b})+1\right] \qquad \textbf{Equation 1.2}$$

$$E_\varepsilon = \varepsilon^2 Yh \qquad \textbf{Equation 1.3}$$

where ν is the Poisson ratio, α is the angle between the Burgers vector b and the dislocation line direction, b_{eff} is the effective Burgers vector, or the interface-plane component of the Burgers vector in the direction of the misfit spacing, Y is the biaxial Young's modulus, and D is the average shear modulus given by,

$$D = \frac{G_o G_s b}{\pi(G_o + G_s)(1-\nu)}$$ **Equation 1.4**

where G_o and G_s are the shear moduli of the overlayer and substrate, respectively. The derived expression for the equilibrium critical thickness is therefore,[11]

$$h_c = \frac{D(1-\nu\cos^2\alpha)(b/b_{eff})\left[ln(\frac{h_c}{b})+1\right]}{2Yf}.$$ **Equation 1.5**

Beyond this critical thickness, defect nucleation becomes energetically favored to achieve strain relaxation across a mismatched interface as illustrated in Figure 1.3. This equation does not take into account the kinetics of dislocation nucleation and glide, which are thermally activated processes. Consequently, experimental critical thicknesses are expected to be larger than the calculated values, as misfit dislocation introduction may be kinetically limited.

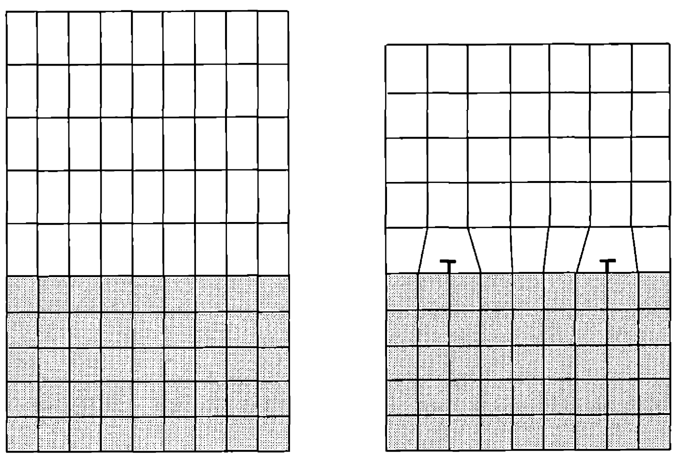


Figure 1.3 **Relaxation of a mismatched heteroepitaxial film. The film is initially pseudomorphically strained to the substrate lattice parameter *(left)*. When the layer thickness exceeds the critical thickness, misfit dislocations provide strain relief *(right)*.**

The diamond cubic and zinc-blende crystal structures have a {111}<110> slip system where the misfit array lies along the orthogonal [110] and $[1\bar{1}0]$ directions for growth on (001) surfaces. The most efficient strain relief occurs when the Burgers vector of these dislocations is along the mismatch interface and perpendicular to the dislocation line. However, such sessile edge dislocations have a {100} glide plane and therefore cannot glide to provide strain relaxation in this system. As a result, glissile 60° dislocations are the dominant type of misfit dislocation observed in structures with low mismatch. These 60° dislocations have a <110> Burgers vector forming a 60° angle with the <110> line direction and have {111} glide planes. In high mismatch systems, edge dislocations will form at the interface, despite their sessile nature.[11]

Since dislocations cannot terminate within a crystal, the formation of misfit dislocations is often accompanied by threading dislocation segments that extend to the film surfaces, as shown in Figure 1.4. The threading segments can glide on the {111} plane and lead to the formation of half-loops consisting of a misfit terminated by threading dislocation segments, as illustrated in Figure 1.5.

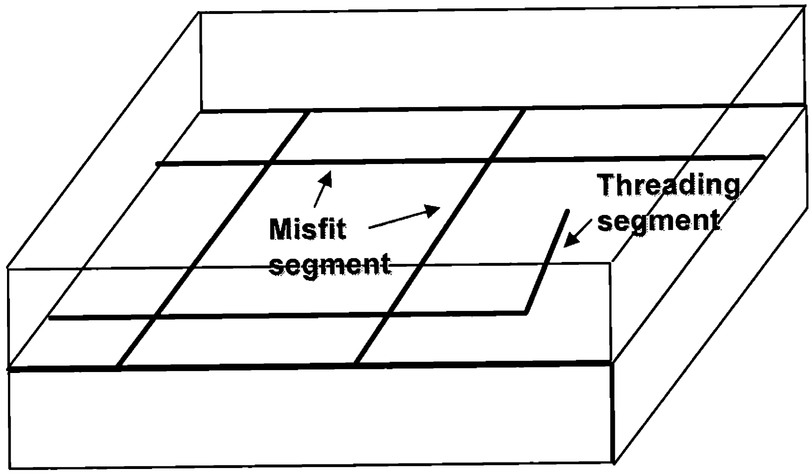


Figure 1.4 **Schematic of misfit and threading dislocations in mismatched heteroepitaxy. Because the misfit segments cannot terminate within the crystal, they lead to the formation of threading segments that extend to the film surfaces.**

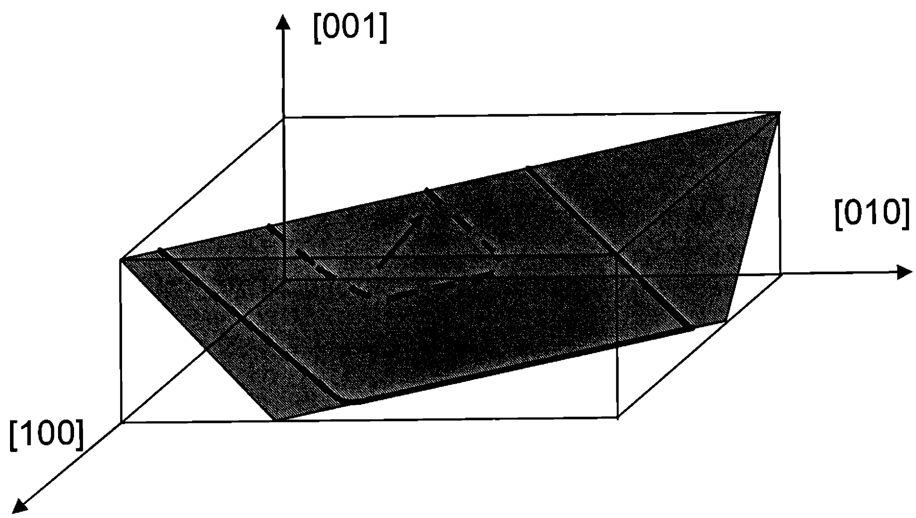


Figure 1.5 **Schematic of half loop formation from misfit segments at the heteroepitaxy interface.**

These threading segments will propagate into the heteroepitaxial film and remain in the film as a by-product of strain relaxation unless they are able to glide to a free surface at the sample edge.[1] When threading segments glide in close proximity of each other, they may interact by repelling, attracting, or blocking one another. Researchers have taken advantage of these characteristics of dislocations in an effort to decrease the final threading dislocation density in high lattice mismatched systems. Methods such as superlattice growth and thermal cycling have been used in the hope that increased dislocation interaction would result in dislocation annihilation.[12,13] Fusion occurs when two dislocation segments with the proper Burgers vectors are attracted to each other to form a single threading segment. Two dislocation segments with opposite Burgers vectors may react to achieve complete annihilation. The benefits of superlattice growth and thermal annealing are limited since the success of these methods requires the presence of a high density of threading dislocations, greater than 10^7 cm^{-2}, to achieve a high probability of dislocation interaction. Note that, due to the dislocation geometry, the probability of dislocation interaction may also be increased simply by growing thicker films.[1] A threading dislocation density on the order of 10^8 cm^{-2} has been achieved with the use of an InGaAs/GaAs superlattice grown between a GaAs layer and the Si substrate.[14] Alternatively, selective area growth (SAG) of GaAs directly on Si has also been attempted. By decreasing the area of growth, the probability of a threading segment reaching an edge and exiting the epilayer region increases, thus decreasing the total threading dislocation density at the top of the epilayer. Again, this method yields limited success, with a final threading dislocation density of 10^8 cm^{-2} for reasonably large areas.[14] By limiting the growth area, the flexibility of device size and design decreases.

Effort has been made to control threading dislocation density in heteroepitaxial films, especially in the active layers of devices, because of the deleterious effects these dislocation segments have on minority carrier devices such as light emitting diodes (LEDs), detectors, and lasers. Dislocation cores form mid-bandgap trap states, or recombination and generation sites that reduce the minority carrier lifetimes in device layers. In addition to the reduction of minority carrier lifetime, dislocations tend to getter metal impurity atoms which leads to excessive leakage current. These combined effects lead to the degradation of the radiative efficiency in light emitting devices, sensitivity in detectors, and energy conversion efficiency.[15,16] Threading segments may also affect the reliability of semiconductor lasers. Waters, *et al.* observed the formation and propagation of dark line defects, nucleating from point defects and threading dislocation segments, into the active regions of AlGaAs lasers, hastening the failure of these lasers.[17] Performance of majority carrier devices may also be affected. High threading dislocation density may result in carrier scattering in field effect transistors (FETs), thereby reducing the mobility and transconductance.[18]

1.2.2 Polar on Non-Polar Epitaxy

In addition to lattice mismatch, problems may arise due to polar on non-polar epitaxy when growing III-V materials on a group IV substrate. Polar on non-polar epitaxy occurs when growing films consisting of more than one type of sublattice on a substrate with only one sublattice.[19, 20] If the III-V/IV interface consists of an unbroken plane of either III-IV or V-IV bonds, a polarized interface is created. Diffusion of the group III or V species across the interface into the group IV substrate, or vice versa, creating a p- or n-doped layer near the interface may easily take place. Group III species are p-type dopants for the group IV substrate, whereas the group V species act as n-type dopants in group IV substrates. At the same time, group IV species are amphoteric in the III-V material, behaving as n-type dopants on group III sites and p-type dopants on group V sites.

In addition to interface polarization and autodoping, polar on non-polar epitaxy can also create antiphase domains separted by antiphase boundaries (APBs) throughout the entire III-V epilayer. Antiphase boundaries result from misalignment of the group III and V atoms at the growth interface. On-axis (001) Si and Ge substrate surfaces display an irregularly spaced single-atomic-layer steps, with each step having a height of a/4[001] along the <110> directions. On neighboring terraces separated by any odd-integer number of steps, the exposed III-V sublattice will shift from one terminating plane to the other, creating domains of misaligned III-V layers. This is shown in Figure 1.6. A X-TEM micrograph of APB formation at a GaAs/Ge interface in Figure 1.7.[1]

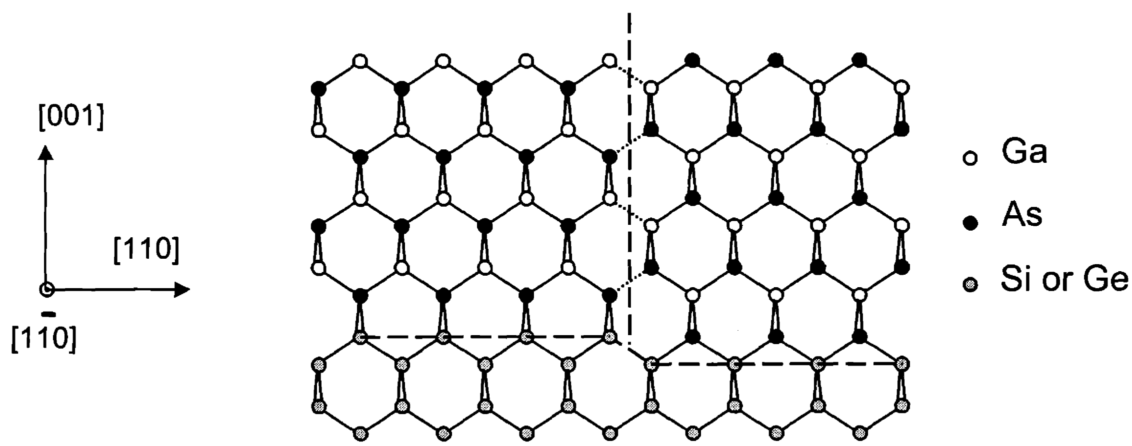


Figure 1.6 **Formation of antiphase boundaries and domains resulting from group III and group V sublattice misalignment at the III-V/IV interface. The sublattice misalignment is produced by the odd-number step structure of the substrate surface.[1]**

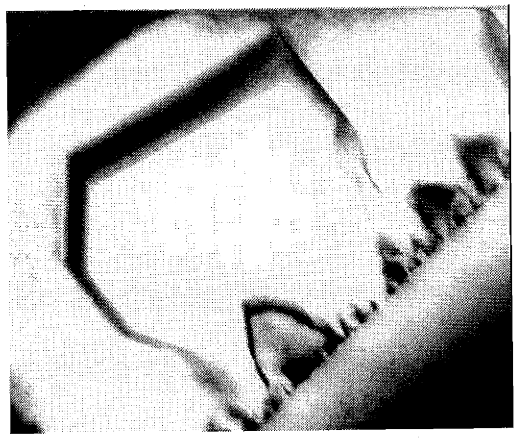


Figure 1.7 X-TEM micrograph of APB at a GaAs/Ge interface.

Similar to threading and misfit dislocation segments, APBs may also act as recombination and generation centers, leading to the degradation of minority carrier lifetime and to majority carrier scattering.[21] Cathodoluminescence (CL) has been utilized to image radiative recombination at APBs at the GaAs/Ge interface.[22] It has also been shown through Hall measurements on GaAs films grown on Si substrates that the Hall mobility decreases with increasing APB density.[23,24] It is therefore important to suppress the formation of APBs during growth of III-V layers used in device fabrication.

1.2.3 Thermal Mismatch

The difference in thermal expansion coefficients between the epilayer and the substrate may lead after a significant temperature change to a large thermal stress in the epilayer. The thermal expansion coefficient (α) values of GaAs, Ge and Si are listed in Table 1.1. Furthermore, α is functions of temperature, as illustrated in Figure 1.8. Under this thermal mismatch stress, crack formation in the epilayer becomes energetically favorable when the epilayer thickness exceeds a critical thickness, much like dislocation nucleation in lattice-mismatched strained layers. The strain resulting from thermal mismatch may be expressed as:

$$\varepsilon_T = (\alpha_o - \alpha_s)\Delta T \qquad \textbf{Equation 1.6}$$

where α_o and α_s are the thermal expansion coefficients of the overlayer and substrate and, respectively, and ΔT is the change in temperature. This expression assumes that the thermal expansion coefficients are not functions of temperature.

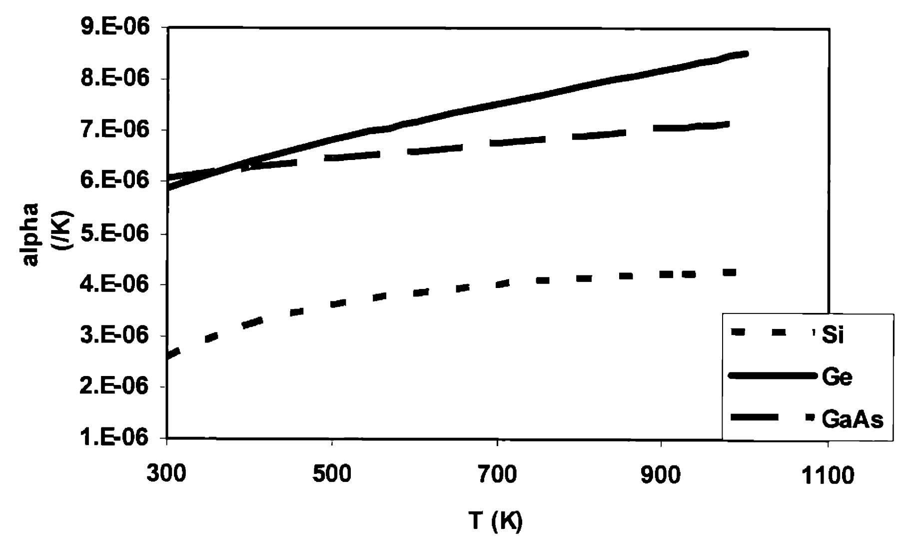


Figure 1.8 **Thermal expansion coefficients of Ge, GaAs, and Si as a function of temperature. Note that the thermal expansion coefficient of Ge increases much faster than the others with increasing temperature.[25, 26]**

Epilayer cracks are undesirable because they not only limit the useful area for device fabrication, but can also create unwanted vertical conducting paths. Consequently, the elimination of cracks from active device layers is crucial. The analytical and experimental determination of critical cracking thickness will be decribed in greater detail in Chapter 4.

1.3 III-V on Si Integration Strategy

Considerable progress has been made toward the integration of III-V materials onto Si. Some notable results in monolithic integration of GaAs on Si with the use of SiGe virtual substrates are shown.[27,28,29,30,31,32,33,34,35] The integration work in this thesis will be based on the utilization of SiGe graded buffer layers, first developed by Fitzgerald, *et al.*[36] In the following sections, this particular integration approach will be discussed.

1.3.1 SiGe Virtual Substrates

The 4.1% lattice mismatch between GaAs and Si makes direct growth of GaAs layers with threading dislocation densities less than 10^8 cm^{-2} nearly impossible. Other substrates with lattice constants similar to GaAs may be used as potential platforms for high quality GaAs growth. One such material is Ge, which has a lattice mismatch to GaAs of only 0.07%. Figure 1.2 shows GaAs and Ge with nearly the same lattice constants, both of which are much greater than the lattice constant of Si. Nevertheless, the ultimate goal remains to have Si as the final substrate for the integration of III-V light emitting materials.

By compositionally grading from Si to Ge via thin alloyed SiGe layers, it is possible to bridge the gap in lattice constant between Si and Ge, thus lattice-engineering a Si substrate with a Ge lattice constant at the top surface. By reducing the lattice mismatch between GaAs and the growth surface, the lattice mismatch between the substrate and the epilayer will greatly reduced, thus reducing the dislocations density needed to relieve the remaining mismatch strain. The single-lens binary phase diagram shown in Figure 1.9 shows that Ge and Si are completely miscible throughout the entire Si_xGe_{1-x} range, making this alloy system an ideal candidate for compositional grading. Using this compositional grading technique, it is possible to create a SiGe virtual substrate. A schematic diagram of the virtual substrate is shown in Figure 1.10. On top of the Si substrate, subsequent layers of increasing Ge content are added until a final composition of pure Ge is reached. The top of the virtual substrate is a layer with 100% Ge.

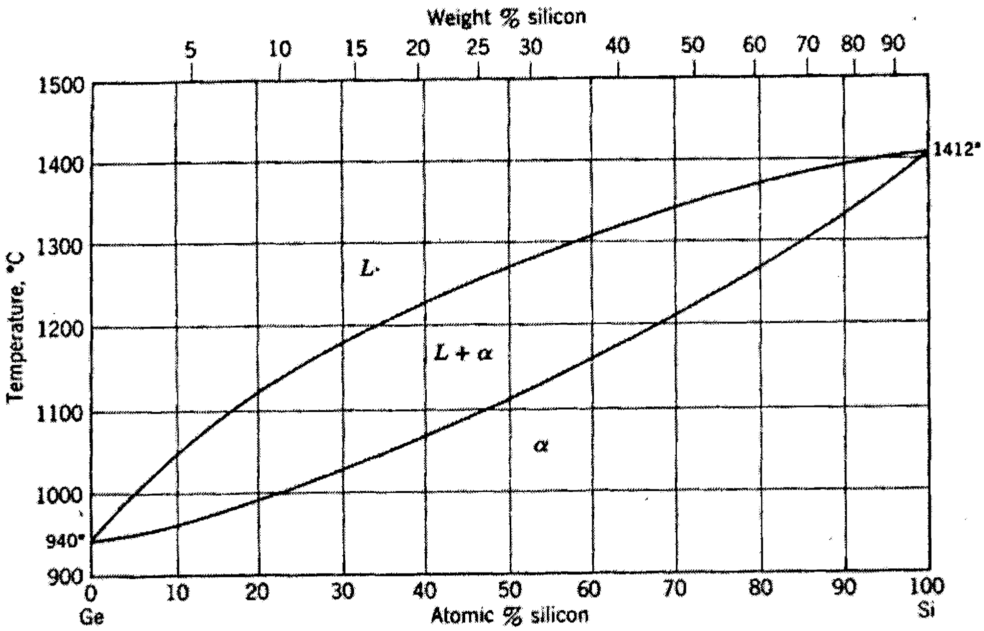


Figure 1.9 Binary phase diagram of Si and Ge showing complete miscibility throughout the composition range.

The threading dislocation density of Ge/Si_xGe_{1-x}/Si substrates may be maintained at a low value with proper control of the growth temperature and the grading rate (the rate at which the Ge composition is increased in each subsequent grading layer). With control of the growth temperature and grading rate, one can ensure the relaxation of strained layers via existing dislocations rather than by the nucleation of new dislocations. By "recycling" existing misfit and threading dislocations, a low final threading dislocation density may thus be maintained wherein strain is relieved by dislocations initiated in the initial layers. This is illustrated in Figure 1.10, where the dislocation is shown to propagate through the graded layers, forming misfit segments along layer interfaces and relieving lattice mismatch strain between each graded layer. It is necessary to carry out the growth of graded layers at a temperature high enough for dislocation glide since dislocation glide is a thermally activated process. Unwanted homogeneous dislocation nucleation is also a thermally activated process, but features a much higher activation barrier than dislocation glide or heterogeneous dislocation nucleation. Thus, by maintaining a slow effective strain rate, it is possible to avoid the onset of homogeneous dislocation nucleation.

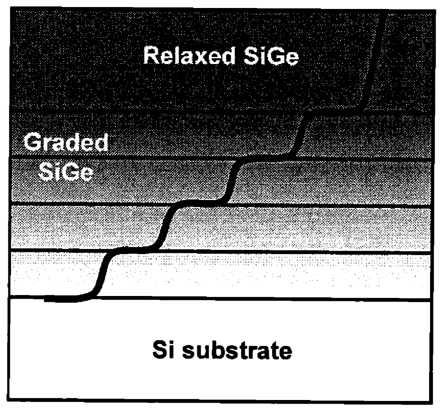


Figure 1.10 Schematic of a relaxed graded buffer. Threading dislocations from initial layers glide to relieve strain at subsequent mismatched interfaces. By reusing the existing defects, nucleation of additional defects may be avoided, thus maintaining a low overall threading dislocation density.

By reusing existing threading dislocations throughout the graded buffer, with a minimal dislocation nucleation rate, a steady-state threading dislocation density is expected in the SiGe virtual substrate, independent of the final Ge content. Nevertheless, in early work, an increase in the threading dislocation density was observed with increasing Ge composition.[12,37,38]

Strain fields from the misfit segments along the in-plane orthogonal <110> directions in the graded buffer layer produce a characteristic cross-hatch surface roughness on the surfaces of the SiGe virtual substrates. The severity of the cross-hatch pattern tends to increase with the final Ge composition.[12,39,40] By adding an intermediate chemical-mechanical polishing (CMP) step halfway through the buffer growth, Currie, *et al.* noted both the decrease in the surface roughness and the threading dislocation density of the final Ge cap layer of the virtual substrate.[12,37] The trenches formed as a result of high surface roughness inhibits the glide of threading dislocation and causes dislocation pile-up to further the relaxation of the buffer layers, as illustrated in Figure 1.11. By removing the excess surface roughness, dislocation segments are once again allowed to glide, thereby reducing the final threading dislocation density. Furthermore, the intermediate CMP step reduced the surface roughness of the final layer.[12]

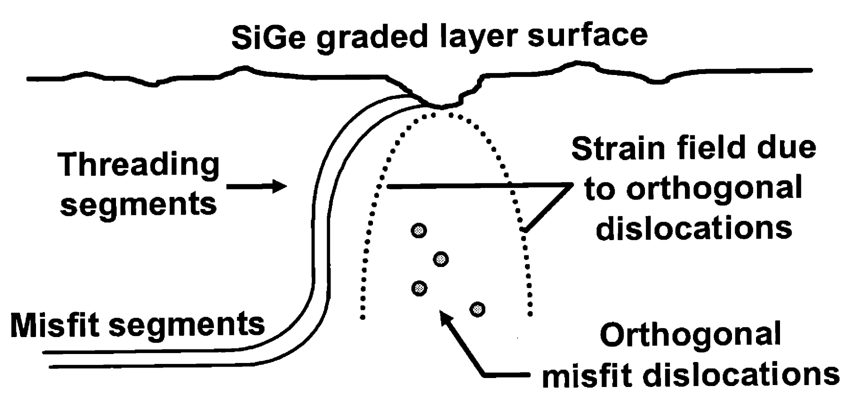


Figure 1.11 Pile-up of threading dislocation segments on trenches at the surface of SiGe graded-buffer. The pile-up inhibits dislocation glide and further strain-relaxation by existing dislocations.[12,38]

The growth of high quality SiGe virtual substrates with a final threading dislocation density of approximately 1×10^{6} cm^{-2} was demonstrated by Currie, *et al* using an ultra-high-vacuum chemical vapor deposition (UHCVD) system.[12] A cross-sectional transmission electron (X-TEM) micrograph of the Ge/Si_xGe_{1-x}/Si virtual substrate is shown in Figure 1.12. The X-TEM does not show any threading segments in the final Ge layer; however, this does not indicate that the layer is dislocation-free. Because the area of view is limited in a X-TEM micrograph, it is not a suitable method for threading dislocation density measurement. More accurate characterization techniques will be discussed in Chapter 2.

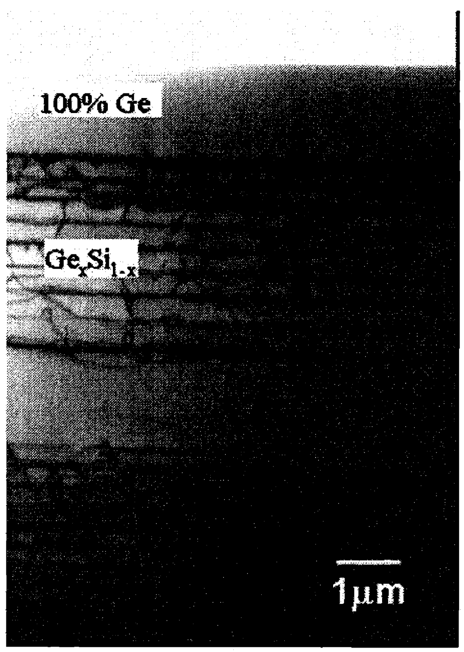


Figure 1.12 X-TEM micrograph of the SiGe virtual substrate. The micrograph shows the graded region of the SiGe substrate and the pure Ge cap layer.

1.3.2 *Suppression of Antiphase Boundaries*

APB-free GaAs growth can be achieved with the use of (001) SiGe virtual substrates offcut toward the [110] direction.[1] Unlike on-axis (001) substrates, offcut substrates exhibit a regular array of single-steps running along the $[1\bar{1}0]$ directions.[1] At high temperature and under ultra-high-vacuum growth conditions, offcut wafers are known to transform from a single-step to a double-step geometry and configuration, therefore allow for APB-free growth. This double-step configuration is shown in Figure 1.13.

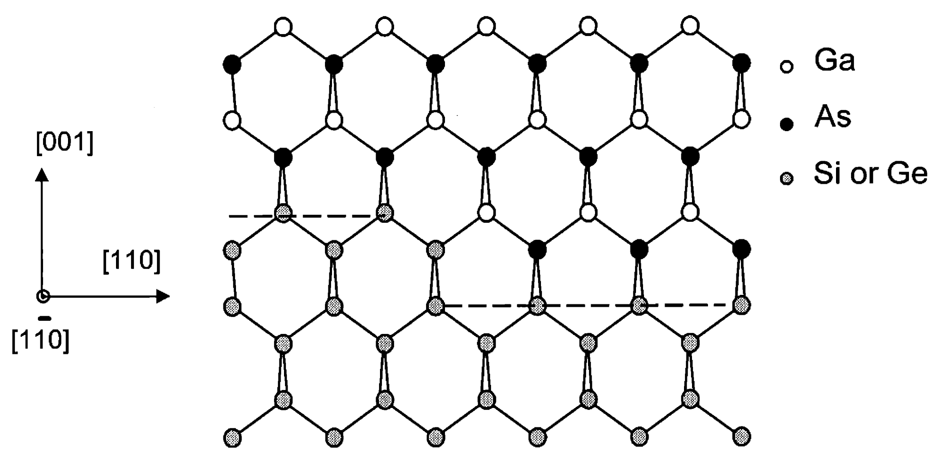


Figure 1.13 Suppression of antiphase disorder in III-V materials using a Si or Ge substrate with a double atomic-layer step surface. The double-step structure decreases the chances of III-V layer sublattice misalignment.[1]

Even without the double-step geometry, self-annihilation of APBs will be promoted by the high single-step density on misoriented substrate surfaces. With an increase in the density of single-steps, the spacing between neighboring APBs is decreased, increasing the probability of small, enclosed domain formation from neighboring APBs. These domains will concentrate near the GaAs/Si interface, as shown in Figure 1.14.[1] With a sufficiently thick GaAs epilayer, devices may be fabricated in areas away from the interface, in the APB-free regions.

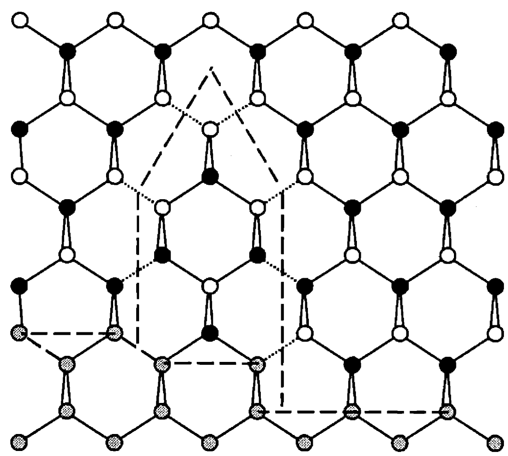


Figure 1.14 Self annihilation of an APB on a Si or Ge surface encouraged by the high density of single-steps. The probability for the coalescence of neighboring APBs increases with the high density of single-steps.[1]

1.3.3 GaAs on Si Integration Results

Ting, *et al.* have demonstrated growth of GaAs on SiGe virtual substrates with a threading dislocation density on the order of 4×10^{6} cm^{-2}.[1] A X-TEM micrograph of a GaAs layer grown on the SiGe substrate is shown in Figure 1.15.

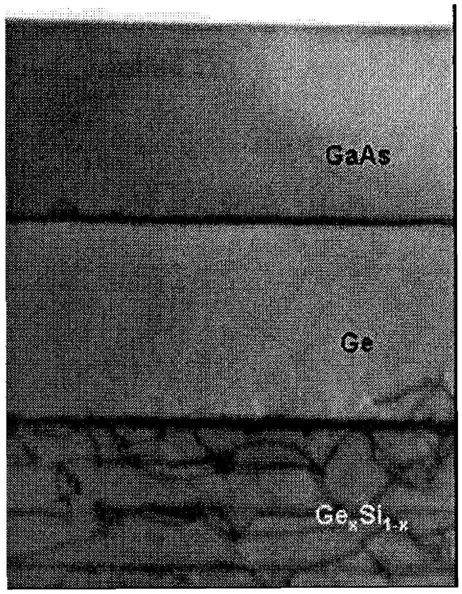


Figure 1.15 X-TEM micrograph of GaAs grown on a SiGe virtual substrate. The micrograph shows misfit dislocation segments at the heteroexpitaxial interface. However, this picture does not show any threading dislocation segments extending into the GaAs layer.

The minority carrier lifetimes of GaAs layers grown on these SiGe virtual substrates have been measured, yielding a record lifetime of 10.5 ns for GaAs on SiGe.[29] Comparing this minority carrier lifetime with that of GaAs on Ge as a function of the threading dislocation density, one can see that, even with a threading dislocation density three orders of magnitude higher than GaAs on Ge, the minority carrier lifetime of GaAs on SiGe approaches that of GaAs on Ge. It can be inferred from Figure 1.16, which shows the minority carrier lifetime as a function of the threading dislocation density, that dislocation densities close to $4x10^6$ cm^{-2} represent dislocation spacings larger than the minority carrier diffusion length. In other words, further improvement in the threading dislocation density of GaAs on SiGe would not lead to a significant increase in the minority carrier lifetime in GaAs, and hence minority carrier device performance. Of course, this assumes that threading dislocations do not have secondary effects on device performance related to degradation and reliability.

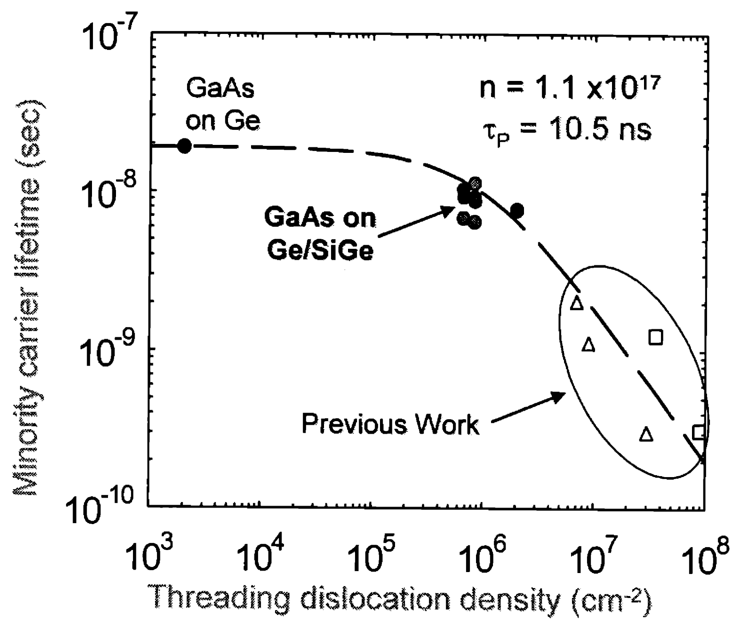


Figure 1.16 Minority carrier lifetime of GaAs as a function of threading dislocation density. The plot shows the lifetime measured for GaAs grown on SiGe substrates approaching the plateau region of the curve, suggesting a decrease in lifetime improvement with further reduction of the dislocation density.[29]

As mentioned in section 1.2, Si and Ge can act as n-type dopants in III-V materials. This becomes a concern for device fabrication with III-V layers on a SiGe virtual substrate, where tight doping control is essential. Ge can be incorporated into the GaAs layer via vapor phase transport as reported by Tobin, *et al.* and Azoulay, *et al.*[41],[42] The secondary ion mass spectroscopy (SIMS) profile shown in Figure 1.17 reveals a substantial Ge diffusion from the substrate into the GaAs layer. The GaAs layer examined in this SIMS analysis was grown via MOCVD on a SiGe virtual substrate at 650 °C. Lowering the growth temperature could potentially alleviate this problem but will potentially create APBs.[1]

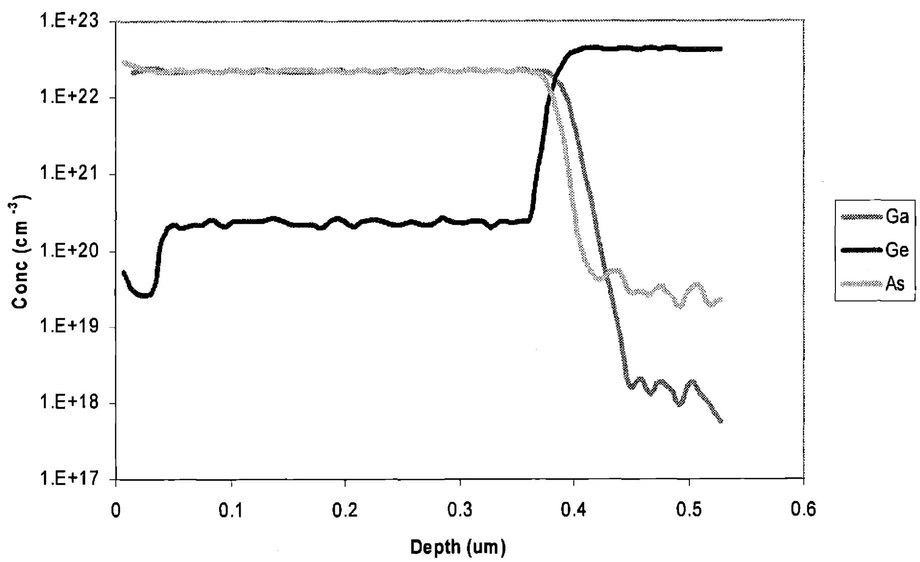


Figure 1.17 SIMS analysis of Ge concentration in a GaAs film grown on a SiGe virtual substrate at 650 °C. The scan shows the diffusion of Ge into the GaAs layer, thus unintentionally changing the doping profile.

Alternatively, the autodoping of Ge in GaAs may be minimized by ensuring GaAs growth in a Ge-free environment. This may be achieved by removing the Ge layer on the backside of the SiGe substrate, an inherent of hot-walled UHVCVD growth, coating the substrate with a thin GaAs layer before device layer growth, and then performing device growth in a clean reactor tube.[43]

1.4 Experimental Synopsis

This work further investigates the feasibility of monolithic integration of GaAs-based thin film devices on Si via the use of SiGe virtual substrates. Chapter 2 will describe the growth and characterization of GaAs-based thin films on SiGe used in this work. To better illustrate the need for SiGe as a platform for III-V/Si integration, an indium arsenide (InAs) avalanche photodiode grown directly on Si has been fabricated without the use of SiGe. The results of the InAs growth and device characteristics are discussed in Chapter 3.

The thermal mismatch issues of GaAs on SiGe integration is discussed in Chapter 4. We have calculated the theoretical critical cracking thickness of GaAs on Si and SiGe and compared the theoretical values to the experimental data. Finite element simulation was also performed to further the understanding of crack formation and its influence on the thermal stress profile close to the cracks in the GaAs layer.

The ultimate demonstration of the III-V integration capability of the SiGe virtual substrate is the fabrication of light-emitting devices such as lasers. However, laser reliability is greatly affected by threading dislocation-induced dark line defects. One way to overcome this obstacle is to add indium to the active region of the laser. Indium gallium arsenide (InGaAs) has been shown to be more resistive than aluminum gallium arsenide (AlGaAs) to degradation due to dark line defects.[17] Before fabricating InGaAs quantum well lasers, we will need to investigate the light emission characteristics of InGaAs quantum wells grown on SiGe substrates. We have studied the luminescent characteristics of InGaAs quantum wells grown on several different substrates including SiGe, Si, Ge, and GaAs. The results are discussed in Chapter 5.

The fabrication of GaAs-based light emitting diodes (LEDs) on SiGe virtual substrates has been previously reported.[1] The next step is the fabrication of a simple optical interconnect system, where the LED is linked to a detector via a waveguide. We have fabricated the first working optical interconnect unit integrated onto Si using a SiGe virtual substrate. This optical interconnection consists of a GaAs LED and detector that are connected by an aluminum gallium arsenide (AlGaAs) waveguide. Device performance will be presented in Chapter 6. The thesis concludes with a summary and recommended future work in Chapter 7.

Chapter 2

Growth and Characterization of III-V Materials on SiGe Virtual Substrates

2.1 Metalorganic Chemical Vapor Deposition Overview

Metalorganic chemical vapor deposition, or MOCVD, relies on the mass transport and pyrolytic decomposition of reactant gases at a heated substrate surface using metalorganic precursors. The entire process of MOCVD entails the pyrolysis of carrier gases, the adsorption of reactants onto the surface of the substrate, the diffusion and reaction of reactants, and finally the incorporation of the resulting material.[44] Pyrolysis of reactants is a thermally activated process with an Arrhenius dependence on temperature. Together with mass transport, this process determines the growth rate and crystal structure of the MOCVD-grown thin film onto the heated substrate.[44] An example of carrier gas decomposition and reaction for GaAs from trimethylgallium and arsine can be described as,

$$Ga(CH_3)_3(g) + AsH_3(g) \rightarrow GaAs(s) + 3CH_4(g) \qquad \textbf{Equation 2.1}$$

This is a simplified description of the reaction; there are actually many intermediate steps involved in this process, both in the gas phase and on the substrate surface.[45]

The precursors used in this work are trimethylindium (TMIn), trimethylgallium (TMGa), trimethylaluminum (TMAl), and arsine (AsH_3). Dopant precursors include 1% silane in H_2 for n-type doping and dimethylzinc (DMZn) for p-type doping. Each precursor requires a different minimum temperature for pyrolysis. In a hydrogen ambient, to achieve a 20% decomposition of TMIn, a reactor temperature of 300°C is needed. Similarly, a reactor temperature of 400°C is required for 20% TMGa and arsine decomposition.[44] The decomposition temperature is related to the bond strength of the precursor molecules.

The growth rate in an MOCVD system features three regimes with different temperature dependences. The first is the kinetically limited regime where the growth rate is limited by the rate of precursor decomposition and occurs when the growth temperature is low. Since the pyrolytic decomposition of precursors is a thermally activated process, the growth rate in this regime is highly temperature dependent. At intermediate growth temperatures, the mass transport of reactants-that is, the diffusion of reactants across the gas boundary layer near substrate surfaces-becomes the limiting factor. Since mass transport is not a thermally activated process, the growth rate in this second regime shows a weak dependence on temperature. The third regime involves growth at even higher temperatures where the growth rate becomes thermodynamically controlled. The high temperature tends to drive the reaction back towards the formation of the reactants through exothermic reactions, thereby decreasing the growth rate with increasing temperature.[46] During the growth of samples used in this work, the growth temperature is in general maintained within the mass transport limited regime; consequently, the flow rate of the group III precursor, not the temperature, determines the growth rates of the III-V material. The flow rate of the group V precursor has nearly no effect on the growth rate because an excess of group V precursor is intentionally maintain and a depletion of the group III precursor over the substrate surface during deposition. Because of the high volatility of the group V material at the growth temperature, a high ratio of group V to group III partial pressure is required. This V/III ratio is given as,[46]

$$\frac{p_V^i}{p_{III}^i} = \frac{K_{III/V}}{a_{III/V}}(p_V^*)^2$$ **Equation 2.2**

where p^i represents the partial pressure of either the group V or group III species at the surface of the substrate, $K_{III/V}$ is the equilibrium constant for the reaction leading to the formation of the III-V compound, $a_{III/V}$ is the activity of the compound, and p_V^* is the partial pressure of the group V element. Note that the partial pressure is proportional to the flow rate. V/III ratios lower than unity usually result in poor surface morphology, the incorporation of impurities, and the formation of group III droplets at the substrate surface.[44,46]

In the mass transport limited regime, the composition of films with more than one group III species, such as $In_xGa_{1-x}As$, is typically linearly dependent on the ratio of the individual group III precursor flow rates.[46] The relative incorporation of the group III species A in solid solution relative to species B can be expressed by a distribution coefficient

$$k_A = \frac{x_A}{p_A^*/(p_A^* + p_B^*)}$$ **Equation 2.3**

where x_A is the mole fraction of A incorporated in the III-V layer.[46]

The dopant fraction, x_D, incorporated in the solid solution is also determined by the partial pressure ratio between the dopant and the group III species if the dopant vapor pressure is low. x_D can be expressed as

$$x_D = \frac{p_D^*}{p_{III}^*}$$ **Equation 2.4**

where p^*_D is the partial pressure of the dopant. The n-type dopant silane would fall into this category. On the other hand, when the partial pressure of the dopant is high, such as for DMZn, x_D becomes

$$x_D = kp^*_D \qquad \textbf{Equation 2.5}$$

where k is the distribution coefficient.[46]

2.2 MOCVD Reactor

A Thomas Swan atmospheric-pressure MOCVD research reactor was used in this study. A schematic diagram of the reactor chamber shown in Figure 2.1 is in the same orientation as the photograph in Figure 2.2. It is necessary to discuss the setup of this reactor since growth conditions are greatly influenced by reactor design.

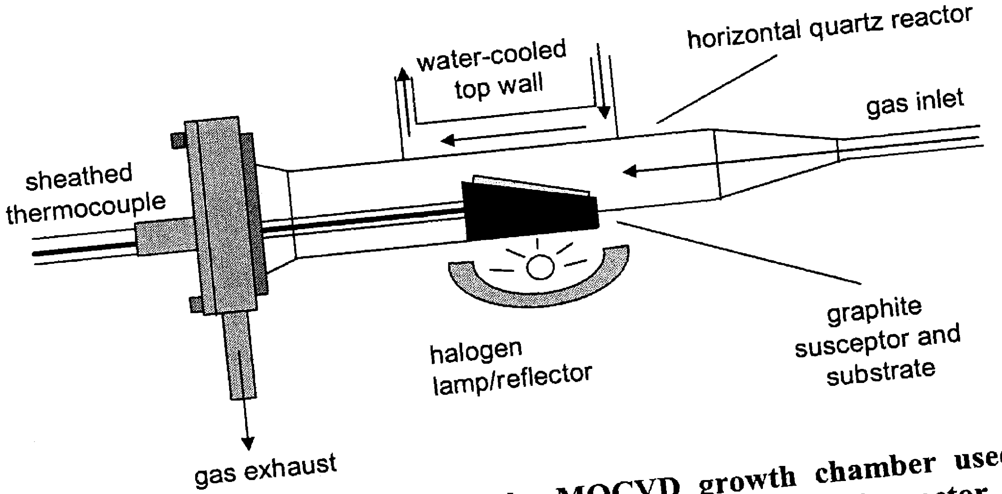


Figure 2.1 **Schematic diagram of the MOCVD growth chamber used in this study. The chamber is a horizontal cold-wall reactor that can accommodate a 1.5×2 cm substrate on the graphite susceptor.**

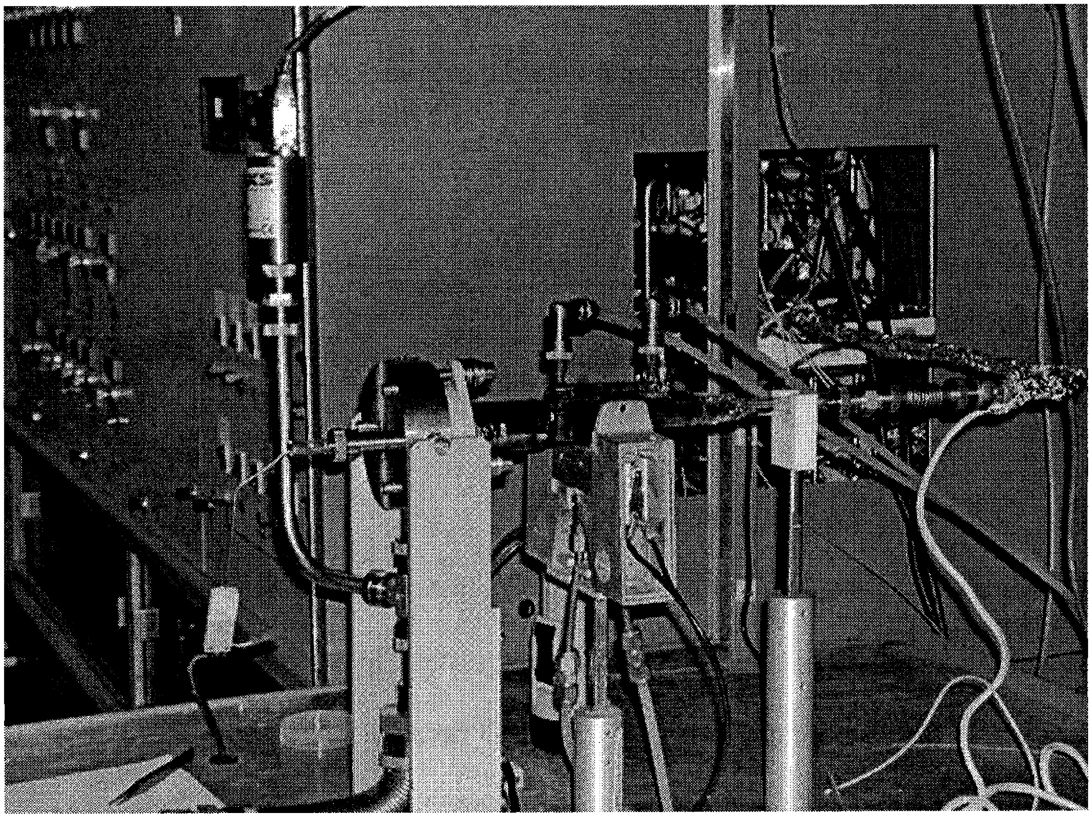


Figure 2.2 Photograph of the atmospheric-pressure MOCVD reactor used in this study.

The Thomas Swan MOCVD is a cold-wall horizontal quartz chamber reactor. This type of reactor configuration provides optimum growth uniformity.[47] The cold-wall condition is maintained by water cooling on the top wall of the tube. Because of the size of the reactor chamber, a sample size no larger than 1.5×2 cm may be accommodated. The sample is placed manually on the tilted graphite susceptor, which is then loaded into the reactor chamber from the open end of the tube. A thermocouple is inserted into the susceptor to provide feedback and control of the growth temperature. A halogen lamp and reflector unit provides the heating for the entire reactor tube.

One advantage of this reactor setup is the ease of maintenance because of the small reactor tube size. However, the small tube size also limits the size of the sample that can be processed. Additionally, because of the limited tube size, thus limiting the samples size, growth non-uniformity becomes a significant problem. As a result of variation in the gas flow direction and the temperature along the sample edges, the deposited layer thickness tends to vary in this region. Recovery from the variation in thickness can take up to 0.5 cm of the sample edge. Unpredictable thickness variations make the useful sample area even more limited.

2.3 Growth Procedure

The substrate materials: Ge, SiGe, GaAs, or Si wafers were first cleaved into smaller pieces to fit onto the graphite susceptor. On-axis GaAs substrates were used in this study, while Ge, SiGe, and Si substrates were offcut 6° to the [110] direction.

Prior to growth the substrate surfaces were cleaned by chemical etching. The Ge and SiGe substrates were cleaned by alternately dipping in 30% hydrogen peroxide (H_2O_2) and a 1:10 hydrofluoric acid (HF) to deionized water solution. Samples were rinsed with deionized water between dips. The final HF dip left the Ge surfaces hydrophobic prior to growth. On the other hand, the Si substrate surfaces were cleaned by a 5 minute piranha solution (1:3 mixture of H_2O_2:H_2SO_4) etch. After the piranha etch, the substrates were treated with an HF dip, leaving the surfaces hydrophobic prior to loading into the MOCVD reactor. The GaAs substrate surfaces were cleaned with a 1 minute dip in a 25:1:1 H_2SO_4:H_2O_2:H_2O solution followed by a 1:10 HCl: H_2O etch that, similar to the HF treatment of Si, left the GaAs surfaces hydrophobic.

Prior to loading of the substrates, the reactor was baked at 850°C for 15 min with a nitrogen gas ambient atmosphere. Unlike this baking step, all other growth steps were carried out under a hydrogen gas ambient. The substrates were first heated up to 350°C for 10 min in order to evaporate water or other volatile contaminants from the surface. The substrate was then subjected to a 5 minute, 700°C thermal oxide desorption step for the SiGe, Ge, and GaAs substrates then followed. With this annealing step to ensure proper surface reconstruction, APB propagation is minimized.[1] A thin layer of approximately 1000 ´ of GaAs was then initiated at 700°C on these three substrate surfaces. The temperature was then brought to the desired level for epitaxy. Growth on the Si substrate varies slightly from the growth procedure described above. Before the initiation of the GaAs layer, a thin Si layer was first deposited onto the Si substrate surface at 550°C to bury surface defects and ensure a more uniform growth surface.

It should be mentioned that for all GaAs growth, arsine was introduced into the reactor chamber before any group III precursor gas flows. With initial arsine exposure at the Si, Ge, or SiGe substrate surfaces prior to growth at temperatures between 400°C and 600°C, reducing the likelihood of APB formation. Ting, *et al.* have shown that at elevated temperature, arsine (AsH_3) exposure leads to the displacement of the Si dimer pair at the Si substrate surface or the Ge dimer pair at the Ge substrate surface by As dimer pairs that are aligned parallel to the surface step edge. Outside of this temperature range, modification of the substrate surface is not energetically favored, resulting in an

As dimer alignment perpendicular to the step edge.[1] Therefore, the growth temperature must be maintained within the same temperature regime. Both of the above mentioned scenarios are shown in Figure 2.3.

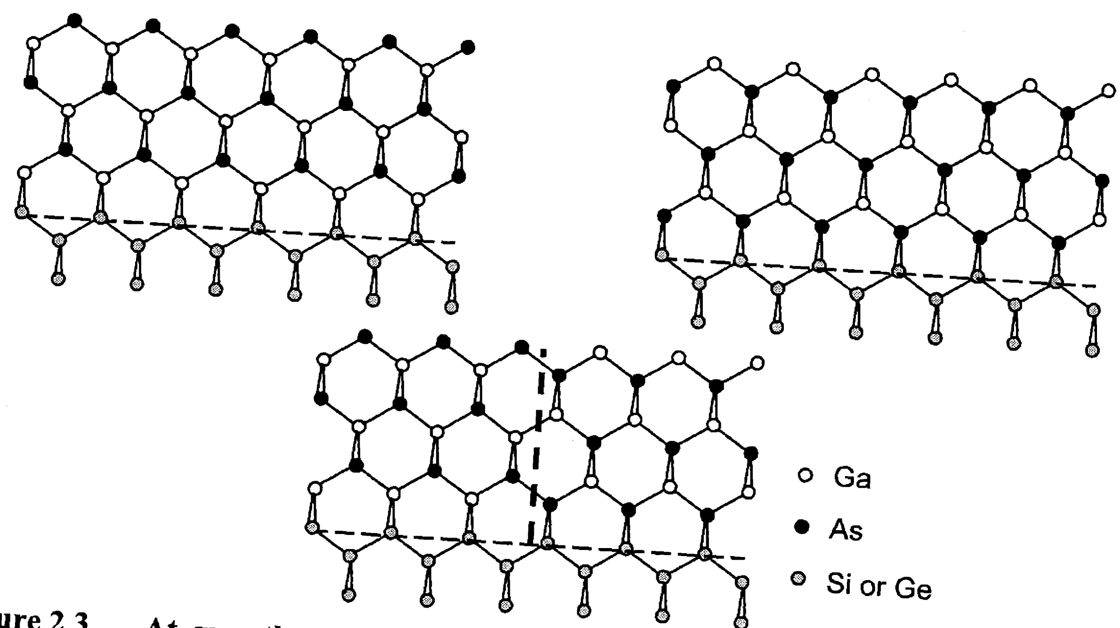


Figure 2.3 **At growth temperatures of 400 to 600°C, arsenic replaces the first monolayer of the single-domain Ge surface (left). Outside of this temperature range, GaAs atoms are only absorbed onto the Ge surface rather than displacing the first monolayer (right). The two diagrams show a rotation in the GaAs aligment.[1]**

2.4 Characterization Techniques

To accurately determine the epitaxial film quality, growth morphology, growth rate, and dislocation density, the use of appropriate characterization techniques is imperative.

Each of the characterization techniques used in this work will be described in the following sections.

2.4.1 Surface Morphology Characterization

Three characterization techniques were used in this study to determine the surface morphology of the epitaxial films. Nomarski optical microscopy can be used to measure the macroscopic surface structures of either the substrate or the III-V film. This technique, combined with the selective etching technique described later in this section, can help determine the threading dislocation density of a thin film. For a more microscopic look at the surface morphology, atomic force microscopy (AFM) and scanning electron microscopy (SEM) are better characterization tools. AFM is capable of providing nanometer-scale surface roughness measurements over areas ranging from 1×1 μm to 100×100 μm. AFM micrographs may often give insights into the growth modes of III-V films. Similarly, SEM may also provide information on the microscopic details of surfaces such as surface crack geometry. Determination of surface cracks with AFM is difficult because of the inherent decrease in accuracy for this technique at sharp changes in the film topography.

2.4.2 Dislocation Characterization

Before a sample can be examined with a transmission electron microscope (TEM), it needs to be thinned to less than 1 μm for electron transparency. Sample thinning is accomplished through polishing and argon ion milling. Cross-sectional TEM, or X-TEM, gives accurate measures of layer thicknesses, and thus the growth rate. X-TEM is also useful for the observation of threading and misfit dislocations; however, it cannot accurately determine dislocation densities of less than 10^8 cm^{-2} because of the limited area of view. On the other hand, plan-view TEM (PV-TEM) serves as a viable method for determining dislocation densities for samples with threading dislocation density on the order of 10^7 cm^{-2} or higher.[11] Again, because of the limited area of view, samples with threading dislocation density lower than 10^7 cm^{-2} must be characterized with the selective etching sample surfaces. Because the etch rate differs near defects, dislocation segments on large sample areas may be revealed. The etching technique used in this study is a light-assisted HF-CrO_3 based etch, termed the $D_{1:x}S_{a/b}L$ etch. x is the DI water dilution D of solution S that consists of a parts HF to b parts 33 wt% aqueous CrO_3 solution. In addition, L denotes the use of light.[48,49] The particular solution used in this study is $D_{1:8}S_{1/2}L$. Photo-generated carriers at defects retard etching to form hillocks, indicating the presence of threading or misfit dislocation segments.[1] This light-assisted etch technique requires a high density of extrinsic carriers and thus can only reveal defects in n-doped samples. An example of a micrograph showing the selective etching of GaAs surface is shown in Figure 2.4.

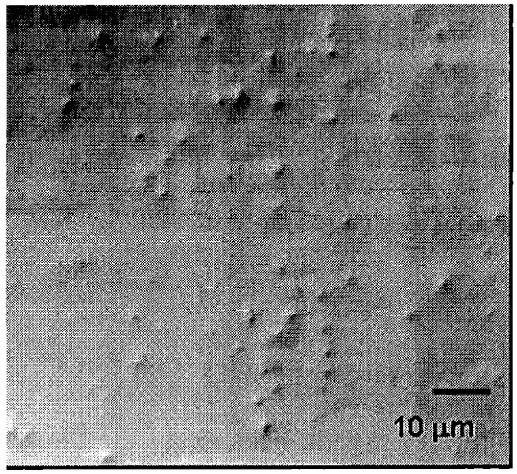


Figure 2.4 Nomarski optical micrograph of a selectively etched GaAs sample surface for analysis of threading dislocation density. Hillocks represent sites where threading dislocations intersect in the film surface.

2.2.3 Other Characterization Techniques

X-ray diffraction (XRD) was used to accurately determine the lattice constant, and thus the composition, of GaAs-based materials grown on SiGe substrates. Together with X-TEM, the MOCVD growth can be calibrated. Once the composition of the film is known, XRD can also be used to determine the degree of strain relaxation.

In addition to XRD, cathodoluminescence spectroscopy (CL) can also be used to determine the composition of III-V films. CL involves the high energy electron beam excitation of carriers. If the composition is known, the luminescence peak position enables the determination of the amount of strain in thin layers. CL has a much higher resolution than XRD for determination of the degree of strain for very thin layers. With dislocation segments acting as recombination sites, the luminescent intensity may also indicate the film quality. The effects of threading and misfit dislocation densities on CL luminescence characteristics are discussed in more detail in Chapter 5.

Chapter 3

Growth and Fabrication of InAs/GaAs/Si Avalanche Photodiodes

3.1 Overview

Heteroepitaxial technology can be applied to the growth and fabrication of avalanche photodiodes, or APDs. An APD is a PIN diode with a net photon collection efficiency, or gain, greater than unity.[50] APDs are, therefore, useful in applications for optical fiber communication systems operating near 1.3 or 1.55 μm,[51] or for optical imaging because of their higher sensitivities compared to conventional PIN receivers. The increase in gain and sensitivity in APD devices is a result of carrier avalanching, or multiplication, resulting from carrier impact ionization at high reverse-bias. Impact ionization occurs in a semiconductor when the energy lost by a carrier in an inelastic collision is sufficient to promote an electron from the valence band to the conduction band.[52,53] In other words, the generation of one primary carrier by an incident photon will lead to the generation of many additional secondary electrons and holes. At the end of this process, a large number of carriers will be present in the high-field region long after the primary electrons have traversed it. The higher the avalanche gain, the longer the avalanche process will persist.[53]

The efficiency at which secondary carriers may be generated depends on the electron and hole ionization coefficients. The ionization coefficients represent the number of electron-hole pairs generated by primary carriers per unit distance traveled by these primary carriers, and are a function of the electric field.[51,53] For example, Wu, *et al.* have determined the electron and hole ionization coefficients of Si to be, respectively,

$$\alpha(E) = 104\,exp(\frac{-1.08x10^6}{E}) \qquad \textbf{Equation 3.1}$$

and,

$$\beta(E) = 77\,exp(\frac{-1.97x10^6}{E}) \qquad \textbf{Equation 3.2}$$

where E, the electric field, is in kV/cm.[54]

Silicon is one of the best material candidates for APD fabrication. It exhibits nearly ideal avalanche multiplication of photoelectrons because of the large ratio, k, between the electron and hole ionization coefficients, α and β. Very large or very small ratios result in APDs with little noise and high gain-bandwidth products.[55] In order to achieve high gain in a stable configuration, the bias must be such that only one type of carrier will undergo impact ionization.[50] Furthermore, if β and α are very different, any fluctuation in density of either type of carrier will lead to a relatively insignificant perturbation, or noise factor.[53] The multiplication gain, M, is given by,

$$M = \frac{(1-\beta/\alpha)\exp[w(1-\beta/\alpha)]}{1-(\beta/\alpha)\exp[\alpha w(1-\beta/\alpha)]} \qquad \textbf{Equation 3.3}$$

where w is the depletion-layer width, which would be the length of the i-region of a PIN diode.[53] One can see that if $\alpha >> \beta$, M would increase, while setting $\alpha \approx \beta$ would decrease the gain value.

For Si, the electron ionization coefficient is much larger than that of the hole. For example, at an electric field of 200 kV/cm, the k ratio is 100, and at an electric field of 500 kV/cm, the k-ratio would be 15.[55] On the other hand, the k ratio for InP is only 1-2, which often leads to a noise factor as high as the gain.[50] Similarly, InGaAs APDs have limited gain-bandwidth product and poor noise because of their low ionization ratio.[51] In addition, the voltage and temperature sensitivity of Si APDs are much lower than that of InP or other III-V superlattice APDs.[54] For example, the typical operating voltage for an InGaAs/InP APD is 75 V, while that for silicon can be as high as 400 V.[50]

Nevertheless, even with the superior avalanche properties of Si, Si APDs do not make efficient near-infrared photodetectors because of silicon's transparency to near-IR wavelengths. Ghiohi, *et al.* attempted to increase the absorption of Si in the near-IR region by heavy doping, thus narrowing the energy gap. However, 1.3 μm photodetectors fabricated using this technique exhibit low quantum efficiency.[56] On the other hand, Hawkins, *et al.* proposed an APD device structure that involves two separate areas, one area made up of a III-V material sensitive to the near-IR wavelength, and the other made of Si as the avalanche material.[51,57] This proposed photodetector is called a Separate-Absorption-Multiplication Avalanche Photodiode, or a SAM-APD. If the Si avalanche region consists of an ideal PIN diode, the width of the avalanche region will be the full intrinsic layer width. In a p-n junction, the avalanche width will be a function of doping and diode geometry. The multiplication will be restricted to a narrow region close to the junction near the peak electric field. For our device, a PIN diode will be used for avalanche multiplication where the width of the avalanche region is predetermined.[53,54] To ensure that no multiplication takes place in the III-V absorption layer of the SAM-APD, the electric field in that area is kept at a low level.[54] In addition, the quantum efficiency of this absorption layer will be a function of the reflectivity R_f, the absorption coefficient α_a, and the thickness of this layer w_a, given as,[54]

$$\eta = (1 - R_f)(1 - e^{-\alpha_a w_a})$$ **Equation 3.4**

The InGaAs-Si SAM-APD fabricated by Hawkins, *et al.* involved the bonding of the III-V material and the Si wafer at high temperature. It was shown that the wafer bonding process did not create oxide barriers at the interface,[55] and that the edge dislocations at the bonding interface did not thread into the epitaxial layers to form threading dislocation segments.[57] Furthermore, no significant charge accumulation or trapping at the bonding interface was observed.[58] This 1.3 μm InGaAs-Si SAM-APD had a gain-bandwidth product comparable to the highest InGaAs-InP APD gain-bandwidth product reported.

The major limitations to the speed of a SAM-APD include: 1) the electron and hole transit time, 2) the avalanche buildup time, which may be improved by keeping the absorption layers and multiplication layers thin, and 3) the time it takes to charge and discharge the built in capacitor at the p-n junction as well as any parasitic capacitance, or RC delay. The RC delay is also a function of the absorption and multiplication layer thicknesses; thinner layers will increase the capacitance and thus the RC delay. However, if the multiplication layer is made too thin, the required electric field to achieve high avalanche gain will increase. Therefore, the challenge for high speed performance is to find the optimal layer thicknesses that minimize the carrier transit time, the capacitance, and the required electric field.[54]

Unlike the integration approach used by Hawkins, *et al.*, we sought to monolithically grow and fabricate a SAM-APD with InAs as the absorption layer that will be sensitive to wavelengths up to 3 μm, and with Si as the multiplication layer.

Monolithic integration is a lower cost and higher throughput approach to putting optical devices onto Si, in addition to the possibility of using highly mature Si technology for III-V devices on a large wafer scale.[59]

3.2 Device Structure

The structure of our SAM-APD is shown in Figure 3.1. The absorption layer used in our device structure is a 2000 Å p-type InAs layer, sensitive to wavelengths up to 3 μm. Carrier multiplication takes place in the Si PIN diode region. The exact doping profile of the Si PIN layers is shown in the spreading resistance scan in Figure 3.2. A 300 Å thick GaAs layer is inserted between the InAs and Si layers in order to reduce the lattice mismatch between each layer. There is an 11% lattice mismatch between InAs and Si, rendering heteroexpitaxial growth of high quality InAs films directly on Si very difficult. With the thin GaAs layer inserted between the InAs and Si layers, the lattice mismatch between each layer is decreased to 7% between GaAs and InAs and 4% between GaAs and Si. The GaAs layer is kept thin in order to minimize the distance traveled by photo-generated carriers from the absorption layer to the multiplication region. These carriers must traverse the GaAs layer before reaching the multiplication region. Graded buffer layers between InAs and Si are not a viable option since grading would further increase the separation between the absorption layer and the multiplication layer, nullifying any benefits from the decreased dislocation density in the InAs layer. Furthermore, these graded layers contain high densities of threading and misfit dislocation segments which would act as carrier recombination sites outside of the multiplication region.

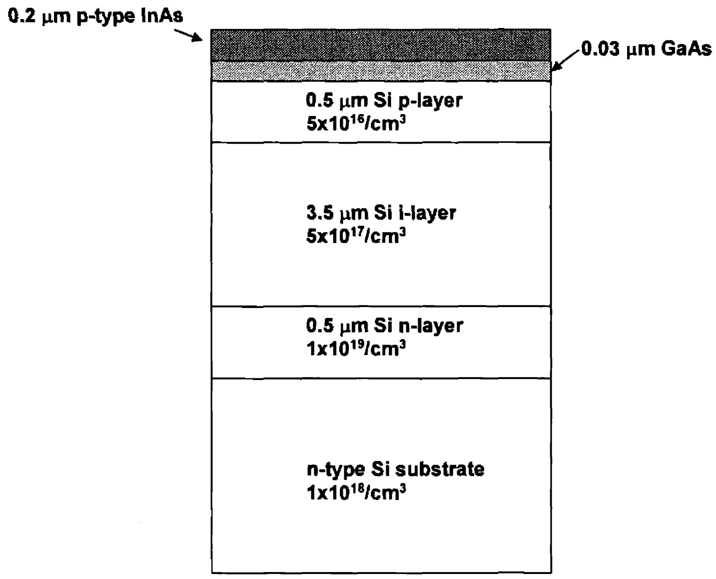


Figure 3.1 **Schematic diagram of the InAs/GaAs/Si SAM-APD device structure. Note a thin GaAs layer is inserted between the InAs absorption layer and Si multiplication layer to reduce the lattice mismatch between the InAs and Si layers.**

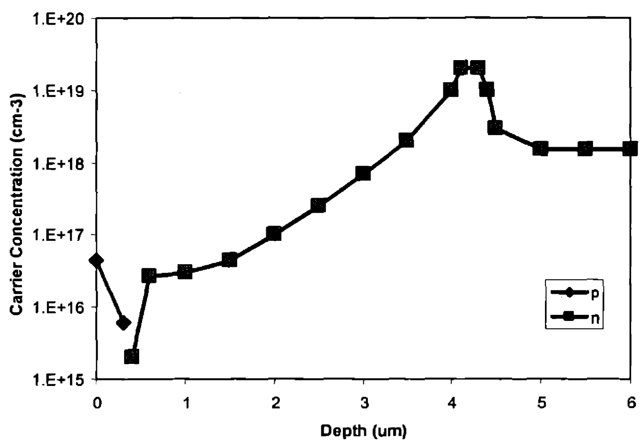


Figure 3.2 **Spreading resistance profile of the doping levels for the Si PIN diode.**

Controlling the distribution of the electric field within each layer is crucial to the operation of a SAM-APD. The field in the absorbing layer must be kept between 2×10^4 and 4×10^4 V/cm. Between these values, the carriers are quickly swept out of the absorption layer without undergoing any multiplication. In addition, by maintaining such a low field value, breakdown can be avoided. On the other hand, a high field of 1×10^6 V/cm should be maintained in the multiplication region to obtain good avalanching behavior.[50] The electric field is controlled by the doping level in each layer. In order to achieve the desired electric fields, the doping in InAs and GaAs should be kept below 1×10^{16} cm^{-3}, the top Si layer should be highly p-doped, and the background doping in the Si layer should be below $1\times10^{11}cm^{-3}$. In the highly-doped cap layer and substrate, the field will be very close to zero while in the i layer, the field will be large enough for avalanche multiplication to take place. On the other hand, if the absorption layer is too highly doped, the field in this layer will become too low, encouraging electron recombination before carriers reach the multiplication layer.[50]

Because of the high density of defects expected in the InAs and GaAs layers, we need to over-dope the top Si p-layer such that the depletion region remains totally within the Si layers. By over-doping, we will prevent the defects in the InAs and GaAs layers from producing significant leakage current. However, this doping profile will lower the electric field in the InAs absorption layer, making diffusion the dominant carrier transfer process. To compensate for this slower and less directional transfer process, the GaAs and InAs layers were lightly doped to lower the potential barrier at the GaAs/Si interface towards Si. Furthermore, this design will avoid the necessity of tight control of the innerlayer doping.[50] The band diagram and the electric field profile of our device are shown in Figure 3.3.

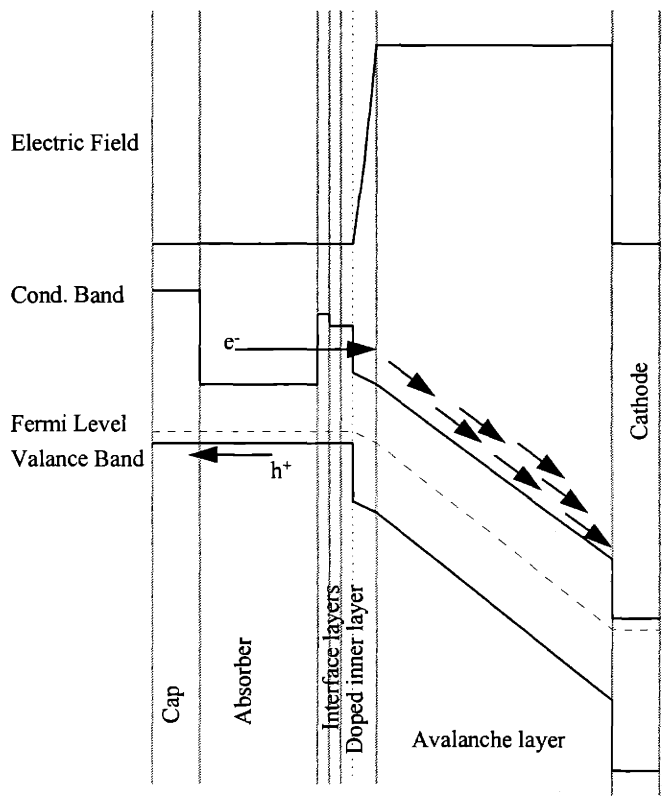


Figure 3.3 Band diagram of the InAs/GaAs/Si SAM-APD. Carriers generated at the absorption layer diffuse toward the Si PIN layers where multiplication takes place.

3.3 Experimental Procedure

Before the growth and fabrication of the SAM-APD, we first studied the InAs and GaAs thin film qualities at different growth conditions for to optimize each layer. This study included growth of InAs thin films on Si and GaAs substrates and GaAs growth on Si substrates. The samples were grown in the atmospheric metalorganic chemical vapor deposition system (MOCVD) as described in Chapter 2. When growing GaAs on a Si substrate, a thin layer of Si was first deposited at 550°C to create bury surface defects. Then a thin GaAs buffer of approximately 2000Å was initiated at 400°C. Following this low temperature growth step, the sample was brought rapidly to 650°C for growth of the GaAs buffer layer. For direct InAs growth on Si substrates, following a thin Si layer deposition, InAs was deposited at 400°C. InAs deposition on GaAs substrates was also performed at 400°C.

The samples were later characterized using X-TEM and AFM. X-TEM was used to verify the various layer thicknesses and AFM was used to characterize the surface morphology of the InAs layers.

3.4 Material Growth Quality

3.4.1 Study of InAs/Si, GaAs/Si, and InAs/GaAs

The InAs film quality will greatly affect the performance of the SAM-APD; thus, growth of InAs and GaAs on Si was studied and characterized. An AFM scan of InAs grown on a Si substrate is shown in Figure 3.4.

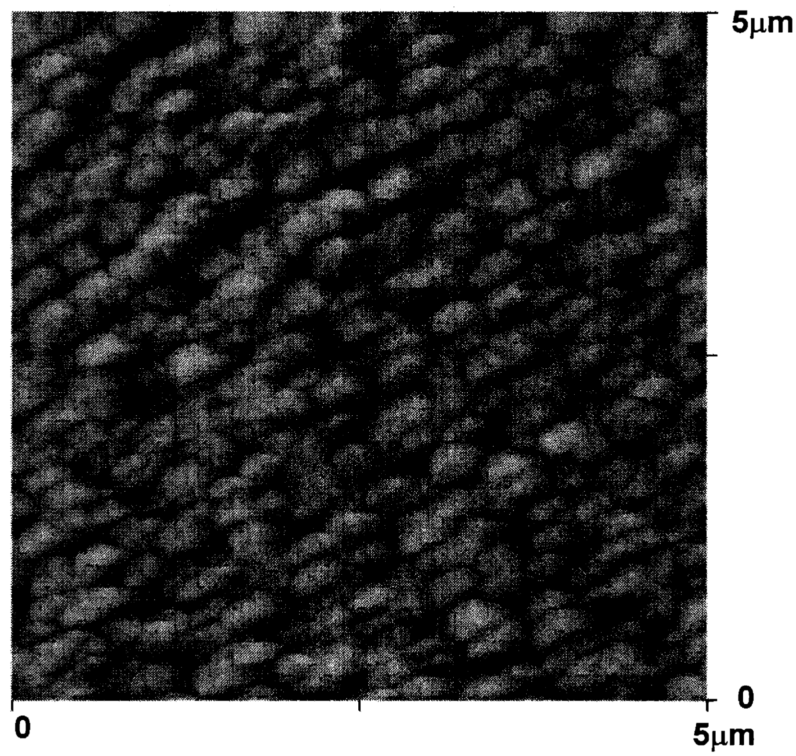


Figure 3.4 **A 5×5 μm AFM micrograph of InAs on a Si substrate. The scan shows 3D growth mode and InAs island formation on the Si surface.**

This scan shows that after 20 minutes of deposition at 400°C, the InAs layer consists mostly of InAs islands. These islands will eventually coalesce to form a continuous layer upon further deposition. This implies that the InAs layer formed in this manner will contain a high density of grain boundaries and defects, as shown in the X-TEM micrograph in Figure 3.5.

Figure 3.5 X-TEM micrograph of an InAs layer on a Si substrate. High densities of defects and grain boundaries are observed.

For most high lattice mismatch systems, growth occurs in the Stranski-Krastanov mode.[59] This means that the initial deposition of the film on the substrate begins with the formation of a stable 2D wetting layer. After this wetting layer reaches a critical thickness, which is a function of the lattice mismatch between the substrate and the thin film, islands begin to form and the island density will increase exponentially with further deposition.[60] However, this critical thickness for InAs on Si is only approximately 5.6 Å, calculated from Equation 1.5. This calculated critical thickness is less than a monolayer of InAs, which means the InAs growth will initiate in a 3-D Volmer-Weber growth mode. Oostra, *et al.* have also shown that As can act as nucleation sites for In islands, further encouraging the 3-D growth mode.[59] It has also been shown by Heinrichsdorff, *et al.* that MOCVD growth of InGaAs of high In composition on GaAs tends to form large plastically relaxed clusters, indicating the strong influence of growth kinetics; for example, long range adatom surface migration.[61] It is therefore not at all surprising to see island formation in our InAs layer on Si.

The AFM scan and the X-TEM micrograph of GaAs grown directly on a Si substrate are shown in Figure 3.6 and Figure 3.7. The X-TEM micrograph shows a single-crystal GaAs film with a high density of threading dislocations and stacking faults. As a result of the high defect density, the surface of the GaAs film has a high RMS roughness value of 23 nm. Similar to the growth of GaAs on Si, InAs films grown on GaAs substrates are single crystal films with high defect densities because of the high lattice mismatch between the two materials, as shown in the X-TEM picture in Figure 3.8. Nevertheless, since growth of InAs on GaAs eliminates the 3-D growth morphology, growth of the InAs absorption layer on GaAs/Si is more desirable than direct InAs growth on Si.

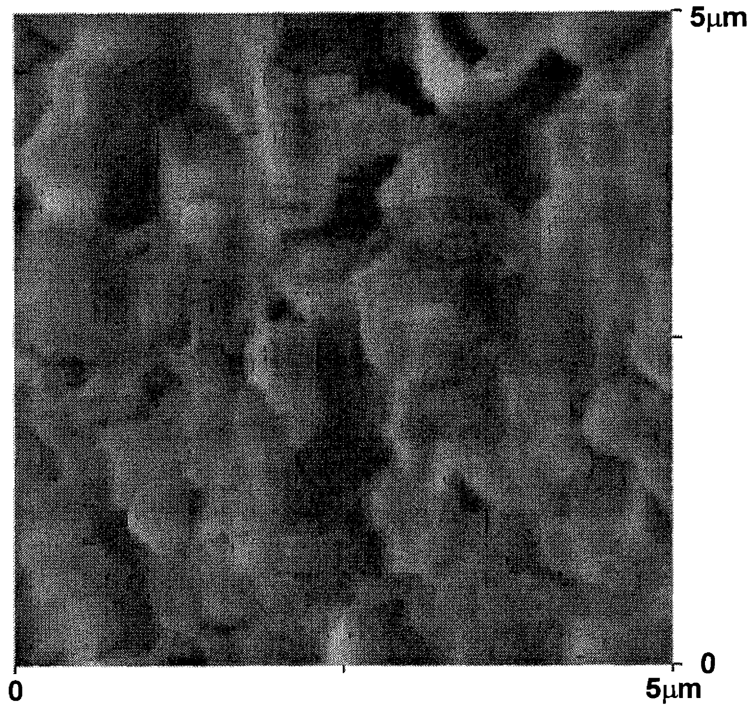


Figure 3.6 **A 5×5 μm AFM micrograph of GaAs surface grown on a Si subtrate. Large surface roughness is a result of the high lattice mismatch between the epilayer and the substrate.**

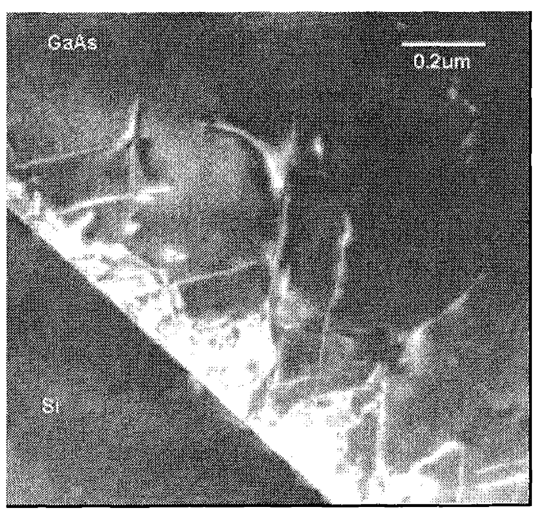


Figure 3.7 **X-TEM micrograph of a GaAs layer grown on a Si subtrate. Due to the high lattice mismatch, a high density of threading and misfit dislocations are present.**

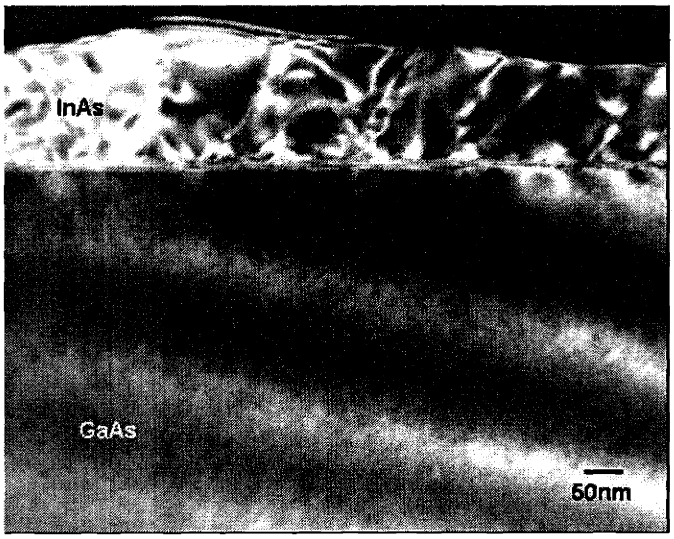


Figure 3.8 X-TEM micrograph of an InAs layer grown on a GaAs substrate. The InAs layer is single crystal but with a high density of defects.

3.4.2 InAs/GaAs/Si APD Structure

Although the thin films of InAs grown on GaAs, and those of GaAs grown on Si, are highly defective, we are able to avoid 3-D growth and islanding of the InAs layer grown on GaAs. For this reason, a thin layer of GaAs was inserted between the InAs absorption layer and the Si multiplication region to ensure single crystal 2-D InAs growth on Si. An X-TEM micrograph of the APD structure is shown in Figure 3.9. By comparing Figure 3.9 and Figure 3.5, we can see that the surface roughness of the InAs film has also decreased upon insertion of the thin GaAs layer.

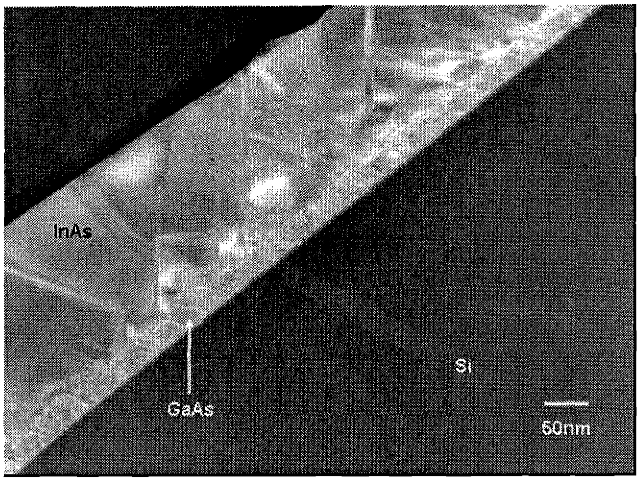


Figure 3.9 **X-TEM micrograph of the InAs/GaAs/Si SAM-APD structure. The surface roughness has been slightly improved compared to direct InAs growth on Si.**

3.5 Device Results

The I-V characteristic of the Si PIN diode, where the multiplication of carriers takes place, was measured and is shown in Figure 3.10. The PIN diode has an ideality factor of 6.2, a leakage current of 1.95×10^{-5} A, and a series resistance at high voltages of approximately 476 Ω. As the SIMS profile in Figure 3.2 shows, the doping profile of the Si PIN diode has a much lower p-type doping level compare to the n-type region. In addition, the i-region has a doping profile that changes gradually from n-type to p-type. All of these attribute can contribute to the non-ideal behavior of the Si PIN diode. The large ideality factor indicates the presence of a high density of traps and space charges where recombination and generation may take place. At reverse bias, these space charge and trap sites also lead to an increase in the leakage current with increasing reverse bias voltage.

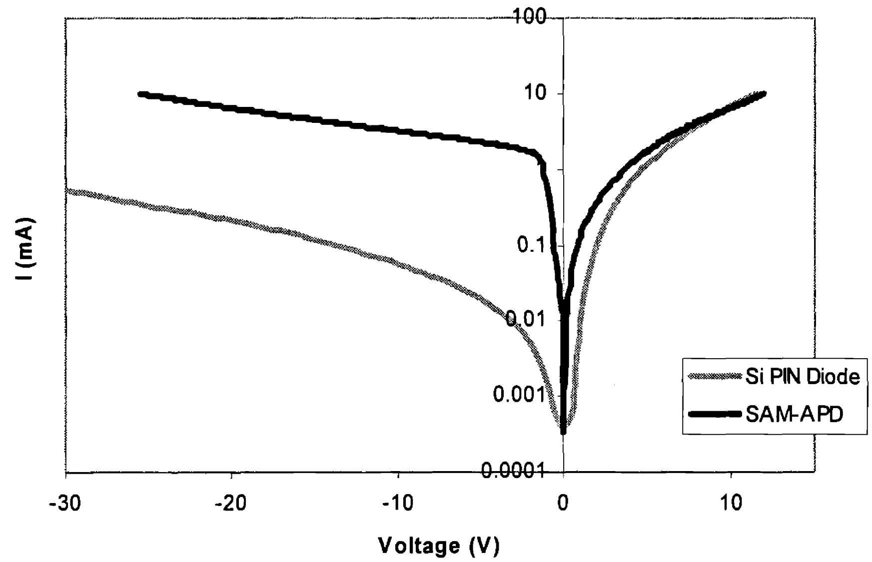


Figure 3.10 I-V characteristic of the Si PIN multiplication region and the InAs/GaAs/Si SAM-APD. Leakage current increases from the Si I-V curve to the SAM-APD I-V curve as a result of increased defect density.

Because of the additional defects introduced by the InAs and GaAs layers, the number of traps and recombination-generation centers increases after deposition of these layers. The effect of these traps is amplified in the I-V curve of the SAM-APD diode as seen in Figure 3.10. The ideality factor is now increased to 13.5, the leakage current is increased to 2.3×10^{-3} A, and the series resistance is increased to 625 Ω. Gopal, *et al.* have also observed that carrier generation may occur at the heterointerface. The most likely agent for generation is the network of misfit dislocations. Free surfaces and defects are known to pin the Fermi level in the conduction band in InAs, thus making the semiconductor degenerate.[62] As mentioned in Section *3.2*, the structure doping profile is designed so that the defects in the InAs and GaAs layers will not contribute to the leakage current. This is done by over-doping the p-type layer of the Si PIN diode. However, the SIMS profile in Figure 3.2 shows that the actual doping level of this layer is only on the order of 10^{16} cm^{-2}, much lower than what was required. Therefore, the misfit and threading dislocations in the InAs and GaAs layers will lead to an increase in the leakage current. These dislocation segments may act as current shorting paths, therefore allowing a higher leakage current. Consequently, an overall degradation of the diode performance is shown.

3.6 Conclusions

We have fabricated a SAM-APD with an InAs absorption layer, Si multiplication layer, and thin GaAs layer inserted between the InAs thin film and the Si substrate to mediate the lattice mismatch between layers. Graded buffer growth between InAs and Si was not feasible since graded layers would not only increase the distance between the absorption layer and the multiplication region, but also contain high densities of threading and misfit dislocation segments. The SAM-APD has an ideality factor of 13.5, a leakage current of 2.3×10^{-3} A, and a series resistance of 625 Ω. Device degradation is a result of increased threading and misfit dislocation densities introduced by the InAs and GaAs layers.

Optical devices and other minority carrier devices are extremely sensitive to these defect densities. It will therefore be impossible to fabricate efficient optical devices with films of high defect density in which the dislocation spacing is less than the minority carrier diffusion length. Therefore, in order to achieve high quality monolithic integration of III-V material on Si, alternative methods such as the use of SiGe virtual substrates become indispensable.

Chapter 4

Critical Thickness for Crack Formation in GaAs Heteroexpitaxial films on Si and SiGe Virtual Substrates

4.1 Overview

During heteroexpitaxial growth, one must consider not only the lattice mismatch between the substrate and the heteroepitaxial film but also the thermal mismatch effects. High thermal mismatch can lead to high density arrays of cracks in the thin film. When the film and the substrate experience different rates of thermal expansion and contraction, the thin film will experience considerable tensile or compressive stress during a temperature cycle, leading to crack formation or delamination. Such crack formation can be observed after a 675°C to 725°C temperature drop from the growth temperature to room temperature in thick GaAs films grown on Si or SiGe virtual substrates. For the GaAs/Si system, the GaAs film has a thermal expansion coefficient, α, of $5.6\times10^{-6}\ ^{\circ}\mathrm{C}^{-1}$, much higher than the thermal expansion coefficients of Si or the SiGe virtual substrate, with α of approximately $2.6\times10^{-6}\ ^{\circ}\mathrm{C}^{-1}$. The thermal expansion coefficient of the SiGe substrate is dominated by that of Si since the Si substrate is much thicker than the total thickness of the graded layers plus the top Ge layer, which is approximately 10 μm. The presence of cracks in thin films is not desirable for device fabrication because they can act as scattering centers for light propagation, resist in-plane electrical current flow, and introduce electrical shorting paths in vertical current flow.[63]

It is therefore necessary to understand crack formation in heteroexpitaxial films. In this chapter, we will present mathematical models for the critical epitaxial film thickness for the onset of crack formation in GaAs films on Si and SiGe substrates and compare these calculations to experimental data. Some finite element modeling results of crack formation using ANSYS 5.7 software[64] are also presented.

4.2 Mathematical Models

Crack nucleation is a heterogeneous process initiated by pre-existing material flaws. Crack propagation is unstable once activated, and will only be arrested when the crack encounters another channel or a sample edge. The crack propagation process depends on two contributing energies; one is the energy released from strain relaxation, and the other is the energy consumed in the creation of the two new surfaces. The two energies act against each other, and as the total energy begins to decrease, the propagation process becomes energetically favorable.[65] Cracks formed will extend their length in the direction of maximum potential energy density.[66] Small initial cracks begin on the upward slope of the energy release rate curve. Once started, they release excess energy in unstable growth. Very quickly they grow to a length where the energy release rate drops below the critical value.[65] For a sufficiently long channel, a steady state is achieved so that the crack front shape and energy release rate remain constant.[67]

When the film and the substrate are thermally mismatched, crack arrays can decrease the overall energy of the system by lowering and relieving local stresses. However, the reduction in stress resulting from crack formation is a localized phenomenon. Stress will be relieved and total strain relaxation will occur only over a finite distance away from the crack plane.[68,69] Typically, 87% of the stress relaxation occurs within a distance t from the crack plane, where t is the thickness of the film.[70] In the GaAs system, cracks may form in the [110] and $[1\bar{1}0]$ directions. However, thin films usually exhibit asymmetric crack arrays at thicknesses closer to the critical cracking thickness, as observed by Murray, *et al.*[69] It was also observed *in situ* by Pompe, *et al.* that as the thermal load increases, existing cracks grow while new ones are formed until a nearly equidistant array has been formed. Propagation cracks are stopped by inhomogeneities as well as by mutual interaction and surface roughness.[65,71]

Thin film cracking patterns may be classified into five different categories: surface cracks, channeling, substrate-damaging, spalling, and debonding.[72] Examples of the different cracking patterns are shown in Figure 4.1. GaAs on Si layers tend to crack without substrate damage. However, substrate damage does occur in the GaAs/SiGe/Si system, where cracks extend into the 100% Ge substrate layer and sometimes even into the SiGe graded buffer region.

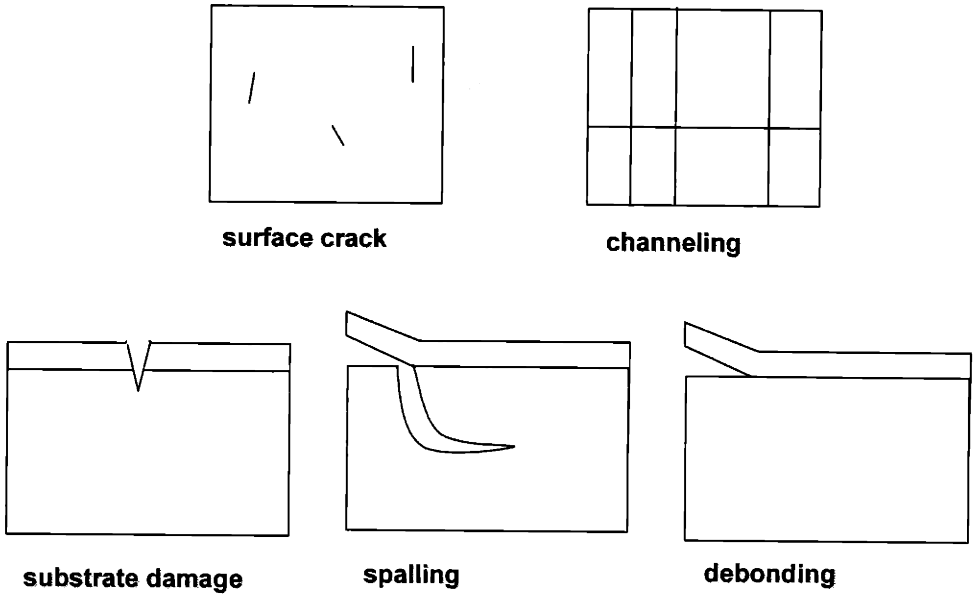


Figure 4.1 **Examples of different cracking patterns that are possible as a result of thermal mismatch between the film and the substrate. This includes surface cracks (plane-view), channeling (plane-view), substrate-damaging (cross-section), spalling (cross-section), and debonding (cross-section).**

4.2.1 Crack Channels with no Substrate Fracture

Hutchinson, *et al.* have formulated a general critical cracking thickness relation for all crack patterns by using a dimensionless driving force number Z such that the energy released per unit area for a crack is

$$G = \frac{Z\sigma^2 t}{\overline{E}_f}$$ **Equation 4.1**

where σ is the stress experienced by the thin film, $\overline{E}_f$ is the biaxial modulus, and t is the film thickness.[72] For a system where the substrate and the film have similar elastic moduli, a typical Z value will range from 2 to 4, depending on the crack type and geometry.[72] The stress is calculated from the difference in the thermal expansion coefficients between the thin film and the substrate

$$\sigma = \overline{E}_f(\alpha_f - \alpha_s)\Delta T$$ **Equation 4.2**

where α_f and α_s are the thermal expansion coefficients of the thin film and the substrate, respectively, and ΔT is the change in temperature.

A critical cracking thickness equation can be derived from Equation 4.1 and Equation 4.2, given that the energy release rate G is equal to the fracture resistance Γ, at the critical thickness for crack formation[72,73]

$$t_c = \frac{\Gamma \overline{E}_f}{Z\sigma^2}$$ **Equation 4.3**

This fracture resistance value is, in our case, twice the surface energy of the film since crack formation creates two new surfaces in the film.[65] The new crack may then propagate and channel through the thin film layer if the elastic energy released by its growth is greater than the energy required to create the fracture surfaces-this is the well-known Griffith criterion.[66]

Another theoretical approach considered by Murray, *et al.* takes into account the balance between the surface energy of the newly formed cracks and the stored elastic energy, similar to the Matthews-Blasklee calculation for the critical thickness of dislocation formation. Using this method, Murray, *et al.* have determined the critical thickness for the onset of crack formation for InGaAlAs thin films grown on InP substrates resulting from lattice mismatch strain in the thin film. The stored energy E_e per unit length of incipient crack is[74]

$$E_e = \overline{E}_f t \varepsilon_T^{\ 2}. \qquad \textbf{Equation 4.4}$$

where t is the thickness of the GaAs film. In our case, the strain resulting from thermal mismatch, ε_T is given by

$$\varepsilon_T = \Delta T(\alpha_s - \alpha_f) \qquad \textbf{Equation 4.5}$$

and the increase in surface energy due to crack formation is:

$$E_s = 2\gamma \qquad \textbf{Equation 4.6}$$

where γ is the (110) surface energy of the epilayer. Using both equations, we can determine the critical thickness t_c for the onset of crack formation a result of thermal mismatch by equating the two energies, obtaining

$$t_c = \frac{2\gamma}{\overline{E}_f \varepsilon_T^{\ 2}} \qquad \textbf{Equation 4.7}$$

Notice that this formulation gives an identical relation to Equation 4.3 if Z=1.

In our case, the Z values of the GaAs/Si or GaAs/SiGe systems are unknown. We therefore need to use a mathematical approach devised by Beuth to calculate the critical thicknesses. In steady-state, the total work released per unit length of crack propagation can be deduced by examining the difference in energy in front of and behind the crack front. This yields a steady-state energy release rate of[67,75]

$$G = \frac{\sigma^2 t}{2\overline{E}_{fp}} \pi g(\alpha, \beta) \qquad \textbf{Equation 4.8}$$

where

$$\overline{E}_{fp} = \frac{E}{1 - \nu^2} \qquad \textbf{Equation 4.9}$$

where $\overline{E}_{fp}$ is the plane strain tensile modulus of the film, ν is the Poisson ratio of the film, and $g(\alpha,\beta)$ is a non-dimensional integral of the crack opening displacement that is dependent on the two Dundurs parameters, α and β.[67,75] This describes a Mode I fracture, or the tensile opening mode where the faces separate in the direction perpendicular to the plane of the crack.[76] The two Dundurs parameters are functions of the material parameters,

$$\alpha = \frac{\overline{E}_{fp} - \overline{E}_{sp}}{\overline{E}_{fp} + \overline{E}_{sp}} \qquad \textbf{Equation 4.10}$$

$$\beta = \frac{\mu_f(1 - 2\nu_s) - \mu_s(1 - 2\nu_f)}{2\mu_f(1 - \nu_s) + 2\mu_s(1 - \nu_f)} \qquad \textbf{Equation 4.11}$$

where $\overline{E}_{sp}$ is the substrate plane strain tensile modulus and μ_f and μ_s are the film and substrate shear modulus, respectively. Values of $g(\alpha,\beta)$ as a function of different α and β are given in reference 54.

4.2.2 *Crack Channels with Substrate Fracture*

With a brittle substrate, cracks initiated from the film can extend into the substrate. If cracks extend into the substrate, the critical thickness for film cracking will be lowered.[77] A dimensionless critical cracking number Ω_c, which is equal to $1/Z$, can be used in calculating the critical cracking thickness for this type of system.[68,77] This critical cracking number is dependent on the film thickness, residual stress, the substrate yielding strength, the ratio of the elastic moduli Σ and the fracture resistances of the film and substrate.[68,77] Assuming a Mode I fracture, the critical cracking number can be expressed as

$$\Omega_c = \frac{1}{Z} = \frac{[1 + \frac{\Gamma_s}{\Gamma_f}(\eta_s - 1)]}{\Sigma}$$ **Equation 4.12**

where Γ_f and Γ_s are the fracture resistance of the film and the substrate, respectively, η_s is the ratio between the length of the crack a_s and the film thickness t, and Σ is the ratio between the plane-strain tensile moduli of the film and the substrate.[77] By substituting this new non-dimensional number back into Equation 4.3, we can calculate the critical cracking thickness of a system where the crack penetrates into the substrate surface.

4.2.3 *Crack Channel Arrays*

Thouless has investigated the equilibrium spacing between crack channels in a system where no substrate fracture takes place.[78] The energy release rate during the propagation of cracks can be obtained, again, by comparing the strain energy in cracked and uncracked portions of the system. The channels will grow if the energy release rate at the front of the crack plane is greater than the fracture energy of the film.[77] The cracks will be more closely spaced if the film stress or the film thickness is increased or if the fracture toughness of the film is decreased.[78] However, in reality, the spacing will be largely determined by pre-existing defects since crack formation is a heterogeneous nucleation process.[65,69,78]

A number of cracks may propagate when the critical stress level is reached. Once the critical thickness is reached, the minimum crack spacing for a film of thickness t will be[78]

$$\lambda = 8t\left(1-\sqrt{1-\frac{t_c}{t}}\right). \qquad \textbf{Equation 4.13}$$

4.3 Analytical Calculations

4.3.1 Crack Channels with no Substrate Fracture

Material parameters used in our calculation are listed in Table 4.1. We must note that the surface energy values for Ge and GaAs given in different references vary greatly, and will therefore affect the accuracy of our calculations.[79,80,81,82,83] The surface energy is also very sensitive to dopant and contamination levels in the material layers. For example, Maucione has found that As-contaminated Ge exhibits lower surface energy than bulk Ge.[80] For GaAs thin films grown on Si, Murray's formulation (Equation 4.7) yields a critical thickness of

$$t_c = \frac{2.37}{\Delta T^2}. \qquad \textbf{Equation 4.14}$$

For ΔT of -675°C, we obtain a critical cracking thickness of 5.2 μm. However, this assumes that the non-dimensional number Z is equal to unity. This critical thickness calculated from Murray's relation could, therefore, be an overestimation.

Table 4.1 Mechanical properties of Ge, Si, and GaAs[25]

	GaAs	Si	Ge
$\overline{E}$ (Nm^{-2})	1.24×10^{11}	1.81×10^{11}	1.39×10^{11}
$\overline{E}_p$ (Nm^{-2})	9.50×10^{10}	1.41×10^{11}	1.10×10^{11}
$\gamma(110)$ (Jm^{-2})	1.5		1
ν	0.31	0.28	0.26
μ (Nm^{-2})	3.29×10^{10}	5.20×10^{10}	4.10×10^{10}
α at RT (°C^{-1})	5.8×10^{-6}	2.6×10^{-6}	5.86×10^{-6}

Using Beuth's formulation (Equation 4.3), we obtain a critical thickness of 3.9 μm for a GaAs layer on Si substrate with a ΔT of -675°C. The Dundurs parameters α and β for GaAs on Si are -0.195 and -0.044, respectively, giving a non-dimensional integral of the crack opening displacement, $g(\alpha,\beta)$, for GaAs on Si of 1.094.[75] The critical thickness value of 3.9 μm is equivalent to a Z value of 1.327. A summary of the different calculated t_c values is given in Figure 4.2 and plotted for various temperature.

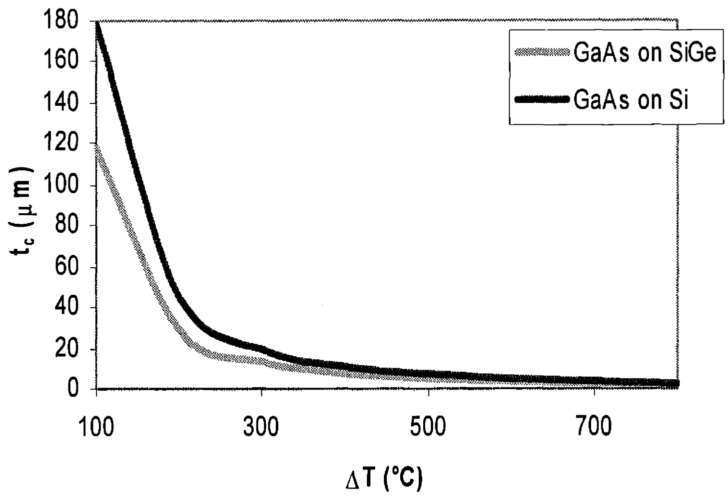


Figure 4.2 **Calculated critical cracking thickness for GaAs on Si and SiGe substrates. Note that the critical cracking thickness for GaAs on SiGe is lower than GaAs on Si because of the additional thermal stress in the Ge layer. Since we have assumed a α_{Ge} value equal to the one for α_{GaAs} in our calculations, the theoretical t_c values for GaAs on SiGe should be slightly lower than the ones shown. The actual α_{Ge} value is slightly higher than α_{GaAs} at the growth temperature.**

Equation 4.4 cannot be used to predict the onset of crack formation in a GaAs film grown on a SiGe substrate, even though the thermal expansion coefficient of this type of substrate is nearly identical to that of Si. The SiGe graded buffer layers are comparatively much thinner than the Si substrate, but their combined thickness does contribute to the formation of cracks in the epilayer. Since the thermal expansion coefficient of Ge is similar to that of GaAs, as shown in Figure 1.8 and Table 4.1, there also exists a thermal stress between the Ge cap layer and the SiGe graded layers within the Si substrate. Therefore, we would expect the critical cracking thickness of GaAs grown on SiGe to be smaller than direct GaAs growth on Si. In addition, unlike GaAs on Si, we have observed substrate damage under X-TEM where cracks penetrate into the SiGe virtual substrate, terminating in the SiGe graded buffer region. We therefore need to treat the GaAs/SiGe system differently. Note that the actual crack tip position observed under X-TEM after growth could have been slightly altered due to the additional stress experienced by the sample during TEM sample preparation.

4.3.2 Crack Channels with Substrate Fracture

The Dundurs parameter α for GaAs on a Ge substrate is -0.073. With a Ge to GaAs surface energy ratio of 0.67, we get a non-dimensional value Ω_c of 0.47, or a Z value of 2.13. This substrate to film surface energy ratio also gives a ratio of the total crack length to the film thickness of approximately 1.5, according to the calculations by Ye, *et al.*[77] The model therefore predicts a critical cracking film thickness for GaAs on SiGe of 2.46 μm and a total crack length of 3.69 μm if ΔT equals -675°C. However, since Ge has a similar thermal expansion coefficient to a GaAs at the growth temperature, and would therefore experience a similar thermal stress from the temperature change, we can assume that the critical thickness calculated above includes both the GaAs film thickness and the Ge cap layer thickness. The SiGe substrates used in this study have a Ge cap layer of approximately 8000Å, consequently decreasing the upper limit of the GaAs film thickness before the onset of crack formation by the same amount. In other words, with a temperature change of -675°C, the GaAs critical thickness will be decreased to 1.66 μm.

Keep in mind that these calculations ignore the thicknesses and thermal expansion coefficients of the individual SiGe graded buffer layers underneath the Ge cap layer, which would further decrease the calculated t_c.

From our calculations of the Z values, we see that a system with substrate damage will have a non-dimensional driving force greater than that with a non-brittle substrate. Z values, therefore, are a representation of the ease of crack formation in the film. The greater the Z value, the lower the barrier to crack formation.

4.3.3 Crack Channel Arrays

We often observe that arrays of crack channels in our GaAs films on Si and SiGe substrates become more pronounced as the film thickness increases. For a film at the critical cracking thickness, the minimum crack spacing will be eight times the film thickness, or 31.5 μm for GaAs on Si with a ΔT of -675°C, according to Equation 4.13. As the film thickness increases, the minimum spacing will decease. However, since crack nucleation is a heterogeneous process, the position and density of cracks depends strongly on the spacing of pre-existing defects and particles at the substrate surface and on the growth conditions.[84] As with all of the analytical calculations presented in this section, we are ignoring the heterogeneous nature of crack nucleation and the kinetics involved in crack formation. Consequently, the observed critical thicknesses should be larger than the predicted values, analogous to the discrepancies between the Matthews-

Blakeslee equilibrium predictions for dislocation formation and the observed critical thickness.[85,86,87]

4.4 Experimental Procedure

GaAs epilayers with thicknesses varying from 1 μm to 8 μm were grown on both Si and SiGe substrates at three different temperatures: 600°C, 700°C, and 750°C. The growth sequences for the two substrate materials varied. For growths on SiGe substrates, after 350°C and 700°C annealing steps in a hydrogen ambient, the GaAs epilayers were grown at either 600°C, 700°C, or 750°C. The procedure for growths on Si substrates varied slightly. Before initiating GaAs growth, thin layers of Si were deposited on the substrate at 550°C to create more uniform Si surfaces.

The samples were later characterized with X-TEM, SEM, and Nomarski optical microscopy. X-TEM was used to determine the GaAs film thickness and to observe the characteristics of the cracks. Because of the high surface roughness of the GaAs epilayers grown on Si substrates, resulting from the large lattice mismatch, use of SEM was necessary to observe cracks in the GaAs/Si samples. Optical microscopy is effective only for smoother surfaces, such as surfaces of GaAs films grown on the SiGe substrates.

4.5 Experimental Results

The observed critical thickness for the onset of crack formation in GaAs epilayers on Si is approximately 7 μm for a ΔT of -575°C, 5.1 μm for a ΔT of -675°C, and 4.9 μm for a ΔT of -725°C. Similar results have been reported by others where they observed the start of crack formation in GaAs layers with thicknesses greater than 5 μm grown on Si after a 675°C temperature drop.[88,89] These values are greater than the theoretical critical thickness value of GaAs on Si, as shown in Figure 4.3. The differences are a result of the kinetics of crack nucleation, which are not captured by the mathematical models. Without consideration of kinetic barriers, the calculated critical thicknesses should be smaller than the actual critical thickness values. However, the difference between the calculated and experimental critical thickness values was not observed in samples of GaAs on SiGe substrates. We have assumed in our calculations, for ease of modeling, that Ge and GaAs have the same thermal expansion coefficients. In reality, Ge has a slightly higher thermal expansion coefficient than GaAs at the growth temperatures (Figure 1.8). Our assumption would therefore slightly overestimate the critical cracking thickness of GaAs on SiGe. Furthermore, the SiGe graded buffer layers also contribute to the overall thermal stress and are also not included in the mathematical model.

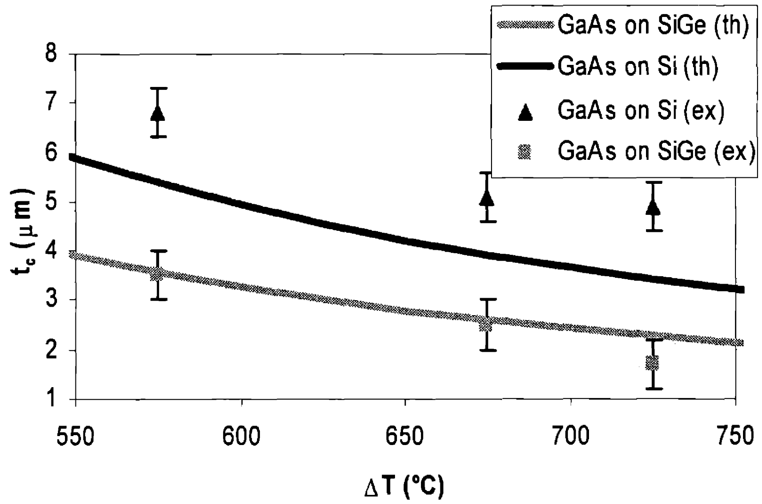


Figure 4.3 **Comparison of calculated and experimental critical cracking thickness for GaAs layers grown on Si and SiGe substrates. The experimental values for GaAs on Si are slightly higher than calculated due to kinetic barriers to crack formation.**

Asymmetric crack array formation was observed in all of our samples. Cracks were observed running in the <110> directions parallel and perpendicular to the substrate offcut direction. However, when the GaAs thickness is close to the critical thickness for crack formation, the cracks run predominantly in only one of the <110> directions. The crack densities for each of the two directions were calculated for the set of samples with GaAs films grown on the SiGe substrates at 750°C, giving a ΔT of -725°C, as shown in Figure 4.4. Notice that the density of cracks in the <110> direction parallel to the substrate offcut direction shows a rapid increase as the film thickness increases, but eventually reaches a plateau value of approximately 130 cm^{-1} for thicknesses greater than approximately two times t_c. A similar trend of sudden increase in density as a function of film thickness was also observed for misfit dislocation formation in mismatch strained layers.[90] The graph also suggests that GaAs film thickness must extend beyond 4 μm before the appearance of substantial crack densities in the orthogonal <110> direction. Optical micrographs of samples with and without cracks in both <110> directions are shown in Figure 4.5. The 4 μm thick GaAs layer in Figure 4.5 was grown at 700°C (ΔT = -675°C) on a SiGe substrate. GaAs films with thicknesses well beyond the critical thickness exhibit high crack densities extending in both crystallographic directions, as shown in Figure 4.5. This figure shows a sample with approximately 8 μm of GaAs grown at 700°C, or ΔT = -675°C, on a Si substrate, a thickness two times the critical cracking value. This same effect has also been noted in InGaAlP/InP systems where a film thickness of approximately four to six times t_c can be grown before cracks in both directions are observed.[74,91]

We have compared the experimental crack density values with the ones predicted by Thouless in Equation 4.13 in Figure 4.4. The calculated values are much higher than the observed densities. Equation 4.13 is modeled after a system where no substrate damage exists. However in our system with GaAs on SiGe substrates, substrate damage is observed. Therefore, the observed discrepancy between analytical and experimental results is not surprising. Furthermore, kinetic barriers to crack formation were not taken into consideration in the theoretical formulation.

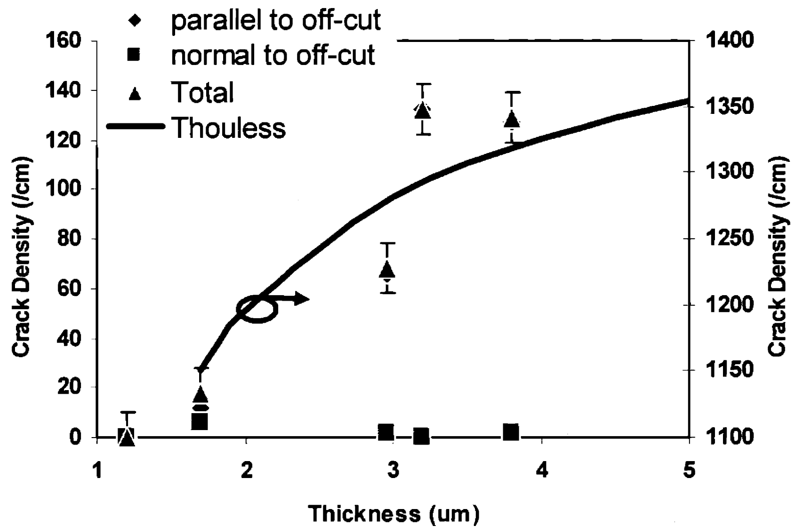


Figure 4.4 Crack spacing as a function of GaAs film thickness grown on Si at 750°C. The spacing shows an exponential increase with the film thickness up to 3 μm for the array parallel to the substrate offcut <110> direction. Past 3 μm, the spacing plateaus to a constant value.

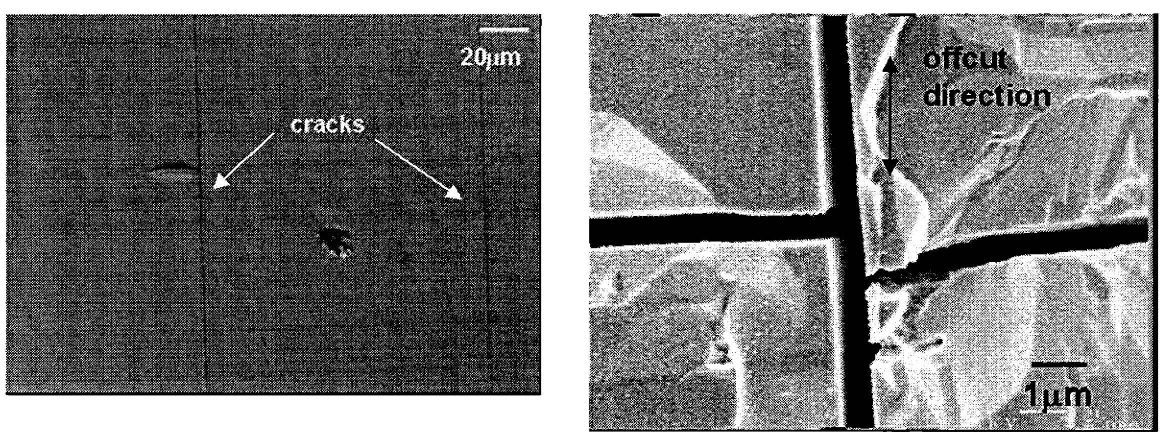
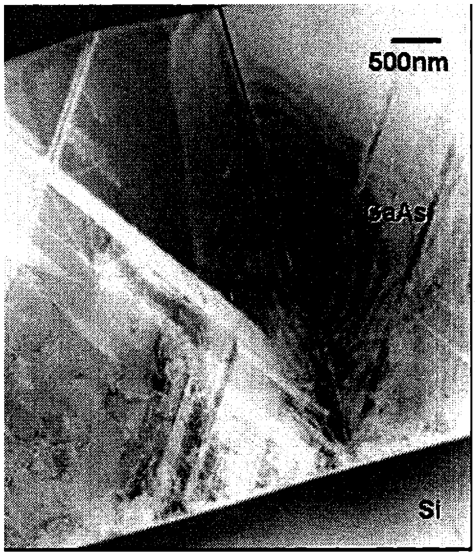


Figure 4.5 Images of cracks in GaAs heteroexpitaxial films. The 4 μm GaAs layer on SiGe has cracks only in the direction parallel to the substrate offcut <110> direction *(left)*. The 8 μm GaAs film on Si has cracks extending in both in-plane <110> directions *(right)*.

Crack channels formed in the GaAs layers grown on Si substrates at 600°C tend to curve instead of propagating along either <110> direction. A possible cause of the curving could be the higher GaAs surface roughness. The growth of GaAs at 600°C results in a higher surface roughness than growth at the two other temperatures. Another possibility is that a lower degree of crystallinity may be affecting crack propagation. X-TEM revealed a higher defect density in the GaAs film grown at 600°C.

The crack tips in the GaAs layers grown on Si tend to terminate at the GaAs/Si substrate interface (Figure 4.6). Unlike cracks formed in the GaAs on Si substrates, the cracks in GaAs layers grown on SiGe substrates extend beyond the GaAs/Ge interface (Figure 4.6). An interesting feature shown in this X-TEM micrograph is the emission of dislocations from a crack tip. The nucleation and propagation of these dislocations at the crack tip can help reduce the overall strain energy in the film.[92]



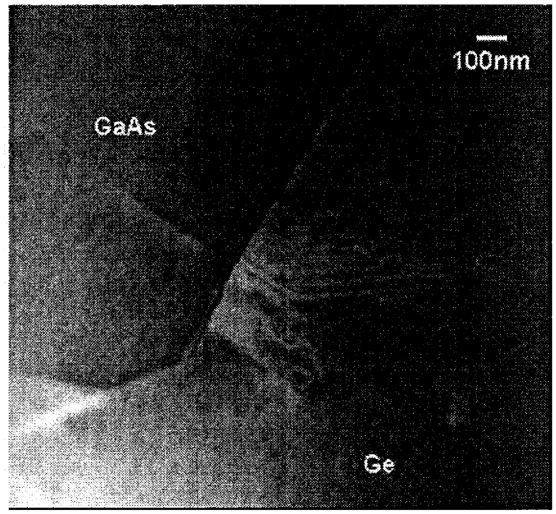


Figure 4.6 **The X-TEM micrograph on the left shows a crack in the GaAs layer grown on a Si substrate where the crack tip terminates at the GaAs/Si interface. The crack tip extends into the Ge layer for GaAs grown on SiGe substrates *(right)*. Dislocation emission from the crack tip is also observed in the X-TEM micrograph.**

A higher density of cracks can usually be observed near the edges of the samples, as also observed by Hayafuji, *et al.* It is most likely a result of higher densities of nucleation sites near the edges[93] and also higher stress experienced at the edges from handling or from growth. For example, the sample with 1.7 μm of GaAs on SiGe grown at 600°C (ΔT = -575°C) has a crack density of 58.2 cm^{-1} at the sample edges, much higher than the density of 17.8 cm^{-1} measured at the center of the sample. In addition, other samples show crack formation only near the edges of the sample. In these GaAs samples, layer thicknesses are close to the critical cracking thicknesses, but because of kinetic limitations, crack formation is not favored at the center of the sample piece. Near the samples edges with abundant nucleation sites, crack formation becomes favorable even if the thickness near the edges can be slightly smaller than in the center of the sample pieces due to uniformity variation.

Since surface scratches and particles present before growth may act as nucleation sites that lead to a higher density of cracks formed, crack density is highly dependent on extrinsic factors. All of our samples exhibit crack spacing much greater than that predicted by Thouless's formulation (Equation 4.3). This formulation does not take extrinsic factors, such as pre-existing defects and particles, into account. It also does not take the kinetics of crack nucleation into account. The crack spacing should exhibit a dependence on the positions of pre-existing defects and will therefore vary across a sample.

4.6 Finite Element Modeling

We have investigated through finite element modeling the characteristics of cracks and their effects on stress distribution in GaAs thin films grown on Si and SiGe substrates. In a 2-D plane strain model, we have created a substrate with a dimension of 500×1000 μm under a thin-film region of thickness varying from 1 to 5 μm and a length of 1000 μm. For the GaAs/SiGe system, a 1 μm Ge layer is sandwiched between the areas representing the GaAs thin film and the Si substrate in order to more accurately depict the epitaxial structure. The degrees of freedom of the system are constrained so that the result of the simulation is not a function of the dimensions of the substrate. In addition, the meshing is arranged so that finer meshes are used near and in the GaAs thin film region.

The systems are then subjected to a -675°C thermal load. Results of the simulation are shown in Figure 4.7.

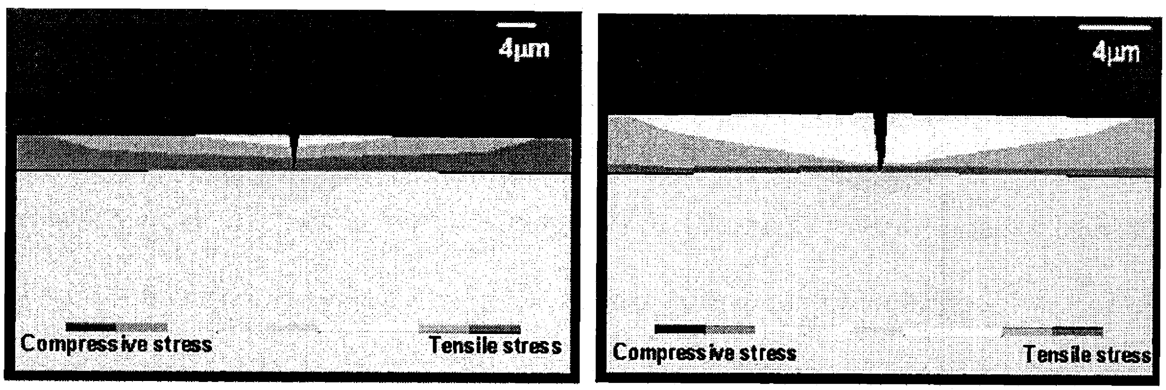


Figure 4.7 **ANSYS simulation of a 5 μm GaAs layer on Si *(left)* and a 3 μm GaAs layer on a 1 μm Ge layer on Si *(right)*. The contours show the evolution of thermal stress around the crack planes.**

Our results show that cracks only help to relieve thermal stress in regions close to the crack planes. Figure 4.7 *(left)* and Figure 4.7 *(right)* show a rapid increase in the stress values as we move away from the crack planes, agreeing with the observations made by Pompe, *et al.*[65] Figure 4.7 *(left)* shows a model with a 5 μm GaAs film on a Si substrate, while Figure 4.7 *(right)* shows one with a 3 μm GaAs film on a SiGe substrate. Both models represent systems in which the GaAs film thicknesses are close to the experimental critical cracking thickness for each substrate type.

As shown in Figure 4.7 *(left)*, approximately 35 μm away from the crack tip, the stress once again returns to the equilibrium value for the GaAs film on a Si substrate. This distance is slightly smaller for the GaAs film on the SiGe substrate, approximately

15 μm. Thus, for GaAs on Si close to the critical cracking thickness, a spacing of approximately 35 μm would be expected between crack channels. At this distance from the crack planes, thermal stress once again reaches a value high enough for crack formation. The predicted value from Equation 4.13 is 31.5 μm, close spacing predicted by finite element modeling. However, both the finite element models and Equation 4.13 ignore the kinetics of crack nucleation that explain the much higher spacing values observed experimentally (Figure 4.4). The spacing of the cracks will most likely be dictated by the positions of pre-existing defects. Similarly, a channel spacing of 15 μm is predicted for GaAs grown on SiGe virtual substrates if the GaAs thickness is close to the critical cracking thickness.

4.7 Conclusion

We have grown GaAs films on Si and SiGe substrates and have determined the critical cracking thickness of GaAs on these two substrates grown at 600°C, 700°C, and 750°C. We have noticed that the cracks in GaAs films grown on Si terminate at the GaAs/Si interface, whereas the cracks in GaAs films grown on the SiGe substrates will extend into the Ge cap layers of the substrates. Ge has a thermal expansion coefficient similar to GaAs, and therefore should also experience a large thermal stress after the temperature change. Thus crack extension into the Ge layer is energetically favorable. These experimentally determined critical cracking thicknesses were compared to the calculated values, which were slightly lower than the experimentally observed critical thicknesses. Kinetic barriers to crack formation are most likely responsible for the deviation. Crack formation is a heterogeneous process, in which the crack spacing is greatly dependent on the position of pre-existing defects. It is also noted that the <110> direction parallel to the substrate off-cut direction is favored in crack propagation over the other <110> direction, much like the behavior observed in other III-V heteroepitaxial systems.

Our finite element modeling shows that stress relief from crack formation is a localized phenomenon, implying that arrays of cracks are needed in order to lower the thermal stress values across a thermal mismatched system. The finite element models predict a crack spacing of 35 μm for GaAs films on Si substrates and 15 μm for GaAs films on SiGe substrates; both values are much smaller than the observed spacing.

Chapter 5

Luminescent Efficiency of InGaAs Quantum Well Structures Grown on Si, GaAs, Ge, and SiGe Virtual Substrates

5.1 Overview

In this chapter, we report the results from a study of substrate-dependent $In_{0.2}Ga_{0.8}As$ quantum well luminescence. The test structure consists of an $In_{0.2}Ga_{0.8}As$ quantum well with GaAs and AlGaAs cladding layers epitaxially grown on SiGe, Ge, GaAs, and Si substrates. InGaAs quantum well laser diodes exhibit lower degradation rates and longer operating lifetimes compared to AlGaAs quantum well laser diodes. With the addition of In atoms to GaAs, the propagation of dark line defects in active laser devices can be prevented.[94,17] Therefore, it is of interest to understand the luminescence characteristics of an InGaAs quantum well on a SiGe virtual substrate. In addition, using this test structure, we can correlate the luminescent efficiency of the InGaAs quantum well to both threading and misfit dislocation densities.

The substrates used in our experiments introduce different lattice mismatch between the III-V layers and the substrate, and thus different mismatch strain between the two layers. To relieve this mismatch strain, an orthogonal set of misfit dislocations will form at the film/substrate interface along the <110> directions, since the slip system of diamond cubic and zincblende crystals is {111}<110>. However, dislocations cannot terminate within a crystal. As a result, threading segments will extend into the deposited thin film and terminate at the film surface.[11] These threading dislocations can act as non-radiative recombination sites, and therefore degrade the performance of light-emitting devices.

In addition to varying the substrate type, we have also varied the InGaAs quantum well thickness, and, consequently, the density of misfit dislocations forming at the interface between the quantum well and the cladding layers. Unlike the misfit dislocation arrays at the substrate/cladding interface, these misfits will have a noticeable effect on the InGaAs quantum well luminescent efficiency because of their proximity to the active layer.

5.2 Experimental Procedure

To compare the quantum well luminescent efficiencies of InGaAs quantum wells grown on the different substrates, an epitaxial test structure was designed. This test structure, shown in Figure 5.1, minimizes the self-absorption of quantum well luminescence, allowing the efficient measurement of light emitted from the quantum well. More importantly, with this structure, CL or PL excitation occurs throughout a large volume of material. The CL excitation volume is a teardrop shape extending approximately 5000 Å below the sample surface. Thus, by placing the quantum well near the surface, the photoexcited or beam-excited electron-hole pairs will interact with a large cross-sectional area of material below the quantum well. In this way, the luminescent intensity should be more sensitive to threading dislocation density. For example, carriers generated deep in the 5000Å GaAs layer must traverse 5000Å of GaAs before reaching the quantum well for light emission. If there is a high density of threading dislocations, a large number of carriers will recombine before reaching the quantum well.

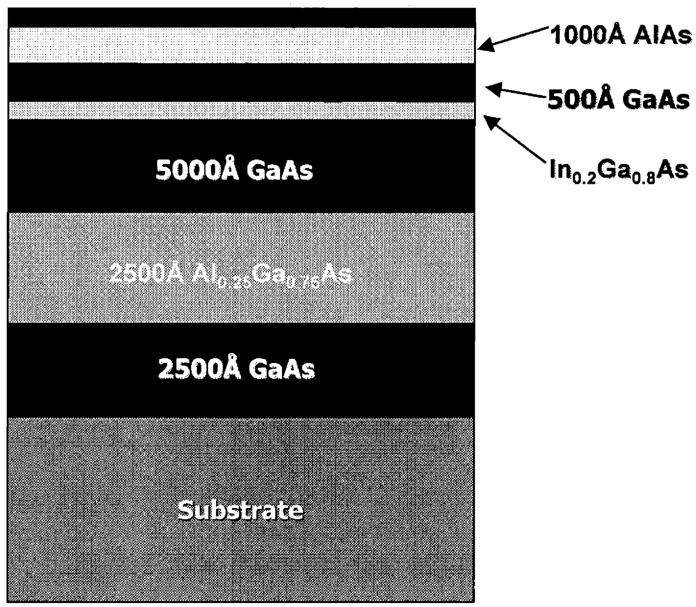


Figure 5.1 Schematic diagram of the InGaAs quantum luminescent test structure. The InGaAs layer is placed close to the surface on top of a relatively thick GaAs layer. This ensures a more accurate measurement of the effects of dislocations on the luminescent efficiency.

Two sets of the InGaAs quantum well structure were grown, one with a quantum well thickness of 120Å and the other with a thickness of 60Å. Growth procedure on the different substrate materials varied. For growths on Ge and SiGe substrates, after 350°C and 650°C annealing steps in a hydrogen ambient, a thin GaAs buffer of approximately 2500Å was initiated at 400°C. Following this low temperature growth step, the sample was brought rapidly to 650°C for growth of GaAs and InGaAs layers. The AlGaAs and AlAs layers were grown at 700°C. It was necessary to grow the aluminum-containing layers at a higher temperature in order to minimize the incorporation of oxygen. Similar growth procedures were followed for structures grown on GaAs substrates, the only difference being the omission of the low temperature initiation step. The procedure for growth on Si substrates was slightly different. Before initiating the low temperature GaAs buffer, a thin layer of Si was deposited on the substrate at 550°C to create a more uniform Si surface.

The samples were later characterized with X-TEM, PV-TEM, selective etching, and CL. X-TEM was used to verify the InGaAs quantum well thicknesses for each sample. Both selective etching and PV-TEM were used to determine the threading dislocation and misfit dislocation densities of the samples. It is necessary to utilize both characterization techniques in order to produce a more accurate value for the dislocation density.[95]

Quantitative CL was used to determine the luminescent characteristics of the various samples. Since the luminescent characteristics of the III-V materials were of interest, we chose a low electron voltage that excited the carriers throughout the 5000Å GaAs region. The excitation depth in μm, in the shape of a teardrop and as a function of CL voltage, can be calculated according to the formula of Kanaya and Okayama:

$$R_e = \frac{0.0276A}{\rho Z^{0.889}} E_b^{1.67} \qquad \textbf{Equation 5.1}$$

where A is the atomic weight in units of g/mol, ρ is the density in units of g/cm^3, Z is atomic number, and E_b is the excitation voltage in keV.[96] A voltage of 5 keV was used in our CL characterization, giving us an excitation depth of approximately 4600Å.

5.3 Luminescence Results

5.3.1 Emission Wavelength Characteristics

Comparative CL analysis at 5 keV and at room temperature was used to quantitatively compare the luminescent efficiencies of the InGaAs quantum well structure epitaxially grown on the various substrates. Figure 5.2a and b shows the room temperature CL plots for the two sets of samples with 60Å and 120Å thick quantum wells. Two peaks are observed for all samples, one near 870 nm and the other at a longer emission wavelength. The peaks near 870 nm correspond to GaAs emission, or emission from the cladding layers, and the longer wavelength peaks correspond to emission from the compressively strained InGaAs quantum well. By integrating the areas under each curve, we can calculate the overall luminescent efficiencies of the samples. Table 5.1 shows these efficiency values from GaAs emission, InGaAs quantum well emission, and the total emission for the various samples.

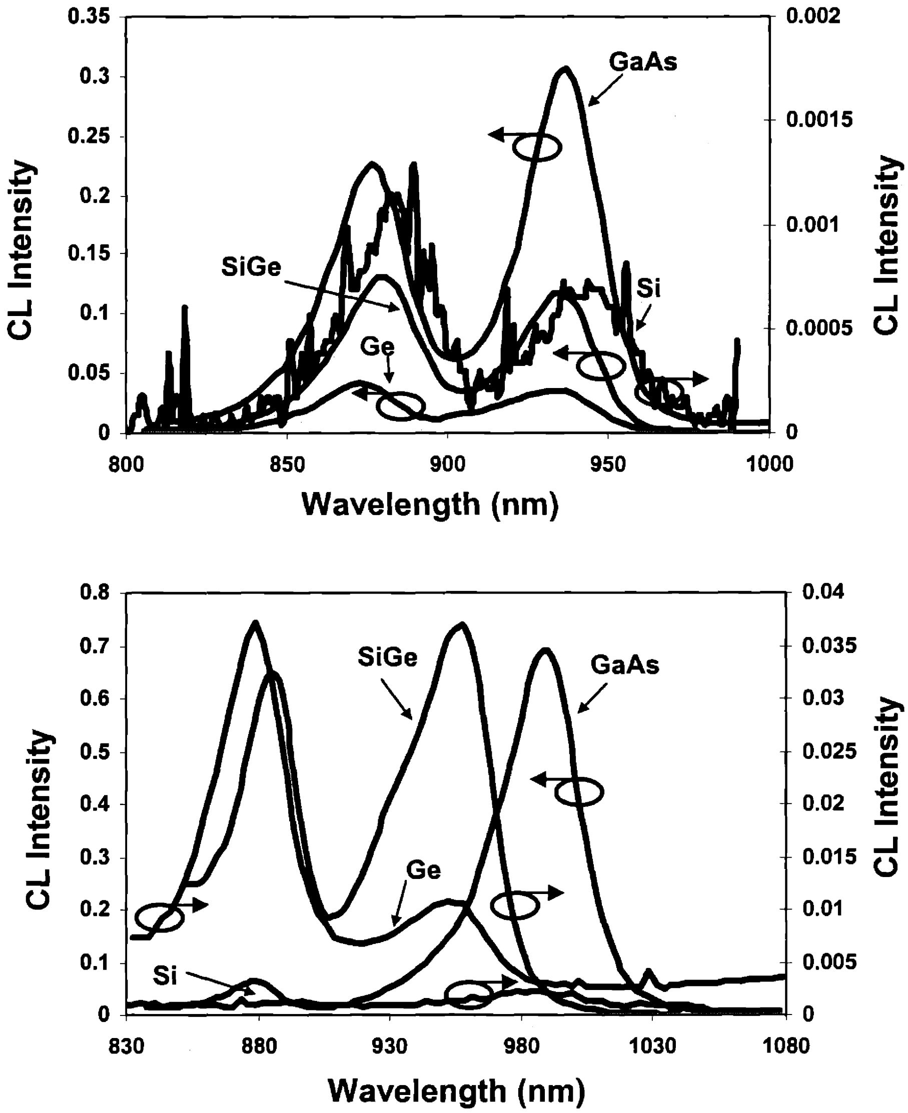


Figure 5.2 **CL spectrum of the test structure grown on GaAs, Ge, SiGe, and Si substrates. The peaks at the longer wavelengths represent emission from the InGaAs quantum well and the peaks at 870 nm end represent GaAs emission. a) shows the spectrum for the 60 Å quantum well set and b) shows the spectrum for the 120 Å quantum well set.**

Table 5.1 **Comparison of integrated luminescent intensity of InGaAs quantum well and GaAs cladding layers for test structures grown on Si, Ge, SiGe, and GaAs.**

QW Thickness	Substrate	Total Normalized CL Area	GaAs Peak Area	QW Peak Area	QW Peak (nm)
60Å	GaAs	0.1094	0.0621	0.0473	937
60Å	SiGe	0.0502	0.0221	0.0282	935
60Å	Ge	0.0173	0.0086	0.0087	934
60Å	Si	0.0004	0.0004	0.0001	943
120Å	GaAs	1	0.9476	0.0524	980
120Å	SiGe	0.0874	0.0649	0.0225	958
120Å	Ge	0.0653	0.0270	0.0383	952
120Å	Si	0.0081	0.0060	0.0021	978

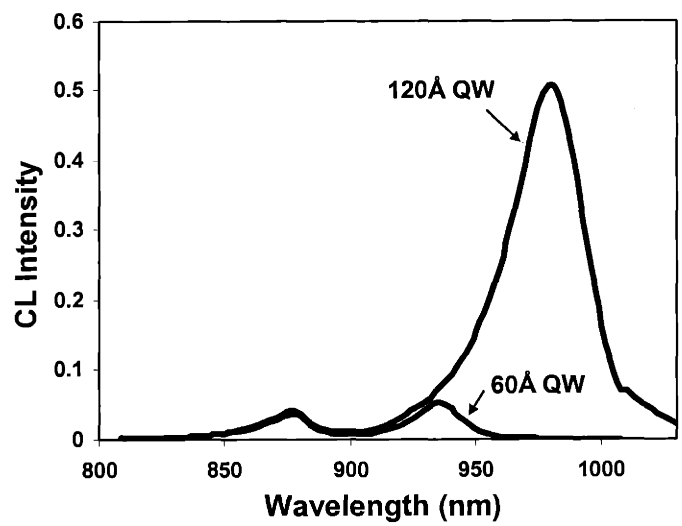


Figure 5.3 **Plot of the CL integrated intensity as as function of wavelength. The intensity increases as the quantum well thickness changes from 60 Å to 120 Å.**

We observe a shift in the quantum well peak wavelength as the quantum well thickness changes. Figure 5.3 compares this shift for the structures grown on GaAs substrates. We can calculate the change in energy levels as a result of the different quantum well thickness by using the finite well model.[97] For an infinite well, the difference between the quantized energy level and the valence or conduction band is given by

$$E_n^{\infty} = \frac{h^2 n^2}{8m^* L^2} \qquad \textbf{Equation 5.2}$$

where h is Planck's constant, n is an integer, m^* is the effective mass for the carriers in the quantum well, and L is the quantum well thickness. However, when the potential barrier V_o decreases from infinity, n is no longer an integer but a real number. This new number n_{QW} can be calculated from the following equations:

$$tan[\frac{\pi}{2} n_{QW}] = \frac{1}{n_{QW}} [n_{max}^2 - n_{QW}^2]^{1/2} \qquad \textbf{Equation 5.3}$$

for the symmetric modes and

$$tan[\frac{\pi}{2} (n_{QW} - 1)] = \frac{1}{n_{QW}} [n_{max}^2 - n_{QW}^2]^{1/2} \qquad \textbf{Equation 5.4}$$

for the antisymmetric modes where

$$n_{QW} = \sqrt{\frac{E_n}{E_1^{\infty}}} \qquad \textbf{Equation 5.5}$$

and

$$n_{max} = \sqrt{\frac{V_o}{E_1^{\infty}}} \, . \qquad \textbf{Equation 5.6}$$

For an unstrained $In_{0.2}Ga_{0.8}As$ quantum well, the ratios of m_e^*/m_o and m_h^*/m_o are 0.058 and 0.44, respectively, where m_o is the electron mass. However, these mass values will decrease with the addition of compressive strain. A 9% decrease in the effective electron mass and a 60% decrease in the hole effective mass for a strained $In_{0.2}Ga_{0.8}As$ layer on GaAs are expected.[98] In our calculations, we also assume a ratio between the conduction band and the valence band offsets of 0.7:0.3.[99] This gives a V_o value of 0.015 eV for the conduction band and 6.42×10^{-2} eV for the valence band. Using these constants and Equation 5.2 through Equation 5.6, we have calculated the theoretical shift in the quantum well peak between the 120Å quantum well and the 60Å quantum well structures to be 0.049 eV. Kolbas, *et al.* have also modeled the transition energy of carriers in InGaAs quantum wells grown on GaAs substrates for different indium compositions and quantum well thicknesses.[99] This model includes the effect of quantum well thickness change and the effect of compressive strain. According to this model, we would expect an approximately 0.05 eV shift in the quantum well peak for our two structures grown on the GaAs substrate. Both our simple calculation and the shift reported by Kolbas, *et al.* agree closely with our experimentally observed shift of 0.058 eV.

Besides the thickness effect, we must also understand the dependence of the quantum well peak shifts on substrates type. The InGaAs quantum well peaks on the Ge and SiGe substrates are expected to shift slightly as compared to peaks on the GaAs substrates, since these quantum well layers are partially relaxed by the misfit array formed at their interfaces, as shown in the X-TEM micrograph in Figure 5.4. This partial relaxation should lead to a slight increase in the effective masses of the carriers compared to those of a strained layer. In order to examine the maximum possible impact of this effect, we have calculated the expected shift assuming the InGaAs quantum well layers on the Ge and SiGe substrates are completely relaxed. Also, by taking into account the increase in the bandgap of bulk InGaAs as a result of compressive strain, we have calculated the expected wavelength difference between the InGaAs quantum well peak on the GaAs substrate and those on the Ge and SiGe substrates. The change in the bulk InGaAs bandgap as a result of strain can be calculated using the following equation,

$$dE_g = \left[-2a(\frac{C_{11} - C_{12}}{C_{11}}) + b(\frac{C_{11} + 2C_{12}}{C_{11}}) \right] \varepsilon \qquad \textbf{Equation 5.7}$$

where dE_g is the change in the bandgap, C_{11} and C_{12} are the stiffness constants with values for $In_{0.2}Ga_{0.8}As$ of 11.17×10^{11} and 5.21×10^{11} dynes/cm^{-2}, respectively, ε is the biaxial strain parallel to [100] and [010] directions, and a and b are the hydrostatic and shear deformation potential constants, with values of –8.84 eV and –1.77 eV, respectively.[100] The maximum peak shift calculated for a strained versus a relaxed quantum well is only 4 nm lower, which is much smaller than the observed difference in CL shown in Figure 5.2. Note also that the quantum well layers on the Ge and SiGe substrates are only partially relaxed, so the shift due to material relaxation would be expected to be even less than 4 nm. Figure 5.2b, however, shows that the 120 Å quantum well peaks on SiGe and on Ge are shifted by a much large energy.

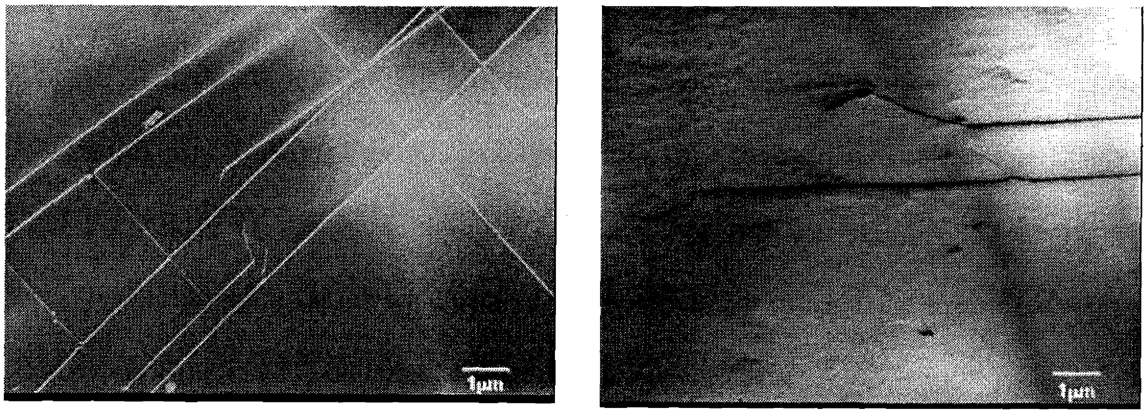


Figure 5.4 PV-TEM of InGaAs quantum well interface on a Ge substrate *(left)* and on a SiGe substrate *(right)*. A higher misfit density is seen in the sample on the Ge substrate.

The InGaAs quantum well peak position is sensitive to both the quantum well thickness and the indium composition of the quantum well. Because of the inherent limitations of this research reactor, slight fluctuations in both the quantum well thickness and the indium composition could be present in these samples. Assuming a worst case scenario of ±10% fluctuation in the quantum well thickness, we would expect a maximum shift of 7 meV for the 120 Å quantum well peak, assuming the unstrained carrier effective masses. These shifts due to thickness change are still too small to explain the approximately 29 meV energy difference between the InGaAs quantum well peaks on the GaAs substrate and the Ge and SiGe substrates.

On the other hand, with a ±2% variation in the indium composition, we would expect a maximum peak shift of 22 meV for the 60 Å quantum well peak and a maximum shift of 40 meV for the 120 Å quantum well peak. This magnitude of fluctuation in the indium composition could explain the 29 meV shift between the peak for the 120 Å quantum well grown on the SiGe and Ge substrates and the one grown on the GaAs substrate. For the 60 Å thick quantum well samples, less noticeable quantum well peak shifts were observed, as shown in Figure 5.2a, suggesting that the origin of the shift in the 120 Å quantum well samples may indeed be a result of In composition fluctuation. A 2% difference in indium composition in the quantum well layers for the 120 Å thick quantum well samples will not affect our analysis of the effects of dislocation densities on the luminescent efficiencies of our structures. We would expect slightly lower luminescence for quantum wells with lower indium composition. However, since the quantum well peak positions for the 120 Å thick quantum well structures grown on the Ge and SiGe substrates are at similar positions, direct comparison of luminescent efficiencies as a function of defect densities between these two samples will be valid.

The GaAs peaks for structures grown on SiGe, Ge, and Si substrates are all at longer wavelengths (lower energies) at room temperature relative to the GaAs peaks for the structures grown on GaAs substrates, as shown in Figure 5.2a and Figure 5.2b. This wavelength shift is due to the thermal strain caused by the different thermal expansion coefficients of the substrates. The thermal expansion coefficients of the various substrates as a function of temperature are shown in Figure 1.8. Upon the temperature decrease from the growth temperature to room temperature, tensile strain will remain in the GaAs layers. This process is summarized in Figure 5.5. In Figure 5.5, the room temperature strain values are calculated by taking the dE_g values from the CL plots. With these dE_g values, we can calculate the amount of strain using of Equation 5.7, where C_{11} and C_{12} of GaAs are equal to 11.88×10^{11} and 5.38×10^{11} dynes/cm^{-2} respectively, and a

and b are equal to –9.8 eV and –1.76 eV respectively.[100] Then the percent strain at elevated temperatures is extrapolated by using the thermal expansion coefficients of the substrates and the GaAs layer. This extrapolation is done assuming no relaxation takes place during the temperature drop.

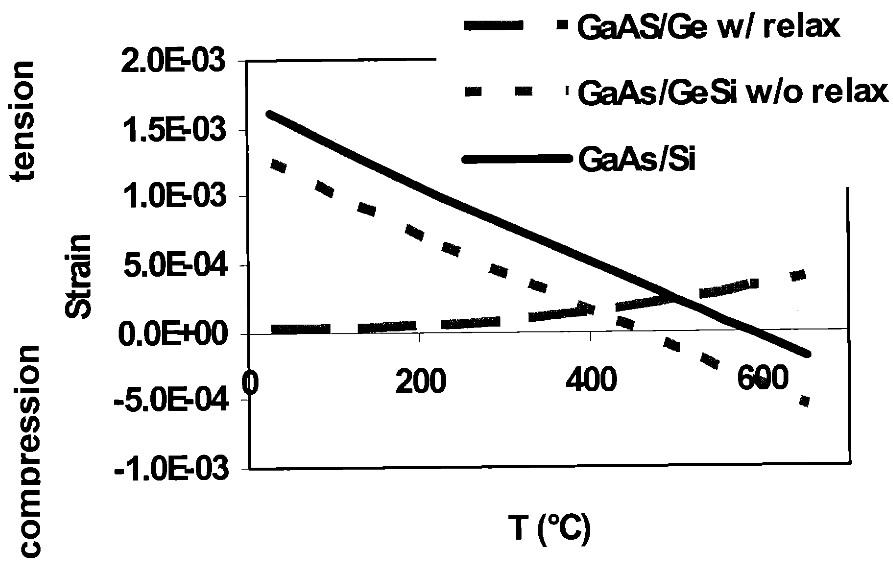


Figure 5.5 **Change in the stress levels in the GaAs cladding layers grown on Ge, Si, and SiGe. The GaAs layer on the SiGe substrate at the growth temperature is nearly stress-free because of the intentional compressive strain added to the SiGe substrate. All GaAs layers are in compression at the growth temperature.**

The calculated strains of GaAs at the growth temperature, as shown in Figure 5.5, have implications on the misfit dislocation densities observed in the InGaAs quantum wells, and will be discussed in section *5.3.3*.

5.3.2 Emission Intensity Characteristics

X-TEM, PV-TEM, and selective etching were used to measure the threading and misfit dislocation densities of the samples. Figure 5.6a shows one of the X-TEM micrographs of the test structure. Because of the limited area that an X-TEM micrograph can cover, it is necessary to look at the PV-TEM micrographs to accurately determine the dislocation densities. An example of a PV-TEM micrograph showing the dislocations is given in Figure 5.4b. The dislocation data gathered from TEM and selective etching are shown in Table 5.2. As shown in Table 5.2, only the misfit dislocation density tends to increase with increased quantum well thickness.

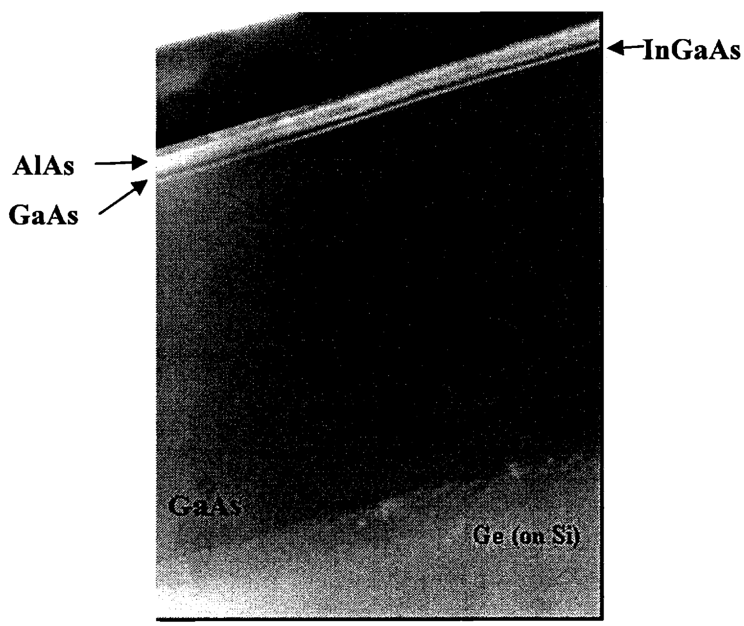


Figure 5.6 **X-TEM micrograph of the luminescence test structure. The InGaAs quantum well is close to the structure surface. Below the quantum well is approximately 5000 Å of GaAs. This test structure is designed for increased sensitivity to dislocations.**

Table 5.2 **Threading and misfit dislocation densities in InGaAs quantum wells grown on the substrates studied in this work.**

QW Thickness	Substrate Type	Threading Dislocation Density (cm^{-2})	Misfit Dislocation Density (cm^{-1})
60Å	GaAs	$2.44x10^4$	0
60Å	SiGe	$7.93x10^6$	$2.50x10^2$
60Å	Ge	$6.74x10^6$	$7.43x10^3$
60Å	Si	$>3.64x10^9$	0
120Å	GaAs	$2.45x10^4$	0
120Å	SiGe	$5.59x10^6$	$1.08x10^3$
120Å	Ge	$7.31x10^6$	$9.01x10^3$
120Å	Si	$>3.60x10^9$	0

As shown in Figure 5.3, plotting the integrated CL intensity versus wavelength of the structures grown on the GaAs substrate for two quantum well thicknesses shows that, in the absence of significant densities of misfit and threading dislocations, a thicker quantum well will give a higher luminescent efficiency due to better carrier confinement in the active region. Selective etching indicates that both the 120 Å and 60 Å quantum well structures on GaAs have similar threading dislocation densities. PV-TEM confirmed that no misfits formed in these two structures. Even though the GaAs cladding layers and substrate emission in the two structures are almost identical, the 120 Å InGaAs quantum well has a much higher luminescent efficiency than that of the 60 Å well. However, this trend is not replicated in structures grown on the other substrates simply because these structures contain misfit and threading dislocations, the densities of which are functions of the quantum well thickness.

Misfit dislocation arrays are formed when the InGaAs layer thickness exceeds the critical thickness defined in Equation 1.5. This expression tells us that an increase in the lattice mismatch between the substrate and the epilayer will decrease the critical thickness. Since Figure 5.5 clearly shows that the GaAs film lattice constant will vary at the growth temperature depending on the different substrate, it can be expected that the critical thickness of the InGaAs quantum well will vary from substrate to substrate. PV-TEM shows that no misfit dislocations form at the InGaAs/GaAs cladding interface for samples on the GaAs substrate. On the other hand, misfit densities in the range of 10^2 to 10^3 cm^{-1} are present at this interface in samples grown on Ge and SiGe.

Higher misfit densities may result in higher threading dislocation densities, especially as the misfit dislocation density becomes large. Our results show that even though the interfaces between the quantum well and the clads grown on Ge substrates have almost an order of magnitude higher misfit density than those on the SiGe substrate, the threading dislocation densities are similar to those on the SiGe virtual substrates. The factor contributing to the additional threading dislocations on the SiGe virtual substrates must be the substrates themselves. The SiGe substrates used in this study have threading dislocation densities of approximately $2x10^6$ cm^{-2}, while the Ge substrates have threading dislocation densities of less than 5000 cm^{-2}. The threading dislocations in GaAs on Ge are due to the creation of misfit dislocations, whereas the threading dislocations in GaAs on SiGe are derived from the residual threading dislocation density in the relaxed buffer.

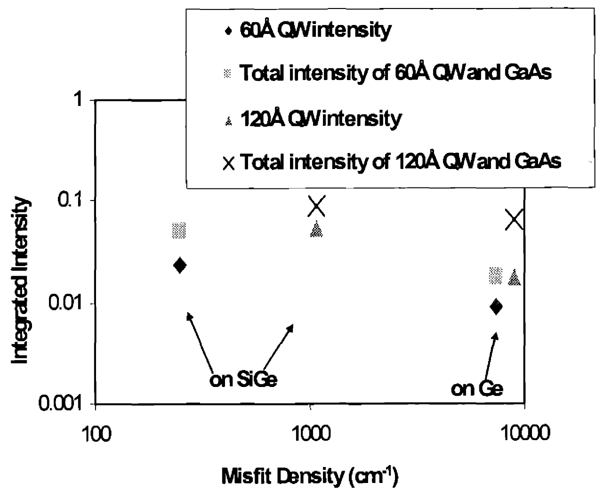


Figure 5.7 The change in the luminescent intensity of the InGaAs quantum well test structures as a function of the misfit dislocation density. The intensity decreases with increases in the misfit density and decreases in the quantum well thickness. Note that data points with misfit densities ranging from 100 to 1000 cm^{-1} are from samples grown on SiGe substrates and the points near 10000 cm^{-1} are from samples grown on Ge substrates.

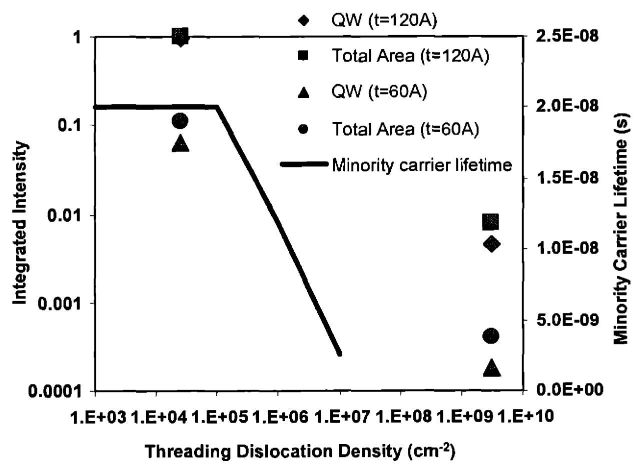


Figure 5.8 The change in the luminescent intensity of the InGaAs quantum well test structure as a function of the threading dislocation density. The intensity decreases with increases in the misfit density and decreases in the quantum well thickness. Note that data points with threading densities near 1×10^{4} cm^{-2} are from samples grown on GaAs substrates and the points near 1×10^{9} cm^{-2} are from samples grown on Si substrates.

Having determined the misfit and threading dislocation densities, we can now correlate the defect densities with luminescent efficiency. The effects of misfit dislocation and threading dislocation densities on the luminescent efficiency of the InGaAs quantum well are shown in Figure 5.7 and Figure 5.8. Note that in Figure 5.7, the relationship between misfit dislocation density and luminescent efficiency is depicted by the data points denoted by SiGe and Ge, since these structures have similar threading dislocation densities, and thus represent a more valid comparison. In Figure 5.8, data from structures on GaAs substrates and on Si substrates were used since the main difference between these samples is the threading dislocation density. As expected, both types of defects have deleterious effects on the overall luminescent efficiency because they can act as non-radiative recombination centers. Misfit dislocations have a greater effect on the luminescent efficiency than do threading dislocations simply because they lie entirely at the quantum well interface, unlike the threading segments that penetrate through the quantum well layer. The probability of carriers reaching a misfit dislocation segment at the edge of the quantum well and non-radiatively recombining is much higher than that of carriers recombining at a threading segment that passes through the well.

Ge autodoping could also have an effect on the luminescent intensity of InGaAs quantum wells grown on both the Ge and SiGe substrates. It was noted that the level of autodoping in structures grown on Ge substrates is higher than ones on the SiGe substrate. This higher doping level could also slightly contribute to the lower quantum well luminescent intensity. However, the affect of misfit dislocation density on the luminescent intensity is assumed to be the dominant factor in the lowering the intensity level.

5.3.3 *Misfit Dislocation Densities in InGaAs on Ge and SiGe Substrates*

As shown in Figure 5.3, we have observed a significantly higher misfit dislocation density at the quantum well/cladding interfaces in structures grown on Ge substrates compared to those on SiGe substrates. We believe the difference in misfit densities originates in the difference in thermal expansion coefficient between the Ge substrates and SiGe substrates, as shown in Figure 1.8. Even though the top surfaces of the SiGe substrates are 100% Ge, the Si substrate controls the thermal expansion characteristics of the entire structure. Ge has a much higher thermal expansion coefficient of $8.28\times10^{-6}\ ^{\circ}C^{-1}$ at the growth temperature of 650°C, whereas the thermal expansion value of Si at 650°C is only $4.2\times10^{-6}\ ^{\circ}C^{-1}$.

To better understand the behavior of the structures during the post-growth temperature drop, we examine the behavior between the growth temperature and room temperature lattice constants of the GaAs cladding layers grown on the Ge and SiGe substrates, as shown in Figure 5.5. In Figure 5.5, compressive strain is represented by negative values and tensile strain by positive values. The method that we have employed for this strain calculation has been described in section *5.3.1*. Assuming no relaxation taking place during cool down and all the misfit formation occuring at the growth temperature, the GaAs layer grown on the SiGe substrate will be more compressively strained at the growth temperature. However, as the temperature decreases, this GaAs layer will start to lose some of its compressive strain and eventually acquire tensile strain due to the thermal expansion difference between GaAs and Si. For the GaAs layer on Ge substrates, as the temperature decreases, the GaAs layer on the Ge substrate will acquire even more compressive strain. Finally, for the GaAs layers grown directly on Si, the GaAs compression at the growth temperature is less than the one experienced by the GaAs layer on the SiGe due to compression stored in the SiGe substrate. This compression is built into the SiGe substrate by completing the graded buffer growth at low temperature, thus trapping in metastable compressive strain from lattice mismatch. The magnitude of this compression is so that-as shown in Figure 5.5-at room temperature, less tensile stress is incorporated into the Ge or GaAs layers.

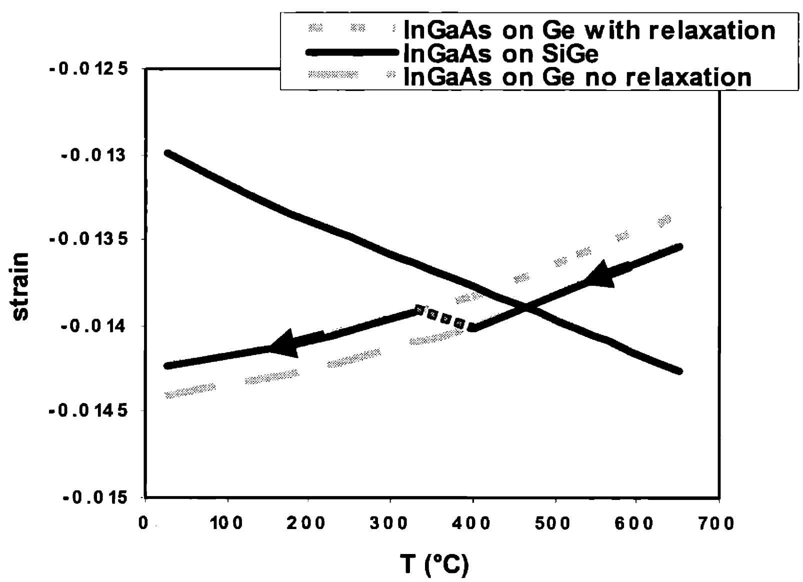


Figure 5.9 **Change in the relative stress levels in the InGaAs quantum well grown on Ge and SiGe. The InGaAs quantum well on a Ge substrate experiences an increase in the compressive strain from the temperature drop while the quantum well on the SiGe substrate experiences a relative tensile strain as the temperature falls.**

Figure 5.9 is a graph of strain in the InGaAs quantum well on the Ge and SiGe substrates based on the GaAs lattice constant information from Figure 5.5 and the misfit dislocation spacing at the InGaAs quantum well interface. In this figure, we have plotted the strain value as a function of temperature with and without the consideration of misfit relaxation at the InGaAs quantum well interface, represented by the two dotted curves, where the upper dotted curve represents the strain values in the InGaAs layer with relaxation and the lower dotted curve represents the strain values without any relaxation. The difference between the two dotted curves is determined by the strain relieved with the creation of misfit dislocations at the interface. With the misfit dislocation spacing, one can calculate the equivalent strain relieved δ from the following expression,

$$\delta = \frac{b}{2S}$$ **Equation 5.8**

where S is the dislocation spacing, which is the inverse of the dislocation density.[11] Relaxation can take place in a InGaAs film at temperatures above 440°C.[101] Dunstan has found that temperature has no effect on relaxation in InGaAs systems in the range between 350°C and 650°C.[102] As the InGaAs layer relaxes via misfit formation, the strain value slowly moves from the higher compressive strain boundary, created by the lower dotted line to the lower compressive strain boundary, created by the upper dotted line. This transition is denoted by the connecting line between the two curves in Figure 5.9. However, we do not know exactly where this transition occurs. We do know that the transition will have to occur at a temperature higher than 440°C.

At the growth temperature, we would not expect a significant density of misfit formation at the InGaAs/GaAs interface for the structure on the Ge substrate, since the in-plane lattice constant of InGaAs would be close to that of the GaAs on Ge substrate. However, cooling down this structure from the growth temperature will induce a relative compression in the quantum well, leading to misfit formation at the quantum well/GaAs interface. For the SiGe substrate, the InGaAs layer will be in compression at the growth temperature because of the smaller in-plane lattice constant of the GaAs layer on the SiGe substrate compare to the GaAs layer on the Ge substrate. During cooling, the GaAs and InGaAs layers on the SiGe substrate will experience a relative tension as shown in Figure 5.5 and Figure 5.9. We assume that the thick GaAs cladding layers relax at the growth temperature prior to the InGaAs layer deposition. The strained InGaAs layer relaxes via the formation of misfit dislocations at the InGaAs layer interface only. Relative tension during cooling is desirable since InGaAs has a larger lattice constant than GaAs, Ge, or Si. Because of this relative tension, there will be no additional misfit formation at the quantum well during cool down.

5.4 Conclusion

We have analyzed and compared an InGaAs quantum well luminescent test structure grown on various substrates, including GaAs, Si, Ge, and SiGe. Our results show promise for the integration of efficient light-emitting devices onto Si using SiGe virtual substrates, as the luminescent intensity from the InGaAs quantum well is near that of the

InGaAs quantum well on GaAs substrates. This result also demonstrates the advantage of InGaAs quantum well growth on SiGe substrates over growth on Ge substrates, namely the lower misfit dislocation density at the InGaAs/GaAs interface. However, strained quantum well structures optimized for growth on GaAs substrates cannot be directly transferred onto either SiGe or Ge substrates because of the differences in thermal expansion coefficient and lattice parameter of the substrates. These differences will change the critical thickness of the InGaAs thin film. Further studies of the dislocation formation, especially during process temperature changes, are needed to eliminate misfit dislocations in InGaAs quantum wells integrated on Si using graded SiGe buffers.

Chapter 6

GaAs-Based Optical Interconnects on SiGe Virtual Substrates

6.1 Overview

The introduction of optical interconnect can alleviate the bottleneck created by electrical interconnects as device size is further decreased. In many case, electrical interconnection delay has already become a dominant factor limiting circuit performance.[5] Therefore, by replacing these electrical interconnects with optical interconnects, it is possible to realize future improvement in interconnect power consumption and an increase in the data transmission rate.[3,4]

There are a variety of optical emitter-to-detector coupling schemes. One such scheme involves transmission through free-space with the use of micro-lenses for alignment.[103,104,105] This approach requires low-tolerance alignment of on-chip devices to external microlenses. Others have investigated the use of three-dimensional micro-optical interconnection systems that require either the bonding of emitters onto the carrier substrate where the waveguide and detector are located,[106] or flip-chip bonding of the emitter onto a transparent substrate.[107] Both of these hybrid integration approaches require alignment and a separate fabrication process for the optical devices. More recently, the employment of plastic optical fibers has been studied.[108,109] This approach requires hybrid integration techniques that are not present in conventional Si processing, as well as a low-tolerance alignment of the plastic optical interconnects. The high cost and complexity of these techniques makes it crucial to study a more feasible design and fabrication process for monolithically integrated on-chip interconnect units that can be processed using existing Si fabrication technologies.

In this chapter, we report working optical links that are monolithically integrated and subsequently fabricated on SiGe virtual substrates. The optical link consists of a GaAs LED emitter, an AlGaAs waveguide, and a GaAs PIN photodetector.

6.2 Design of GaAs-Based Optical Interconnects

The design of our optical link structure is shown in Figure 6.1. A GaAs PIN-LED acts as the light source and an identical PIN diode acts as the detector. These devices are vertically coupled to an $Al_{0.15}Ga_{0.85}As$ waveguide that is cladded with SiO_2 and $Al_{0.9}Ga_{0.1}As$. The advantages of having an emitter and detector of the same structure include ease of fabrication and the formation of a bi-directional link structure. We have chosen a vertical coupling scheme into the waveguide structure, in which both the emitter and the detector sit directly on top of the waveguide.[110] This type of coupling scheme has been studied and employed by others in the fabrication of devices such as InP MSM photodetectors and optical interconnect units on GaAs substrates.[111,112,113,114] Vertically coupled devices require no regrowth and are inherently self-aligned. With this configuration, additional regrowth steps may be avoided, simplifying fabrication and processing. However, because of the weak guide-absorber coupling, the detector length must be increased in order to absorb a significant fraction of the waveguided light. Alternatively, an edge-coupling scheme may be used. However, this approach will require the regrowth of waveguide materials on the cladding layers. The two types of coupling schemes are shown in Figure 6.2.

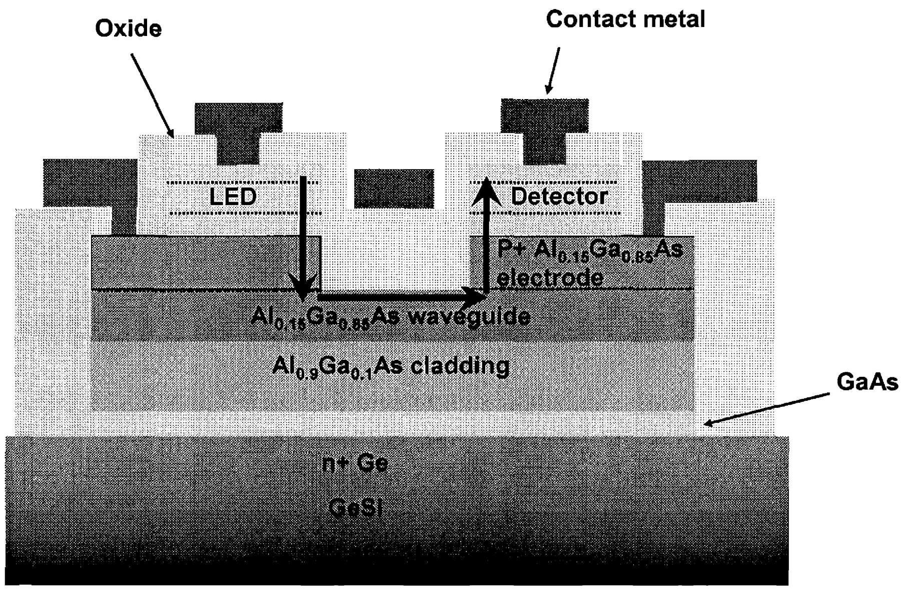


Figure 6.1 **Schematic diagram of the optical interconnect unit with an GaAs LED as the light emitter connected by the waveguide to a GaAs detector. The two devices are butt-coupled to the $Al_{0.15}Ga_{0.85}As$ waveguide. Oxide and metal layers act as the top cladding layers for the waveguide and an $Al_{0.9}Ga_{0.1}As$ layer acts as the bottom cladding layer.**

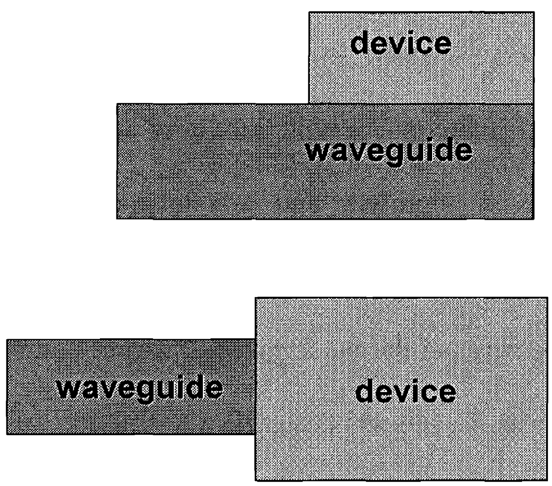


Figure 6.2 **Two types of waveguide coupling schemes. The one on top shows a butt, or vertical, coupling scheme. In this coupling method, the device sits directly on top of the waveguide. On the bottom, an edge coupling scheme is shown where the waveguide connects directly to the active region of the device.**

The waveguide thickness was chosen so that

$$a \geq \frac{\lambda}{2}$$ **Equation 6.1**

where λ is the wavelength of the emitted light and a is the thickness of the waveguide. In order to contain the mode of the light signal, the waveguide thickness must be at least half the length of the signal wavelength. If a approaches $\lambda/2$, the phase velocity of the guided wave becomes imaginary. This signifies that propagation through the waveguide is no longer possible.[115] Therefore, the thickness of our waveguide must be at least 4250 Å to contain the 850 nm signal emitted by the GaAs LED. The thickness of the cladding underneath the waveguide must also be carefully chosen. We have chosen a cladding thickness such less than 1% of the optical power at the waveguide-cladding interface penetrates beyond the cladding layer. The light signal power in the waveguide core can be calculated from the following equation:

$$Power = |A|^2 = e^{-x/dL}$$ **Equation 6.2**

where A is the amplitude of the signal, x is the thickness of the cladding, and dL is the decay length given by [116]

$$\frac{1}{dL} = \gamma_m = n_2 k_o \left(\frac{\cos^2 \theta_m}{\cos^2 \bar{\theta}_c} - 1\right)^{1/2}$$ **Equation 6.3**

where γ_m is the extinction coefficient, n_2 is the index of refraction of the cladding layer, k_o is the wave vector, θ_m is the bounce angle of the mode, and θ_c is the complimentary critical angle and is equal to n_2/n_1, with n_1 as the index of refraction for the waveguide material. This equation is valid for a single-mode light signal with a symmetric cladding structure. However, in our structure, we have an asymmetric waveguide, where one side is cladded with $Al_{0.9}Ga_{0.1}As$ and the other side with SiO_2. We also have a multi-mode light emitter. Nevertheless, the above equation serves as a good approximation for our structure, understanding that the actual confinement will be decreased as a result of the shift in the light mode toward the $Al_{0.9}Ga_{0.1}As$ cladding, which has an index of refraction higher than that of SiO_2. With the 99% power criterion, we have chosen an $Al_{0.9}Ga_{0.1}As$ cladding layer thickness of 7000 Å for the $Al_{0.15}Ga_{0.85}As$ waveguide.

Our device structure included an additional PN junction below the optical devices. Together with the PN junction formed by the optical devices, an additional PN junction will limit the electrical leakage through the substrate and waveguide and improve electrical isolation between the devices. An equivalent circuit diagram of the interconnect unit is shown in Figure 6.3.

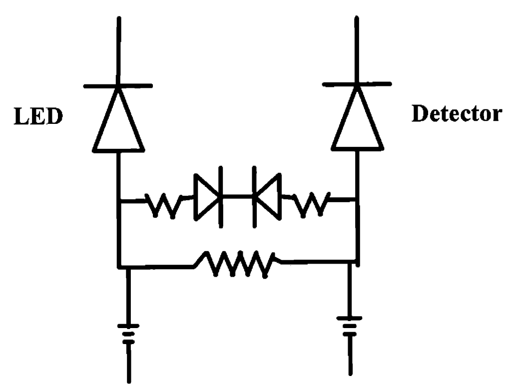


Figure 6.3 Equivalent circuit diagram of the optical interconnect unit. The additional PN junction under the devices minimizes electrical leakage through the substrate.

A variety of waveguide lengths and geometries are incorporated in our fabrication mask set, including Y-junctions and bends of different angles. All Y-junction and bending waveguides have lengths of 100 μm for ease of comparison. In addition to 100 μm straight waveguides, straight waveguide lengths of 10 μm, 1000 μm and 1 cm are also included in our mask set for comparison. A schematic diagram of the mask designed for the fabrication of these devices is shown in Figure 6.4.

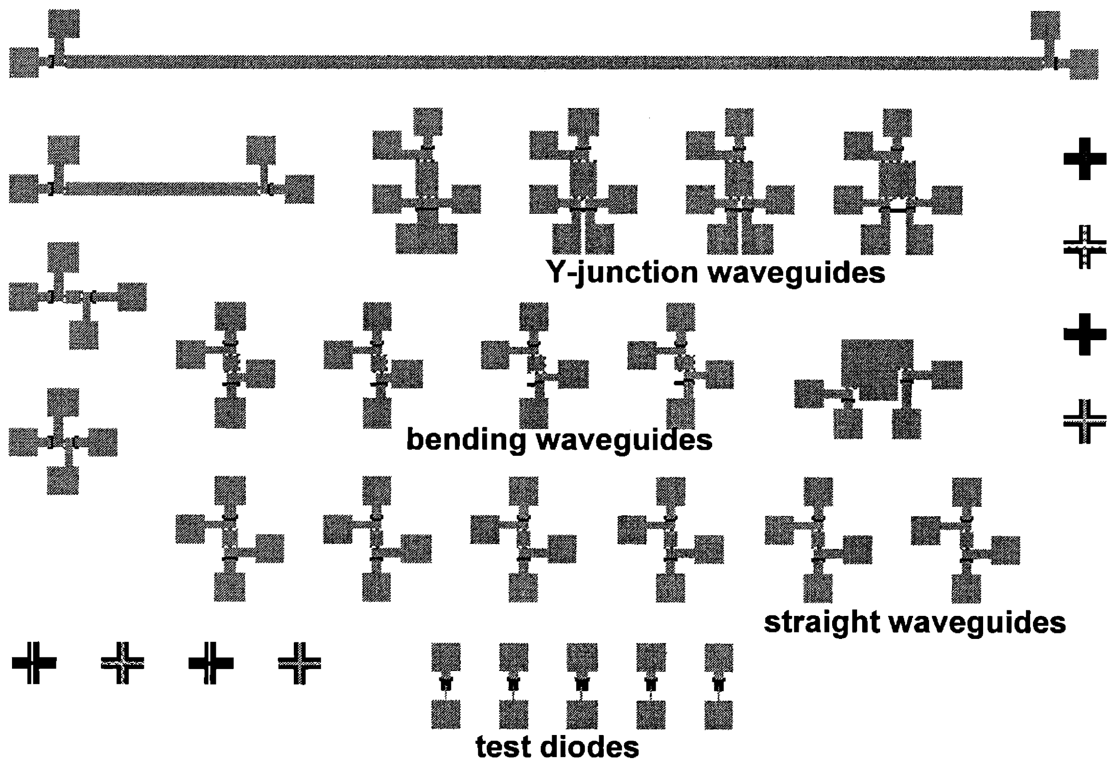


Figure 6.4 **Top view of the mask design used for the fabrication of the optical interconnects. Different interconnect geometries are shown, including Y-junctions with different junction angles, bends with various bend angles, and straight waveguides ranging from 10 μm to 1 cm in length.**

6.3 Device Growth and Fabrication

The device layers were grown at 650°C, except for layers containing aluminum. These layers were grown at 700°C in order to minimize oxygen incorporation. The doping level in both the p^+- and n^+-electrode layers was on the order of 7×10^{18} cm^{-2}.

The fabrication of the optical interconnect unit was performed in the Microsystems Technology Laboratory at MIT. The interconnect unit mesas were first etched using 1:1:20 mixture of H_3PO_4:H_2O_2:H_2O. Then the waveguide area was time-etched using the same etching solution after a photolithography step. This etching step exposures the surface of the waveguide by removing the layers above it. Following the waveguide etch step, a photolithography step was performed and the layers on top of the p^+-electrode area were removed. A 6000 Å SiO_2 layer was then deposited over the entire sample surface. After this, contact holes over the p^+- and n^+- electrodes were etched using reactive ion etching (RIE). Then metal contact pads were deposited and formed using e-beam deposition followed by a lift-off step. Finally, the devices were annealed in a rapid thermal annealing (RTA) chamber at 400°C for 30 seconds.

6.4 Device Results

The various waveguide geometries are shown in Figure 6.5. Figure 6.5 *(left)* shows an optical link with a straight 10 μm waveguide, Figure 6.5 *(center)* shows a link with a 5° Y-junction, and Figure 6.5 *(right)* shows a link with a 30° bend in the waveguide. The optical links were characterized using a HP 5145b semiconductor parameter analyzer, and a typical optical link diode I-V behavior is shown in Figure 6.6. The I-V curve shows low leakage current of 8×10^{-3} A/cm^2. However, these diodes have very high series resistance, likely resulting from the thin 7000 Å p+ electrode. The p+ electrode was purposely kept thin to minimize the propagation distance of the light signals into the waveguide and also into the detector. Moreover, p-doped AlGaAs layers are intrinsically more resistive than their n-doped counterparts. In addition, given that Ge auto-doping is a prevalent problem in GaAs-based materials, we would expect the actual doping profile to be quite different from the intended doping levels, meaning that the effective p-type doping level could be lower than expected. This may also contribute to the high series resistance of the devices.

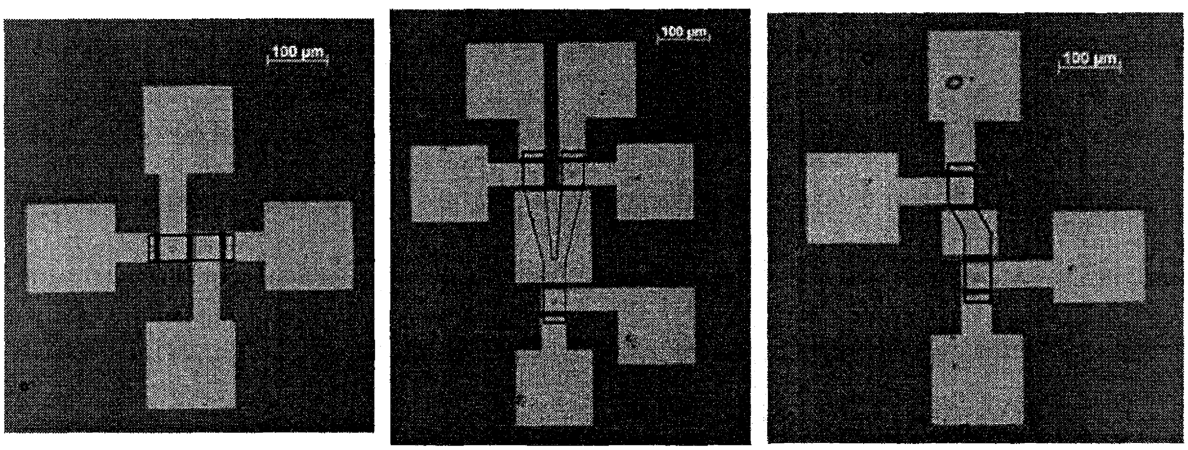


Figure 6.5 Optical micrographs of fabricated optical interconnect units. Interconnects with a 10 μm straight waveguide *(left)*, Y-junctions *(center)*, and bends *(right)*.

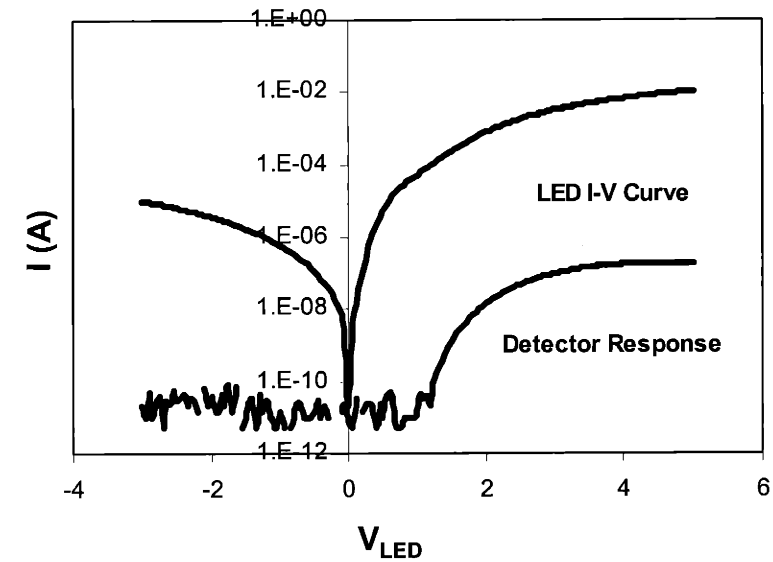
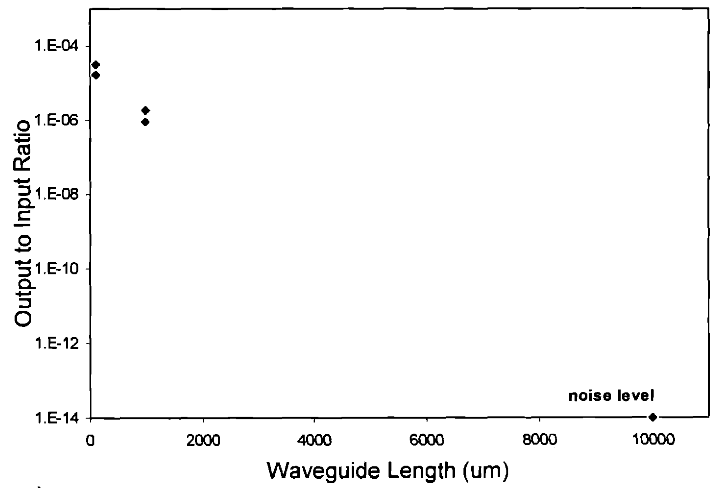
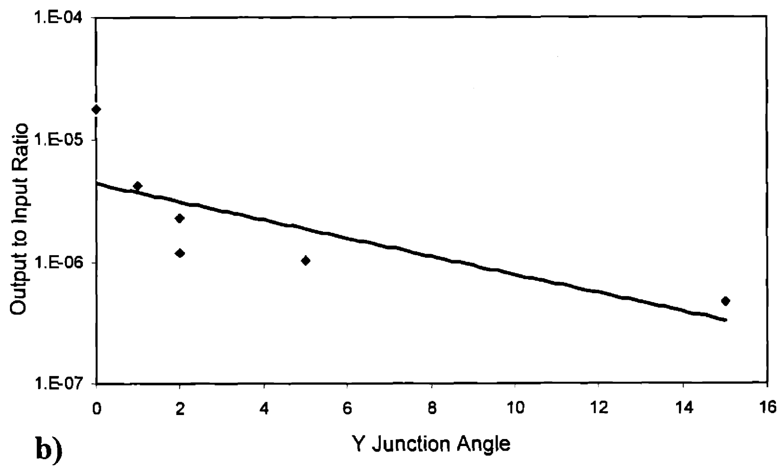


Figure 6.6 **I-V characteristic of the LED and the detector response to the LED emission. The detector diode output current starts increasing as the LED is turned on. The I-V curves show high series resistance at high voltages.**

The detector responses of optical links with various waveguide geometries were measured and compared. As expected, the signal detected by the detector decreases with increasing waveguide length. For a straight waveguide of 100 μm, a ratio of emitter current to detector response current of 1.8×10^{-5} was obtained. As the waveguide length increased to 1 cm, the detector could not detect signals greater than the noise level of the detector signal (< 5 nA), as shown in Figure 6.7a. For waveguides of 100 μm, but with different Y-junction angles, we found that as the angle increased the detected signal decreased, as illustrated in Figure 6.7b.



a)



b)

Figure 6.7 The detector output current to LED input current ratio of the optical interconnect unit as a function of length a). Note that only a noise level signal is detected for a waveguide length of 1 cm. This ratio also decreases with increasing Y-junction angle b).

Our straight waveguides show a loss of 144 dB/cm. We suspect the greatest contributor to the loss is waveguide leakage. With the vertical-coupling scheme, successful light transmission into the waveguide depends greatly on the index difference between the waveguide and the cladding layers. Our links have an index difference of 0.6 between the waveguide and the cladding, which leads to a critical angle for insertion of only 56° and thus a relatively small numerical aperture.[26] In addition, the light emitter used in this interconnect unit is an LED, which means that light emission is non-directional. A large portion of the emitted light will not be directed into the waveguide.

Waveguide and electrode layer absorption could be additional contributors to the loss. Before light signals can reach the waveguide, they must first traverse the p+ electrode. The greater valence band complexity of p-doped III-V materials causes significantly higher absorption.[117] Loss of 56 dB/cm for both GaAs- and InP- based alloys with p-doping on the order of $10^{18}/cm^3$ at λ = 1.3 μm has been reported in the literature.[117] Undoped GaAs-based materials may also introduce great losses as a result of near-midgap energy levels. Martin observed a 3.5 dB/cm loss in undoped GaAs at λ = 1.5 μm.[118]

Scattering is the third factor that could contribute to our waveguide loss. Because the waveguide layer was defined with wet etching, considerable roughness at the waveguide walls may exist. Furthermore, the SiGe substrates on which the optical link structures are monolithically grown have RMS roughness values of 15 nm on a typical 50×50 μm scan, adding to the non-planarity of the waveguide layer.

6.5 Conclusion

We have successfully demonstrated the first working optical links consisting of an LED, a PIN detector, and a vertically coupled waveguide fabricated on SiGe virtual substrates via monolithic epitaxial integration. Our optical links have a loss of 144 dB/cm, most likely due to of high coupling loss and waveguide leakage. Performance could be improved by replacing the LED with a laser as the emitter. Together with an edge-coupled waveguide, this replacement could help reduce both the coupling loss and the waveguide leakage. However, this coupling scheme would require regrowth steps and greater processing complexity. In addition, fabricating the optical link on proposed GaAs on SiGe-on-insulator (SiGeOI) technology,[119] could ensure better isolation between the devices via complete elimination of electrical conduction through the substrate.

Chapter 7

Conclusions and Suggestions

With the flexibility provided by the SiGe virtual substrate, it is now possible to integrate onto Si high-quality GaAs-based materials and optical devices with a threading dislocation density lower than 4×10^{6} cm^{-2} onto Si. Most of the threading dislocation segments in the GaAs layers are inherited from the substrate. Furthermore, there is no fundamental barrier to further lowering the threading dislocation density in the substrate, thereby improving the defect densities in the GaAs-based expitaxial layers.

7.1 Summary of Experimental Work

In this work, we have fabricated InAs SAM-APDs grown directly on Si. The results show that direct growth of high quality lattice mismatched films onto Si is difficult because of the presence of dislocations in high densities. We then investigated the GaAs critical cracking thickness on Si and SiGe substrates, studied InGaAs quantum well luminescent efficiency and characteristics on Si, Ge, SiGe and GaAs substrates, and fabricated the first working optical interconnects on Si via the use of SiGe virtual substrates. We have found that the critical cracking thickness is directly proportional to the GaAs surface energy and inversely proportional to a non-dimensional driving force value *Z*. In addition, the critical thickness on the SiGe substrate is half the value of that for GaAs grown on a Si substrate because of the additional residual thermal stress contributed by the Ge layer of the SiGe substrate. Asymmetric <110> crack arrays have also been observed in GaAs layers on both SiGe and Si substrates. Moreover, the crack density tends to increase with increasing film thickness. This crack formation behavior shares many similarities with misfit dislocation formation at mismatched heterointerfaces.

We have also shown in this work the dependence of InGaAs quantum well luminescent efficiency on both misfit and threading dislocation densities. Interestingly, it was shown that SiGe is a better integration platform for InGaAs quantum well devices than Ge because of its smaller thermal expansion coefficient. Since InGaAs has a larger lattice constant than both Ge and SiGe, when the InGaAs layer cools down from the growth temperature, it experiences a smaller compressive stress on SiGe compared to that on Si. Thus the InGaAs quantum well on the SiGe virtual substrate will have a smaller misfit density near the quantum well interface. This unexpected advantage provides even more incentive for the integration of InGaAs-based devices onto Si via SiGe virtual substrates, joining a long list of other benefits intrinsic to the Si platform.

With the advances made in the growth of GaAs-based materials on Si using SiGe virtual substrates, we have used this technology to demonstrate the first working optical interconnect on Si. The simple design of this optical interconnect is aimed at a device structure that can be processed easily in a short period of time; it is not designed to optimize the device performance. Nevertheless, we were able to measure the effect of waveguide length and Y-junction angles on the detector signal decay.

7.2 Suggestions for Future Work

With the understanding of crack formation dynamics gained herein, methods for the control of these crack arrays can be suggested. For example, by limiting the area of

GaAs film growth to a dimension approximately the size of the expected crack spacing, the frequency of crack formation can be lowered to a value that ensures the survival of most of the devices fabricated in the GaAs layer. This method would be similar to the proven selective area growth of GaAs layers on Si for dislocation control.[14] Others have also combined this method with the intentional introduction of cracks at predetermined sites through substrate defects or SAG mask defects. With the predetermined crack sites, devices can then be fabricated at the crack-free regions.[120,121] One way to extend the thickness limitation dictated by crack formation in the GaAs layers grown on a SiGe substrate is to minimize the Ge layer thickness. Wafer bonding thin Ge layers onto Si could be one such approach.[122] With the removal of the SiGe graded layers, the remaining Ge layer will have a lesser affect on the thermal mismatch between the Si substrate and the heteroepitaxial films, thus increasing the critical cracking thickness of the GaAs layer.

The current optical interconnect device structure was designed as a demonstration of feasibility rather than for optimal performance. To improve the signal transmission from the emitter to the detector via the waveguide, a different coupling geometry will be required. Figure 7.1 shows the schematic of a possible interconnect unit architecture. In this new design, the LED light source is replaced by a laser edge-coupled to the waveguide end. The advantages of using a laser include the highly directional emission of the light signal. In addition, the interconnect is cladded with a higher index material such as SiO_2, forming an interface with a greater index change and better confinement. However, this design will require regrowth steps, one of which is the growth of the waveguide material on SiO_2. This particular growth technique may have to be optimized to minimize scattering at the waveguide/clad interfaces. Nevertheless, this new architecture should provide a lower signal loss through the waveguide along with higher signal power, via the use of a laser.

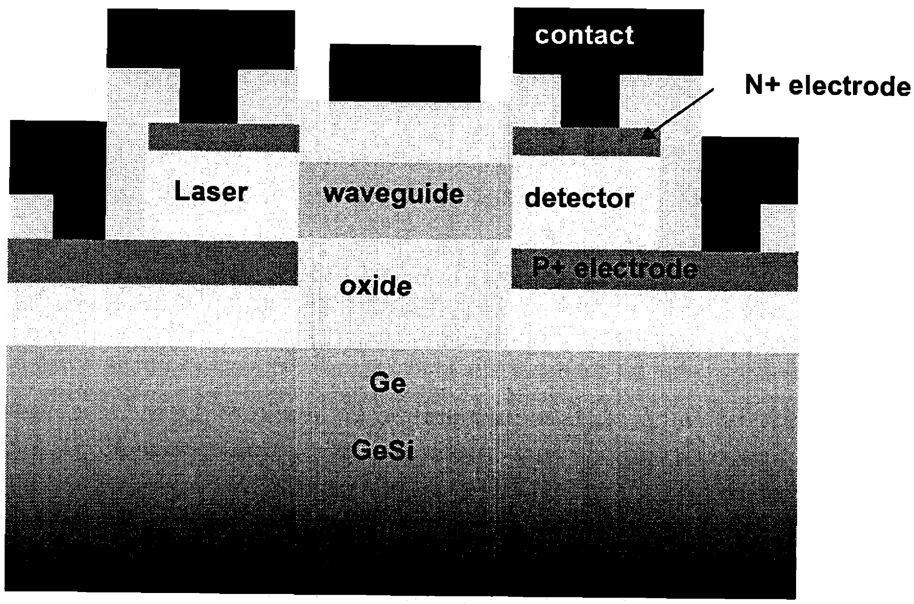


Figure 7.1 **Schematic diagram of an alternative optical interconnect structure. The light emitter is a laser which is edge-coupled to the interconnect. The interconnect is surrounded by SiO_2 cladding layers to increase the index contrast.**

Bibliography

1 S. M. Ting, "Monolithic Integration of III-V Semiconductor Materials and Devices with Silicon," Ph.D. Thesis, Massachusetts Institute of Technology (1999).

2 A. F. J. Levi, "Optical Interconnects in Systems." Proceedings of the IEEE **88**, 750 (2000).

3 D. V. Plant and A. G. Kirk, "Optical Interconnects at the Chip and Board Level: Challenges and Solutions." Proceedings of the IEEE **88**, 806 (2000).

4 M. R. Feldman, S. C. Esener, C. C. Guest and S. H. Lee, "Comparison Between Optical and Electrical Interconnects Based on Power and Speed Considerations." Appl. Optics **27**, 1742 (1988).

5 D. A. B. Miller, "Rationale and Challenges for Optical Interconnects to Electronic Chips." Proceedings of the IEEE **88**, 728 (2000).

6 J. Gourlay, T. Yang, M. Ishikawa and A. C. Walker, "Low-Order Adaptive Optics for Free-Space Optoelectronic Interconnects." Appl. Optics **39**, 714 (2000).

7 C. Fan, D. W. Shih, M. W. Hansen, D. Hartmann, D. van Blerkom, S. C. Esener and M. Heller, "Heterogeneous Integration of Optoelectronic Components." The Internation Society for Optical Engineering, 2 (1997).

8 T. Nakahara, H. Tsuda, K. Tateno, S. Matsuo and T. Kurokawa, "Hybrid Integration of GaAs Pin-Photodiodes with CMOS Transimpedance Amplifier Circuits." Elec. Lett. **34**, 1352 (1998).

9 H. Wada and T. Kamijoh, "Wafer Bonding of InP to Si and its Application to Optical Devices." Japanese J. Appl. Phys. Part 1 **37**, 1383 (1998).

10 J. W. Matthews and A. E. Blakeslee, "Defects in Epitaxial Multilayers. I. Misfit Dislocation." J. Cryst. Growth **27**, 118 (1974).

11 E. A. Fitzgerald, "Dislocations in Strained-Layer Epitaxy: Theory, Experiment, and Applications." Mater. Sci. Reports **7**, 87 (1991).

12 M. T. Currie, "SiGe Virtual Substrate Engineering for Integration of III-V Materials, Microelectromechanical Systems, and Strained Silicon MOSFETs with Silicon," Ph.D., Massachusetts Institute of Technology (2001).

13 C. W. Leitz, M. T. Currie, A. Y. Kim, J. Lai, E. Robbins, E. A. Fitzgerald and M. T. Bulsara, "Dislocation Glide and Blocking Kinetics in Compositionally Graded SiGe/Si." J. Appl. Phys. **90**, 2730 (2001).

14 K. Kawaguchi, T. Tomiyoshi and K. Sasakura. personal communication. Yokohama, Japan (2001).

15 J. C. Zolper and A. M. Barnett, "The Effect of Dislocations on the Open-Circuit Voltage of Gallium Arsenide Solar Cells." IEEE Trans. Elec. Devices **37**, 478 (1990).

16 M. Yamaguchi and C. Amano, "Efficiency Calculations of Thin-Film GaAs Solar Cells on Si Substrates." J. Appl. Phys. **58**, 3601 (1985).

17 R. G. Waters, D. P. Bour, S. L. Yellen and N. F. Ruggieri, "Inhibited Dark-Line Defect Formation in Strained InGaAs/AlGaAs Quantum Well Lasers." IEEE Photonics Tech. Lett. **2**, 531 (1990).

18 A. Bartels, E. Peiner and A. Schlachetzki, "The Effect of Dislocations on the Transport Properties of III/V-Compound Semiconductors on Si." J. Appl. Phys. **78**, 6141 (1995).

19 H. Kroemer, "Polar-on-Nonpolar Epitaxy." J. Cryst. Growth **81**, 193 (1987).

20 D. K. Biegelsen, R. D. Bringans and J. E. Northrup, "Heteroepitaxial Growth of Polar Semiconductors on Nonpolar Substrates." Mat. Sci. Eng. B **B14**, 317 (1992).

21 D. B. Holt, "Antiphase Boundaries in Semiconducting Compounds." J. Phys. Chem. Sol. **30**, 1297 (1969).

22 S. Chu, S. Nakahara, S. J. Pearton, T. Boone and S. M. Vernon, "Antiphase Domains in GaAs Grown by Metalorganic Chemical Vapor Deposition on Silicon-on-Insulator." J. Appl. Phys. **64**, 2981 (1988).

23 A. Georgakilas, J. Stoemenos, K. Tsagaraki, Ph. Komninou, N. Flevaris, P. Panayotatos and A. Christou, "Generation and Annihilation of Antiphase Domain Boundaries in GaAs on Si Grown by Molecular Beam Epitaxy." J. Mater. Res. **8**, 1908 (1993).

24 H. Noge, H. Kano, M. Hashimoto and I. Igarashi, "Antiphase Domains in GaAs Grown on a (001)-Oriented Si Substrate by Molecular-Beam Epitaxy." J. Appl. Phys. **64**, 2246 (1988).

25 O. Madelung, Ed. *Landolt-Bornstein: Numerical Data and Functional Relationships in Science and Technology*. Intrinsic Properties of Group IV Elements and III-V, II-VI, and I-VII Compounds. Berlin, Springer-Berlag (1961).

26 S. Adachi, "GaAs, AlAs, AlxGa1-xAs: Material Parameters for Use in Research and Device Applications." J. Appl. Phys. **58**, R1 (1985).

27 R. M. Sieg, J. A. Carlin, S. A. Ringel, M. T. Currie, S. M. Ting, T. A. Langdo, G. Taraschi, E. A. Fitzgerald and B. M. Keyes, "High Minority-Carrier Lifetimes in GaAs Grown on Low-Defect-Density Ge/GeSi/Si Substrates." Appl. Phys. Lett. **73**, 3111 (1998).

28 S. A. Ringel, J. A. Carlin, R. M. Sieg, J. J. Boeckl and E. A. Fitzgerald, "Highly-Controlled GaAs/Ge Interfaces and Implications for III-V Optoelectronic Integration onto Group IV substrates." First International Workshop, Warrendale, PA, 87 (1999).

29 J. A. Carlin, S. A. Ringel, E. A. Fitzgerald and M. Bulsara, "High-Lifetime GaAs on Si using GeSi Buffers and its Potential for Space Photovoltaics." Solar Energy Mat. Solar Cells **66**, 621 (2001).

30 J. A. Carlin, S. A. Ringel, E. A. Fitzgerald, M. Bulsara and B. M. Keyes, "Impact of GaAs Buffer Thickness on Electronic Quality of GaAs Grown on Graded Ge/GeSi/Si Substrates." Appl. Phys. Lett. **76**, 1884 (2000).

31 J. A. Carlin, S. A. Ringel, E. A. Fitzgerald and M. Bulsara, "High Quality GaAs growth by MBE on Si Using GeSi Buffers and Prospects for Space Photovoltaics." Prog. Photovolt. Research Appl. **8**, 323 (2000).

32 J. A. Carlin, M. K. Hudait, S. A. Ringel, D. M. Wilt, E. B. Clark, C. W. Leitz, M. Currie, T. Langdo and E. A. Fitzgerald, "High Efficiency GaAs-on-Si Solar Cells with High Voc
Using Graded GeSi Buffers." 28th IEEE Photovoltaic Specialists Conference, Piscataway, NJ, 1006 (2000).

33 S. M. Ting, M. Bulsara, V. Yang, M. Groenert, S. Samavedam, M. Currie, T. Langdo, E. Fitzgerald, A. Joshi, R. Brown, X. Wang, R. Sieg and S. Ringel, "Monolithic Integration of III-V Materials and Devices on Silicon." SPIE-International Society for Optical Engineering, USA, 19 (1999).

34 M. Groenert, C. Leitz, A. Pitera, V. Yang, H. Lee, R. Ram and E. A. Fitzgerald, "Room Temperature cw GaAs/AlGaAs Quantum-Well Lasers on Si Sbustrates." Materials Reseach Society, Boston, MA(2001).

35 M. Groenert, V. Yang and E. A. Fitzgerald, "AlGaAs and InGaAs-Based Light Emitters on Si via Relaxed Graded GeSi Buffer Layers." Electronic Materials Conference, Santa Barbara, CA(1999).

36 E. A. Fitzgerald, Y-H. Xie, M. L. Green, D. Brasen, A. R. Kortan, J. Michel, Y-J. Mii and B. E. Weir, "Totally Relaxed Ge_xSi_{1-x} Layers with Low Threading Dislocation Densities Grown on Si Substrates." Appl. Phys. Lett. **59**, 811 (1991).

37 M. T. Currie, S. B. Samavedam, T. A. Langdo, C. W. Leitz and E. A. Fitzgerald, "Controlling Threading Dislocation Densities in Ge on Si Using Graded SiGe Layers and Chemical-Mechanical Polishing." Appl. Phys. Lett. **72**, 1718 (1998).

38 S. B. Samavedam, "Growth and Characterization of High Quality Relaxed Graded Silicon Germanium Layers for Integrated Photodetectors," Ph.D., Massachusetts Institute of Technology (1998).

39 E. A. Fitzgerald and S. B. Samavedam, "Lin, Point and Surface Defect Morphology of Graded, Relaxed GeSi Alloys on Si Substrates." Thin Solid Films **294**, 3 (1997).

40 J. W. P. Hsu, E. A. Fitzgerald, Y. H. Xie, P. J. Silverman and M. J. Cardillo, "Surface Morphology of Related Ge_xSi_{1-x} Films." Appl. Phys. Lett. **61**, 1293 (1992).

41 S. P. Tobin, S. M. Vernon, C. Bajgar, V. E. Haven and S. E. Davis, "MOCVD Growth of AlGaAs and GaAs on Ge Substrates for High Efficiency Tandem Cell Application." Eighteenth IEEE Photovoltaic Specialists Conference, Las Vegas, Nevada, 134 (1985).

42 R. Azoulay, N. Draidia, Y. Gao, L. Dugrand and G. Leroux, "Autodoping of GaAs Grown by Organometallic Vapor Phase Epitaxy on Silicon Substrates." Appl. Phys. Lett. **54**, 2402 (1989).

43 M. E. Groenert, C. W. Leitz, A. J. Pitera, V. K. Yang, H. Lee, R. J. Ram and E. A. Fitzgerald, "Strategies for Direct Monolithic Integration of $Al_xGa_{1-x}As/In_xGa_{1-x}As$ LEDS and Lasers On Ge/GeSi/Si Substrates Via Relaxed Graded Ge_xSi_{1-x} Buffer Layers." Fall 2001 MRS, Boston, MA(2001).

44 G. B. Stringfellow, *Organometallic Vapor-Phase Epitaxy: Theory and Practice*. San Diego, Academic Press (1999).

45 T. J. Mountziaris and K. F. Jensen, "Gas-Phase and Surface Reaction Mechanisms in MOCVD of GaAs with Trimethyl-Gallium and Arsine." J. Electrochemical Soc. **138**, 2426 (1991).

46 M. T. Bulsara, "Materials Issues with the Integration of Lattice-Mismatched $In_xGa_{1-x}As$ Devices on GaAs," Ph.D. Thesis, Massachusetts Institute of Technology (1998).

47 L. J. Giling, "Gas Flow Patterns in Horizontal Epitaxial Reactor Cells Observed by Interference Holography." J. Electrochemical Soc. **129**, 634 (1982).

48 J. Weyher and J. van de Ven, "Selective Etching and Photoetching of (100) Gallium Arsenide in CrO_3-HF Aqueous Solutions. I. Influence of Composition on Etching Behaviour." J. Cryst. Growth **63**, 285 (1983).

49 J. van de Ven, J. L. Weyher, J. van den Meerakker and J. J. Kelly, "Kinetics and Morphology of GaAs Etching in Aqueous Cr3-HF Solutions." J. Electrochemical Soc. **133**, 799 (1986).

50 A. Joshi, "1.5 to 2.5 Micron Wavelength, Monolithic, InGaAs-on-Silicon, Infrared Avalanche Photodiodes." 18, Discovery Semiconductors, (1997).

51 W. Wu, A. R. Hawkins and J. E. Bowers, "Frequency Response of Avalanche Photodetectors with Separate Absorption and Multiplication Layers." J. Lightwave Tech. **14**, 2778 (1996).

52 T. P. Pearsall, "Impact Ionization Rates for Electrons and Holes in $Ga_{0.47}In_{0.53}As$." Appl. Phys. Lett. **36**, 218 (1980).

53 S. M. Sze, *Physics of Semiconductor Devices*. New York, John Wiley and Sons (1981).

54 W. Wu, A. R. Hawkins and J. E. Bowers, "Design of Silicon Hetero-Interface Photodetectors." J. Lightwave Tech. **15**, 1608 (1997).

55 A. R. Hawkins, W. Wu, P. Abraham, K. Streubel and J. E. Bowers, "High Gain-Bandwidth-Product Silicon Heterointerface Photodetector." Appl. Phys. Lett. **70**, 303 (1997).

56 M. Ghioni, A. Lacaita, G. Ripamonti and S. Cova, "All-Silicon Avalanche Photodiode Sensitive at 1.3um with Picosecond Time Resolution." IEEE J. Quantum Elec. **28**, 2678 (1992).

57 A. R. Hawkins, T. E. Reynolds, D. R. England, D. I. Babic, M. J. Mondry, K. Streubel and J. E. Bowers, "Silicon Heterointerface Photodetector." Appl. Phys. Lett. **68**, 3692 (1996).

58 B. F. Levine, A. R. Hawkins, S. Hiu, B. J. Tseng, C. A. King, L. A. Gruezke, R. W. Johnson, D. R. Zolnowski and J. E. Bowers, "20 GHz High Performance Planar Si/InGaAs p-i-n Photodetector." Appl. Phys. Lett. **70**, 2449 (1997).

59 D. J. Oostra, R. V. Smilgys and S. R. Leone, "Initial Stages of Heteroepitaxial Growth of InAs on Si (100)." Appl. Phys. Lett. **55**, 1333 (1989).

60 R. Leon, C. Lobo, T. P. Chin, J. M. Woodall, S. Fafard, S. Ruvimov, Z. Liliental-Weber and M. A. Steves Kalceff, "Self-Forming InAs/GaP Quantum Dots by Direct Island Growth." Appl. Phys. Lett. **72**, 1356 (1998).

61 F. Heinrichsdorff, A. Krost, N. Kirstaedter, M. H. Mao, M. Grundmann, D. Bimberg, A. O. Kosogov and P. Werner, "InAs/GaAs Quantum Dots Grown by Metalorganic Chemical Vapor Deposition." Jpn. J. Appl. Phys. **36**, 4129 (1997).

62 V. Gopal, E. P. Kvam, T. P. Chin and J. M. Woodall, "Evidence for Misfit Dislocation-Related Carrier Accumulation at the InAs/GaP Heterointerface." Appl. Phys. Lett. **72**, 2319 (1998).

63 S. Zamir, B. Meyler and J. Salzman, "Thermal Microcrack Distribution Control in GaN Layers on Si Substrates by Lateral Confined Epitaxy." Appl. Phys. Lett. **78**, 288 (2001).

64 ANSYS . ANSYS is a registered trademark of ANSYS, Inc. (2000).

65 W. Pompe, A H, Bahr, G. Gille, W. Kreher, B. Schultrich and H. J. Weiss, "Mechanical Properties of Brittle Materials Modern Theories and Experimental Evidence." *Current Topics in Materials Science*. Ed. E. Kaldis. New York, Elsevier Science Publishers. **12:** 208 (1985).

66 G. C. Sih, *Mechanics of Fracutre Initiation and Propagation*. Dordrecht, Kluwer Academic Publishers (1991).

67 Jr. J. L. Beuth, "Cracking of Thin Bonded Films in Residual Tension." Int. J. Solids Structures **29**, 1657 (1992).

68 M. S. Hu and A. G. Evans, "The Cracking and Decohesion of Thin Films on Ductile Substrates." Acta Metall. **37**, 917 (1989).

69 M. Natali, D. De Salvador, M. Berti, A. V. Drigo, L. Lazzarini, G. Salviati, G. Rossetto and G. Torzo, "Crack Formation in Tensile InGaAs/InP Layers." J. Vac. Sci. Tech. B **18**, 2527 (2000).

70 S. M. Hu, "Film-Edge Induced Stress in Substrates." J. Appl. Phys. **50**, 4661 (1979).

71 R. Fu, Z. Q. Mu, C. W. Lung and S. H. Ai, "Fracta l Characterization of Fatique Threshold." Scripta Metallurgica et Materialia **25**, 1647 (1991).

72 J. W. Hutchinson and Z. Suo, "Mixed Mode Cracking in Layered Materials." *Advances in Applied Mechanics*. Ed. T. Y. Wu. Boston, Academic Press, Inc. **29:** 126 (1992).

73 M. Zhao, R. Fu and T. Zhang, "Multiple Cracks of a Thin Film on a Ductile Substrate." Mat. Res. Soc. Symp. Proc. **695**, L9.3.1 (2002).

74 R. T. Murray, C. J. Kiely, M. Hopkinson and P. J. Goodhew, "Crack Formation in Tensile Strained III-V Epilayers Grown on InP Substrates." Institute of Physics Conference, 207 (1995).

75 Jr. J. L. Beuth, "Fracture Analyses of Thin Films, Interfaces, and Joints," PhD, Harvard University (1992).

76 S. Suresh, *Fatigue of Materials*. Cambridge, UK, Cambridge University Press (1998).

77 T. Ye, Z. Suo and A. G. Evans, "Thin Film Cracking and the Roles of Substrate and Interface." Int. J. Solids Structures **29**, 2639 (1992).

78 M. D. Thouless, "Crack Spacing in Brittle Films on Elastic Substrates." J. Am. Ceram. Soc. **73**, 2144 (1990).

79 R. G. Linford and L. A. Mitchell, "Interplanar Potential Functions for Calculation Surface Energies and Cohesive Stress." Surf. Sci. **27**, 142 (1971).

80 C. Maucione, "Surface Energy of Germanium," M. S., Michigan Technological University (1967).

81 J. J. Gilman, Ed. *Fracture*. New York, Wiley (1959).

82 J. E. Sinclair and B. R. Lawn, "An Atomistic Study of Cracks in Diamond-Structure Crystals." Proc. Royal Soc. London, Ser. A **329**, 83 (1972).

83 C. L. Reynolds, P. R. Couchman and F. E. Karasz, "On the Relation Between Surface Energy, Melting temperature and Interactomic Separation for Metals." Phil. Mag. **34**, 659 (1976).

84 Y. Honda, Y. Kuroiwa, M. Yamaguchi and N. Sawaki, "Growth of GaN Free from Cracks on a (111)Si Substrate by Selective Metalorganic Vapor-Phase Epitaxy." Appl. Phys. Lett. **80**, 222 (2002).

85 G. L. Price, "Critical-Thickness and Growth-Mode Transitions in Highly Strained $In_xGa_{1-x}As$ Films." Phys. Rev. Lett. **66**, 469 (1991).

86 M. Putero-Vuaroqueaux, N. Burle and B. Pichaud, "Metastability in the Matthews-Blakeslee Mechanism for Semiconductor Film Relaxation." Phil. Mag. Lett. **81**, 519 (2001).

87 J. R. Downes, D. J. Dunstan and D. A. Faux, "Analysis of the Shortcomings of the Matthews-Blakeslee Theory of Critical Thickness at Higher Strains." Phil. Mag. Lett. **76**, 77 (1997).

88 T. Nishimura, K. Kadoiwa, N. Hayafuji, M. Miyashita, K. Mitsui, H. Kumabe and T. Murotani, "Surface Morphology Improvement of GaAs-on-Si Using a Two-Reactor MOCVD System and an AlAs/GaAs Low Temperature Buffer Layer: an Approach to Crack-Free GaAs-onSi." J. Cryst. Growth **107**, 468 (1991).

89 B. G. Yacobi, S. Zemon, P. Norris, C. Jagannath and P. Sheldon, "Stress Variations Due to Microcracks in GaAs Grown on Si." Appl. Phys. Lett. **51**, 2236 (1987).

90 X. W. Liu, A. A. Hopgood, B. F. Usher, H. Wang and N. St J Braithwaite, "Formation of Misfit Dislocaitons During Growth of $In_xGa_{1-x}As/GaAs$ Strained-Layer Heterostructure." Semicond. Sci. Technol. **14**, 1154 (1999).

91 R. T. Murray, C. J. Kiely and M. Hopkinson, "Crack Interactions in Tensile-Strained Epilayers." Institute of Physics Conference, 169 (1997).

92 G. Schoeck, "Dislocation Emission from Crack Tips as a Variational Problem of the Crack Energy." J. Mech. Phys. Solids **44**, 413 (1996).

93 N. Hayafuji, H. Kizuki, M. Miyashita, K. Kadoiwa, T. Nishimura, N. Ogasawara, H. Kumabe, T. Murotani and A. Tada, "Crack Propagation and Mechanical Fracture in GaAs-on-Si." Jpn. J. Appl. Phys. Pt 1 **30**, 459 (1991).

94 S. L. Yellen, A. H. Shepard, R. J. Dalby, J. A. Baumann, H. B. Serreze, T. S. Guido, R. Soltz, K. J. Bystrom, C. M. Harding and R. G. Waters, "Reliability of

GaAs-Based Semiconductor Diode Lasers: 0.6-1.1um." IEEE J. of Quantum Elec. **29**, 2058 (1993).

95 E. A. Fitzgerald, A. Y. Kim, M. T. Currie, T. A. Langdo, G. Taraschi and M. T. Bulsara, "Dislocation dynamics in relaxed graded composition semiconductors." Mater. Sci. and Eng. **B67**, 53 (1999).

96 B. G. Yacobi and D. B. Holt, "Cathodoluminescence Scanning Electron Microscopy of Semiconductors." J. Appl. Phys. **59**, R1 (1986).

97 L. A. Coldren and S. W. Corzine, *Diode Lasers and Photonic Integrated Circuits*. New York, John Wiley and Sons, Inc. (1995).

98 J. Singh, Ed. *Properties of Lattice-Matched and Strained Indium Gallium Arsenide*. London, INSPEC, the IEE (1993).

99 R. M. Kolbas, N. G. Anderson, W. D. Laidig, Y. Sin, Y. C. Lo, K. Y. Hsieh and Y. J. Yang, "Strained-Layer InGaAs-GaAs-AlGaAs Photopumped and Current Injection Lasers." IEEE J. of Quantum Elec. **24**, 1605 (1988).

100 U. Das and P. K. Bhattacharya, "Variation of Refractive Index in Strained $In_xGa_{1-x}As$-GaAs Heterostructures." J. Appl. Phys. **58**, 341 (1985).

101 R. Beresford, J. Yin, K. Tetz and E. Chason, "Real-Time Measurements of Stress Relaxation in InGaAs/GaAs." J. Vac. Sci Technol. B **18**, 1431 (2000).

102 D. J. Dunstan, "Strain and Strain Relaxation in Semiconductors." J. Mater. Sci.: Mataer. Electron. **8**, 337 (1997).

103 G. Li, E. Yuceturk, D. Huang and S. C. Esener, "Analysis of Free-Space Optical Interconnects for the Three-Dimensional Optoelectronic Stacked Processor." Opt. Com. **202**, 319 (2002).

104 M. W. Haney, M. P. Christensen, P. Milojkovic, J. Ekman, P. Chandramani, R. Rozier, F. Kiamilev, Y. Liu and M. Hibbs-Brenner, "Multichip Free-Space Global Optical Interconnection Demonstration with Integrated Arrays of Vertical-Cavity Surface-Emitting Lasers and Photodetectors." Appl. Optics **38**, 6190 (1999).

105 D. V. Plant, M. B. Venditti, E. Laprise, J. Faucher, K. Razavi, M. Chateauneuf, A. G. Kirk and J. S. Ahearn, "256-Channel Bidirectional Optical Interconnect Using VCSELs and Photodiodes on CMOS." J. Lightwave Tech. **19**, 1093 (2001).

106 A. V. Krishnamoorthy and K. W. Goossen, "Optoelectronic-VLSI: Photonics Integrated with VLSI Circuits." IEEE J. Selec. Topics in Quan. Elec. **4**, 899 (1998).

107 D. W. Prather, S. Venkataraman, M. Lecompte, F. Kiamilev, J. N. Mait and G. J. Simonis, "Optoelectronic Multichip Module Integration for Chip Level Optical Inteconnects." IEEE Photonics Tech. Lett. **13**, 1112 (2001).

108 B. Wittmann, S. Lehmacher, S. Kopecz and A. Neyer, "Optical Interconnects on and in Printed Circuit Boards." Int. J. Electron. Commun. **55**, 319 (2001).

109 H. Thienpont, C. Baukens, B. Volckaerts, H. Ottevaere, C. Debaes, M. Vervaeke, P. Tuteleers, P. Vynck, A. Hermanne, M. Hanney, M. Brunfaut, J. M. van Campenhout and I. Veretennicoff, "Free-Space Micro-Optical Modules: the Missing Link for Photonic Interconnects to Silicon Chips." Opto-Electronics Review **9**, 238 (2001).

110 R. J. Deri, "Monolithic Integration of Optical Waveguide Circuitry with III-V Photodetectors for Advanced Lightwave Receivers." J. Lightwave Tech. **11**, 1296 (1993).

111 St. Kollakowski, E. H. Böttcher, Ch. Lemm, A. Strittmatter, D. Bimberg and H. Kräutle, "Waveguide-Integrated InP-InGaAs-InAlGaAs MSM Photodetector with Very-High Vertical-Coupling Efficiency." IEEE Photonics Tech. Lett. **9**, 496 (1997).

112 W. Döldissen, F. Fiedler, R. Kaiser and L. Mörl, "Butt Coupled Photodiodes Integrated with Y-Branched Optical Waveguides on InP." Elec. Lett. **25**, 35 (1989).

113 J. B. D. Soole, Y. Silberberg, A. Scherer, H. P. LeBlanc, N. C. Andreadakis, C. Caneau, H. Schumacher and U. Erben, "Fast High-Efficiency Integrated Waveguide Photodetectors Using Novel Hybrid Vertical/Butt Coupling Geometry." Appl. Phys. Lett. **61**, 13 (1992).

114 J. L. Merz and R. A. Logan, "Integrated GaAs-$Al_xGa_{1-x}As$ Injection Lasers and Detectors with Etched Reflector." Appl. Phys. Lett. **30**, 530 (1977).

115 G. Bekefi and A. H. Barrett, *Electromagnetic Vibrations, Waves, and Radiation*. Cambridge, MA, The MIT Press (1979).

116 B. E. A. Saleh and M. C. Teich, "Fundamentals of Photonics. Ed. New York, John Wiley & Sons, Inc: 248 (1991).

117 R. J. Deri and E. Kapon, "Low-Loss III-V Semiconductor Optical Waveguides." IEEE J. Quantum Elec. **27**, 626 (1991).

118 G. M. Martin, "Optical Assessment of the Main Electron Trap in Bulk Semi-Insulating GaAs." Appl. Phys. Lett. **39**, 747 (1981).

119 G. Taraschi, Z. Cheng, M. T. Currie, C. W. Leitz, T. A. Langdo, M. L . Lee, A. Pitera, E. A. Fitzgerald, J. L. Hoyt and D. A. Antoniadis, "Relaxed SiGe on Insulator Fabricated Via Wafer Bonding and Layer Transfer: Etch-Back and Smart-Cut Alternatives." Electrochemical Society Tenth International Symposium on Silicon-on-Insulator Technology and Devices, Washington, D. C.(2001).

120 A. Ackaert, L. Buydens, D. Lootens, P. van Daele and P. Demeester, "Crack Formation and Selective Growth in MOVPE-GaAs on Si and its Application to OEICs." 19th European Solid State Devices Research Conference, Berlin, 397 (1989).

121 A. Ackaert, L. Buydens, D. Lootens, P. Van Daele and P. Demeester, "Crack Formation and Thermal Stress Relaxation of GaAs on Si Growth by Metalorganic Vapor Phase Epitaxy." Appl. Phys. Lett. **55**, 2187 (1989).

122 A. J. Pitera. personal communication (2002).